universität freiburg

Intelligent Embedded Systems Lab

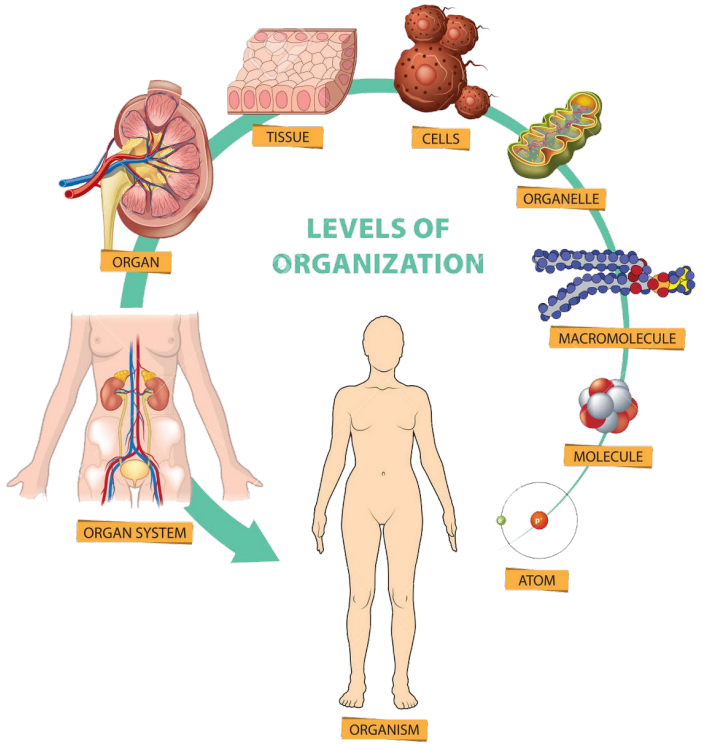
Overview AI Methods and Applications in Healthcare

Digital Health

IES Lab
Dr. Mario Cypko
Freiburg, 02. November 2026

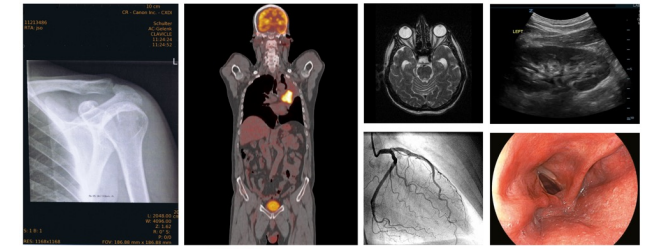
Reminder

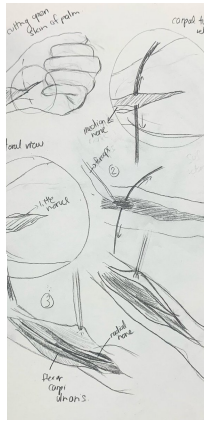
Reminder – Medical data

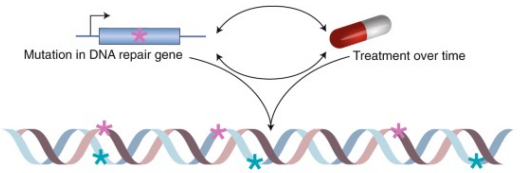


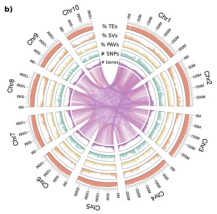
Patient's last name:	First:	Middle:	❑ Mr. ❑ Mrs.	❑ Miss ❑ Ms.	Marital status (circle one) Single / Mar / Div / Sep / Wid
Is this your legal name? ❑ Yes ❑ No	If not, what is your legal name?	(Former name):	Birth date:	Age:	Sex: ❑ M ❑ F
Street address:	Social Security no.:	Home phone no.:			

KLINISCHE CHEMIE					
3555	Calcium im Serum	2.46	mMol/l 1)	2.00-2.65	[----*--]
3556	Chlorid im Serum	101	mMol/l	95-108	[---*---]
3620	Eisen im Serum	84	µg/dl	70-150	[-*-----]
3557	Kalium im Serum	5.25	mMol/l	3.60-5.50	[-----*-]
3558	Natrium im Serum	137	mMol/l	134-152	[-*-----]
3580H1	Phosphat im Serum	1.13	mMol/l	0.80-1.50	[---*---]

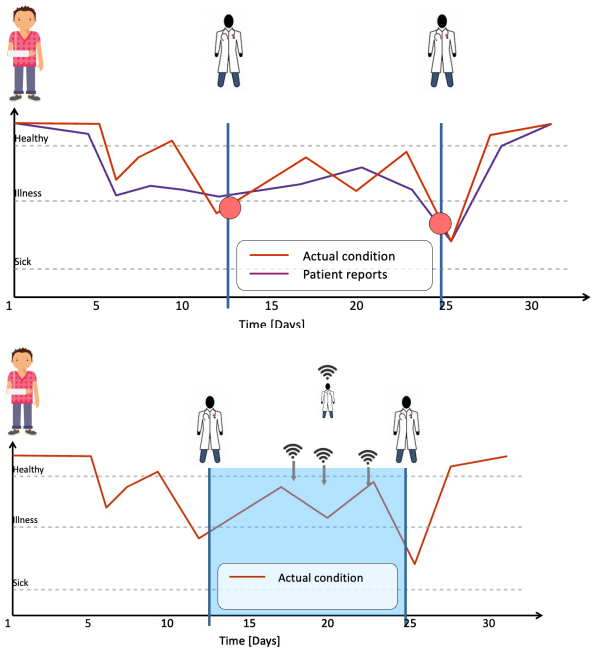








Reminder – Mobile data



mart ring (Oura)

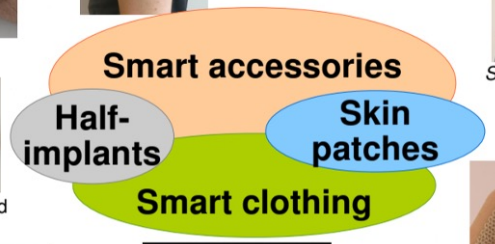
Smart watch (~2011)

QBIC: Q-Belt Integrated Computer (2004)

Smart phoes

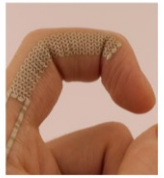
SimpleSkin: Textile sensor building blocks (2017)

BodyMedia arm band (~2010)

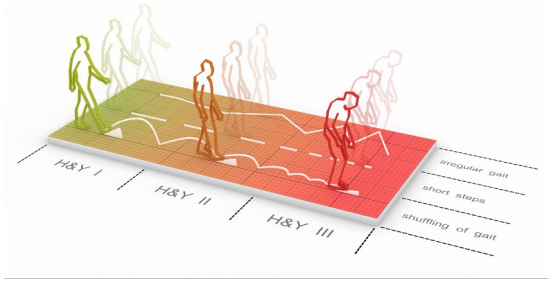


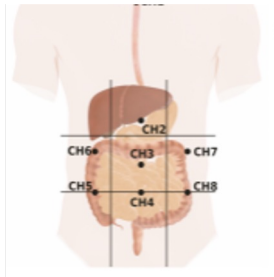
Widex in-ear hearing aid (2015)

Smart eyeglasses (2015-17)


Printed interactive skin electronics (2017, Courtesy Prof. Steimle).

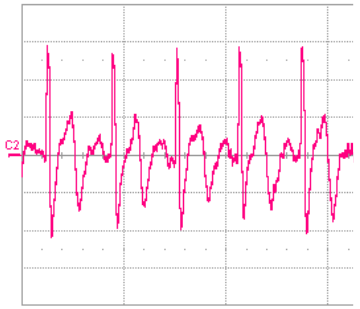
Amft & v Learhoven, IEEE Perv Comput 16(4), '17

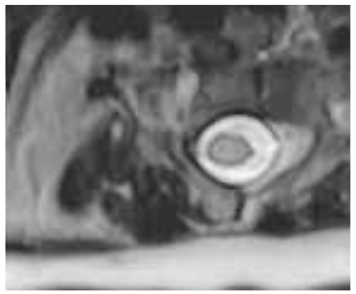


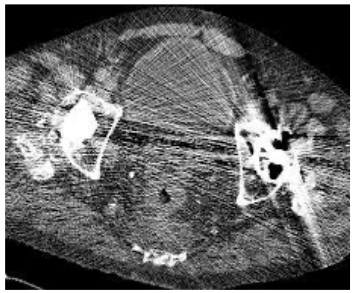


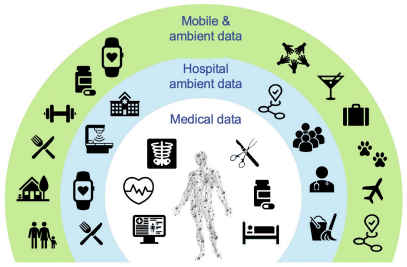
Reminder – Ambient data

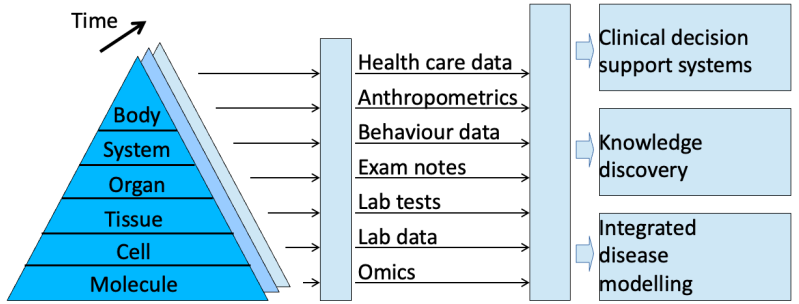
Reminder – Challenges / Opportunities

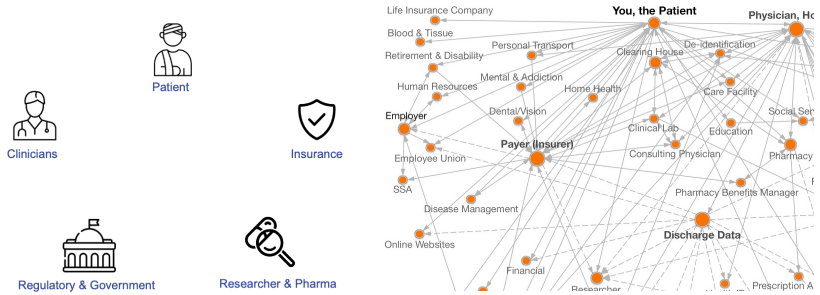


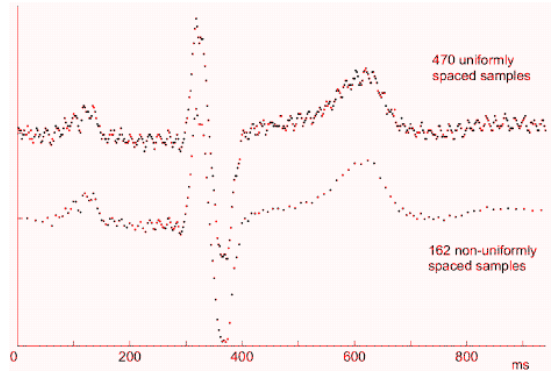


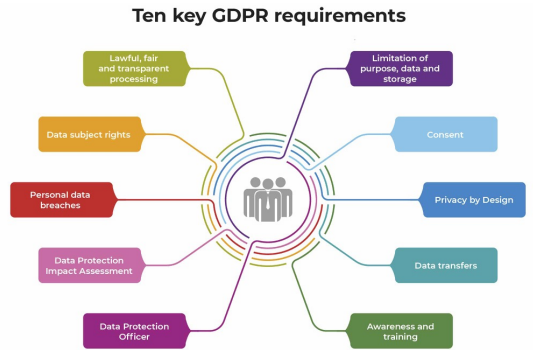






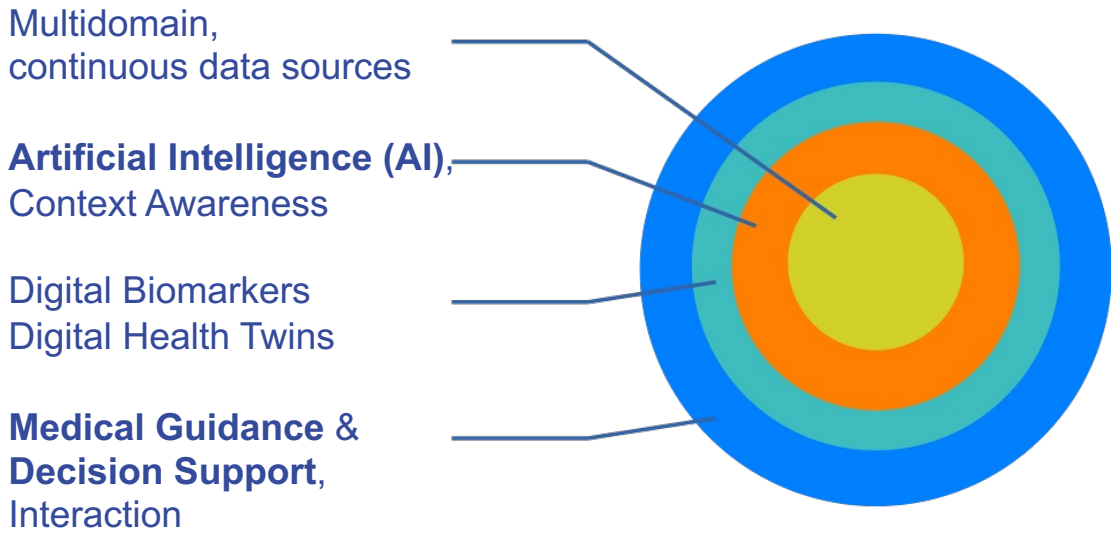






Agenda

1. Reminder
2. Vision and overview
3. Traditional ML – Clustering
4. Deep Learning with CNNs & GANs
5. RNNs & LSTMs
6. LLMs & VLMs
7. Explainable AI
8. Supervised ML and knowledge-based AI
9. Bayesian Networks
10. Data Centric AI



Understanding you.

1. How many AI-related lecutres have you attended?

2. What AI methods have you applied sofar?

3. In what domains have you applied AI methods?

4. How deep is your AI understanding?
(1: new to AI ←→ 7: AI developer)

pingo.coactum.de/815896

Vision and Overview

MedAIM - Medical AI and Modeling Group

Medical Support in Remote (Extreme) Environments

Primary Patient Care Facility

Highly Specialized Private Clinics

Central City Hospital

Clinical Decision-Support Command Suite

Patient Care at individual Departments, Patient Ward, ORs and ICU

MedAIM - Medical AI and Modeling Group

Medical Support in Remote (Extreme) Environments

Continues and Situational - Mobile and Ambient Health Tracking

Home

Patient Check-in

Ambulance & Car

Highly Specialized
Private Clinics

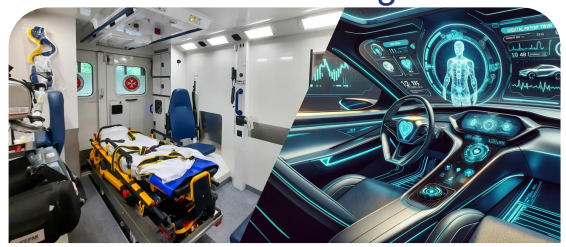
Clinical Decision-Support Command Suite

Patient Care at individual Departments,
Patient Ward, ORs and ICU

MedAIM - Medical AI and Modeling Group

Medical Support in Remote (Extreme) Environments

Continues and Situational - Mobile and Ambient Health Tracking



Motivation for AI in healthcare

- Alerts
- Monitoring
- Diagnostic
- Risk prediction
- Decision and workflow support and optimization
- (Drug) Discovery
- Personalization
- Fraud detection

Primary Patient Care Facility

Highly Specialized Private Clinics

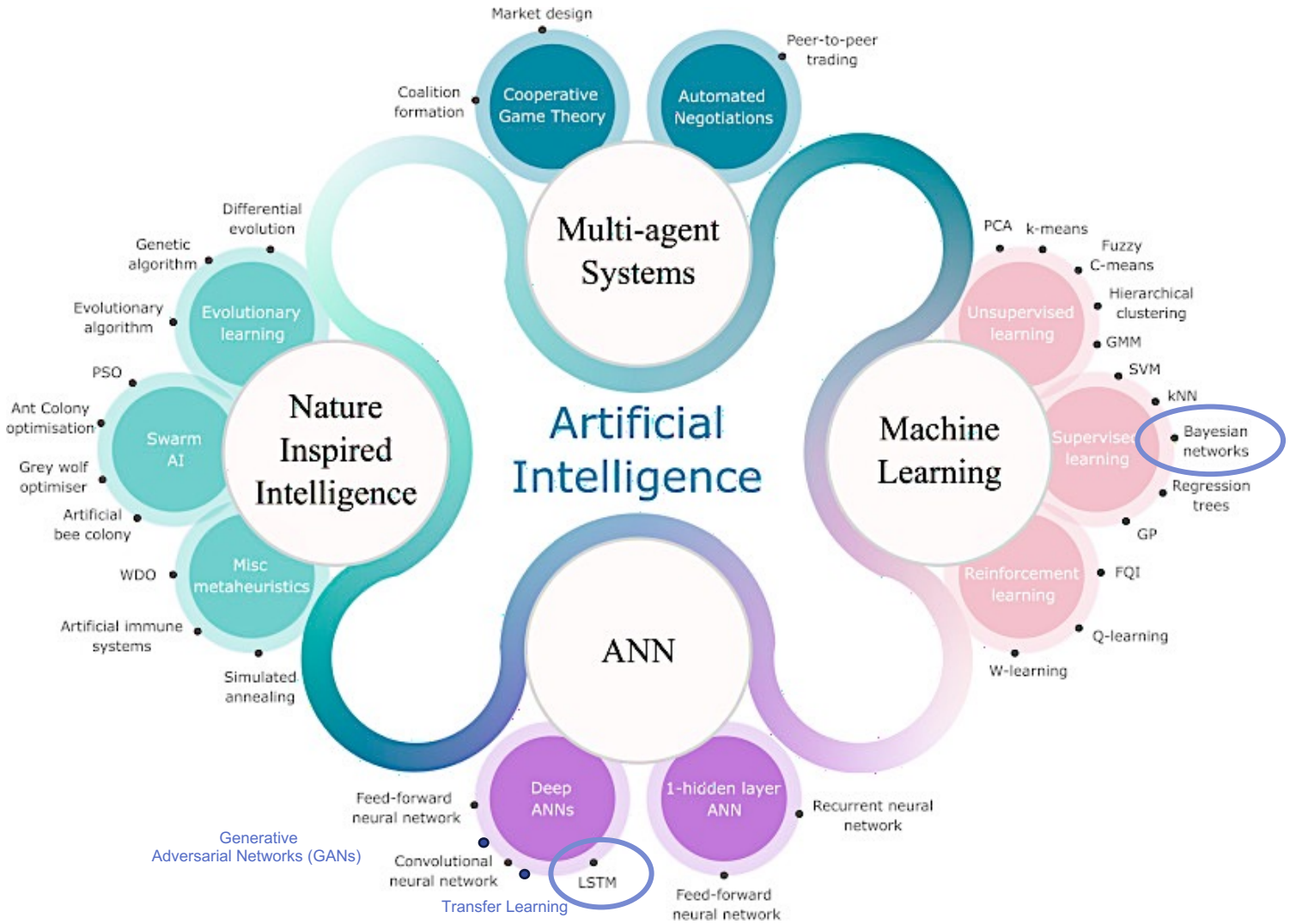
Central City Hospital

Credit: Carlos Armato

Clinical Decision-Support Command Suite

Patient Care at individual Departments, Patient Ward, ORs and ICU

AI Methods



Motivation for AI in healthcare

- Alerts
- Monitoring
- Diagnostic
- Risk prediction
- Decision and workflow support and optimization
- (Drug) Discovery
- Personalization
- Fraud detection

Modified from Antonopoulos, Ioannis, et al. "Artificial intelligence and machine learning approaches to energy demand-side response: A systematic review." *Renewable and Sustainable Energy Reviews*130 (2020): 109899.

AI Methods

Patient: XXX Creation Date: 08/09/2016 Record: XXX
Birth Date: XXX Gender: Female
Exam requested: US Bilateral Breast
Applicant name: XXX

Breast ultrasound with Axillary Extension Evaluation

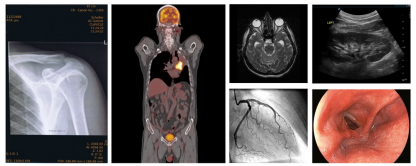
The exam was performed using a multifrequential linear transducer Frequency of 7.5 MHz. Normal thickness Skin with ecotexture. Subcutaneous cellular tissue and retrograde musculature without alterations. Biotextura mammary tissue of homogeneous background, with predominance of adipose tissue. Presence of vertical nodule, irregular shape, microlobulated margins, hypoechoic echogenicity in relation to fat, posterior acoustic findings without

Unstructured text data

KLINISCHE CHEMIE

3555	Calcium im Serum	2.46	mMol/l 1)	2.00-2.65	[----*--]
3556	Chlorid im Serum	101	mMol/l	95-108	[---*---]
3620	Eisen im Serum	84	µg/dl	70-150	[-*------]
3557	Kalium im Serum	5.25	mMol/l	3.60-5.50	[------*-]
3558	Natrium im Serum	137	mMol/l	134-152	[-*------]
3580H1	Phosphat im Serum	1.13	mMol/l	0.80-1.50	[---*---]

Structured data



Medical Imaging Data

Decision Tree
NLP techniques
Bayesian network
Markov random fields
SVM
Time Series Analysis
CNN
GAN
RNN
LSTM
Clustering

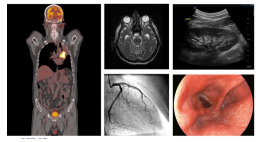
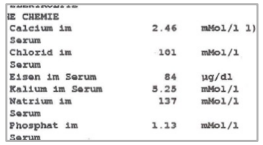
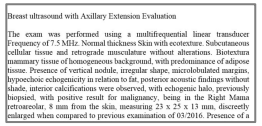
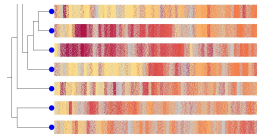
Sequence data

Time Series Data

Motivation for AI in healthcare

- Alerts
- Monitoring
- Diagnostic
- Risk prediction
- Decision and workflow support and optimization
- (Drug) Discovery
- Personalization
- Fraud detection

AI Methods

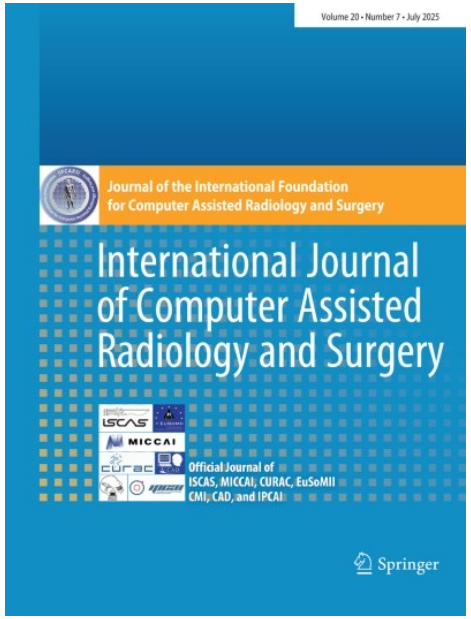
	Data Type	Examples from Healthcare	Typical AI Methods	Recommended Applications/Challenges
	**Imaging Data**	MRIs; CT Scans; X-rays; Ultrasound	CNN; “Transfer Learning”, VLMs	Classification; Segmentation; Anomaly Detection; Predictive Modeling
	**Structured / Tabular Data**	Electronic Patient Records; Lab test results, Drug interactions; Patient medication histories	Decision Trees; SVM; Neural Networks; Bayesian Networks	Risk Prediction; Outcome Prediction; Survival Analysis; Clustering
	**Unstructured Text Data**	Doctor’s notes; Medical records, Medical transcripts	NLP techniques; RNN; LSTM, LLMs	Information Extraction; Sentiment Analysis; Risk Stratification; Text Classification
	Time Series Data /Audio Data	Heart rate from wearables; Activity levels; GPS data from mobile devices; Patient interviews; Coughing, breathing sounds	RNN; LSTM; Time Series Analysis	Time-to-Event Analysis; Predictive Modeling; Anomaly Detection; Forecasting
	**Sequence Data**	DNA Sequences; Protein Sequences	RNN; LSTM; Bayesian Networks	Sequence Prediction; Genomic Data Analysis; Mutation Prediction; Clustering

! This list is an example and by far not complete (neither in context of data types nor AI methods)

AI Methods | Examples in IJCARS 2024

Model types in IJCARS 2024, Vol. 19, Issue 1 and 2:

- 2D U-Net CNN model with Inception-ResNet-v2
- 3D model
- 3D printing model
- 3D ResNet and 3D DenseNet
- 3D U-Net
- 4D classification model
- Age prediction model
- AI phase recognition model
- AR model
- Bayesian neural networks (BNNs)
- Biomechanical model
- cGAN-based network
- Compensation model
- Convolutional neural networks
- Deep neural network (DNN) model
- Endoscopy model
- Ex vivo and in vivo OCT models
- Ex vivo middle ear models
- Inception-ResNet-v2
- Large language models
- RCNN model
- (SOTA) models
- Transformer-based artificial Intelligence (AI) model
- Temporal segmentation model
- Vascular model
- Vision transformers (ViTs) models
- YOLO v5 network

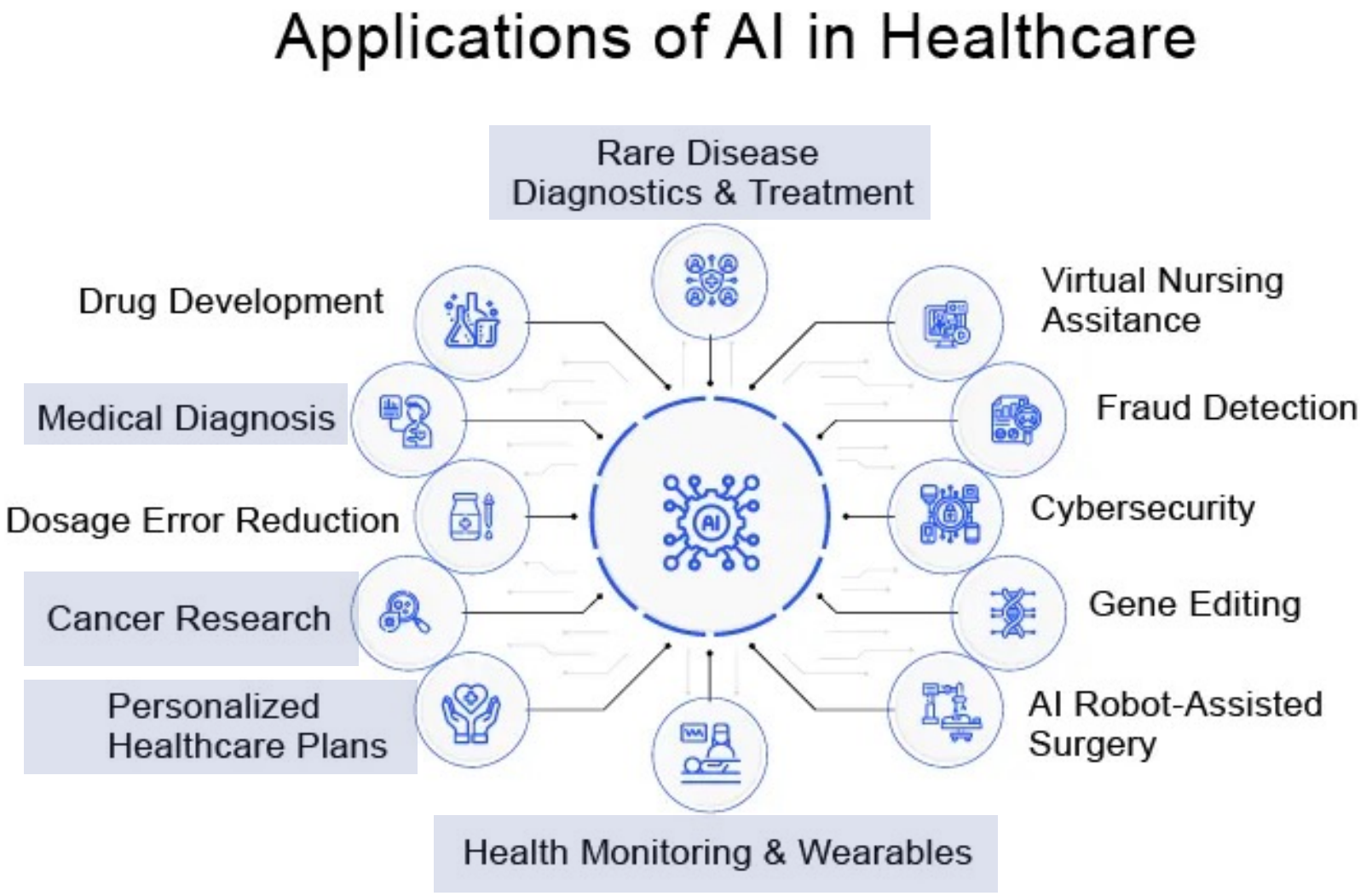


Model types in IJCARS 2024, Vol. 19, Issue 6:

- Computational fluid dynamics model
- DeepLabV3 model
- Diffusion probabilistic model
- Ellipsoidal model
- Free-form deformation model
- Frequency domain-based models (FF-ViT)
- Fully supervised models
- I3D model
- LaMa, fast Fourier convolution-based model
- LightGBM model
- Model with FP32 precision
- Multi-task model PitSurgRT
- Patient-specific models
- Personalized brightness-aware model
- Point-to-box teacher model
- PyTorch model
- Regression model
- Reinforcement learning (RL) models
- ResNeXt-50 model
- Segment anything model (SAM)
- Spline-based motion models
- State-of-the-art (SOA) ML models
- Supervised graphical models
- Multimodal models
- Tiny model I3D
- Video Swin models
- YOLO-based model

H.U. Lemke 2025

AI in healthcare beyond patient care



Reference: Top 12 AI Applications in Healthcare by https://www.delveinsight.com

Tasks in healthcare that benefit from AI

1. Large Data Volumes: Processing and analyzing vast amounts of data that would be overwhelming for humans.
2. Pattern Recognition: Identify patterns in data that are difficult for humans to see.
3. Automation: Performing repetitive tasks quickly and efficiently with / without human (co-op) intervention.
4. Real-time Analysis: Responding swiftly to data or signals in real-time, such as in the intensive care unit example.
5. Predictions: Forecasting future events or trends based on historical data.
6. Complex Problem Solving: Addressing complex problems that challenge conventional algorithms or are particularly complicated and require transparency.
7. Personalization: Tailoring services or products to individual users based on their preferences or behaviors.
8. Image and Video Analysis: Image recognition, facial recognition, and video surveillance.
9. Research and Development: Support and accelerate research in the discovery of new materials, medicines, or solutions to complicated scientific challenges.
10. Language Processing: Voice recognition and translating between languages.
11. Virtual Assistants: Assisting users in their daily tasks through chatbots or virtual personal assistants.
12. Security: Detecting anomalies or security breaches in networks.

Reference: Top 12 AI Applications in Healthcare by https://www.delveinsight.com

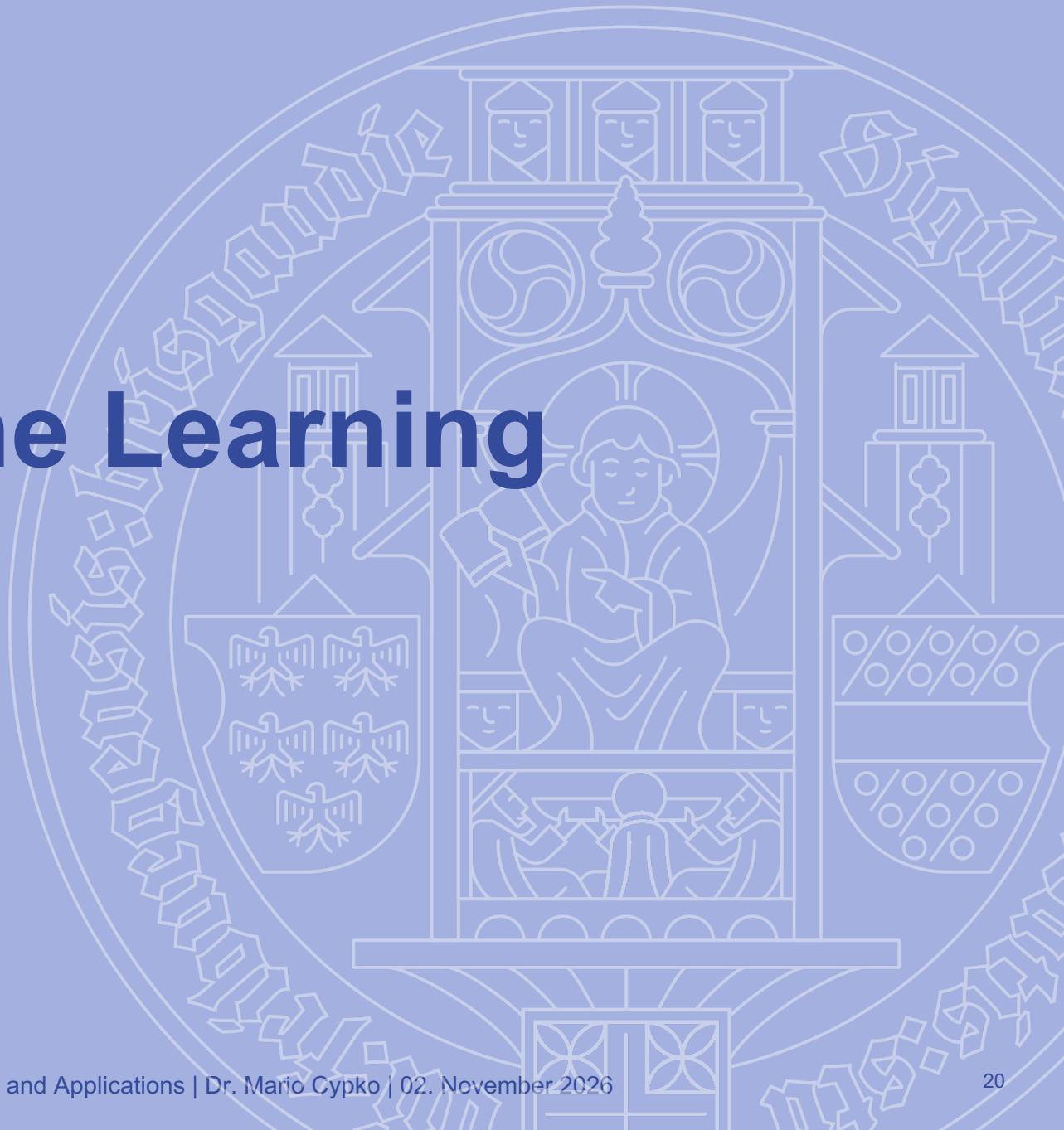
Activation!

5-9. I know quite well…

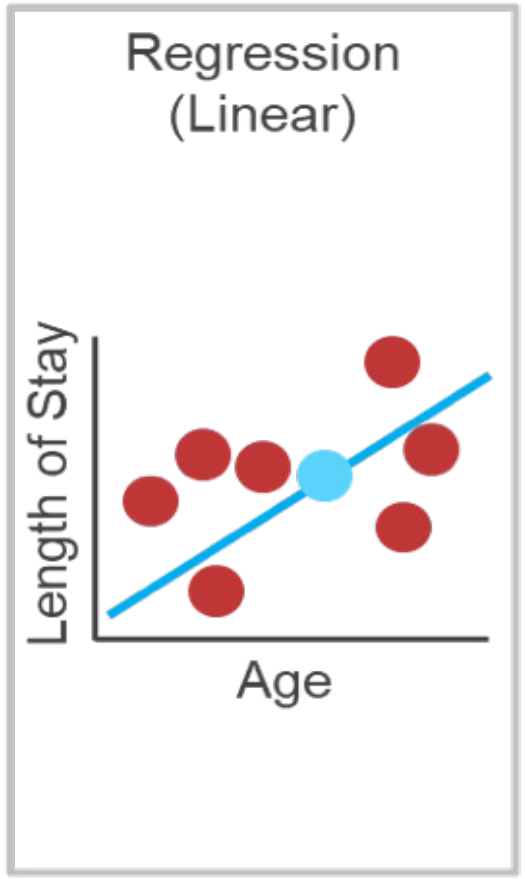
- machine learning quite well (regression/clustering)
- ANN & Deep Learning
- RNN & LSTM
- LLM, VLM (incl. RAG, prompt engineering, etc.)
- Knowledge-based AI
- Data-centric AI

pingo.coactum.de/815896

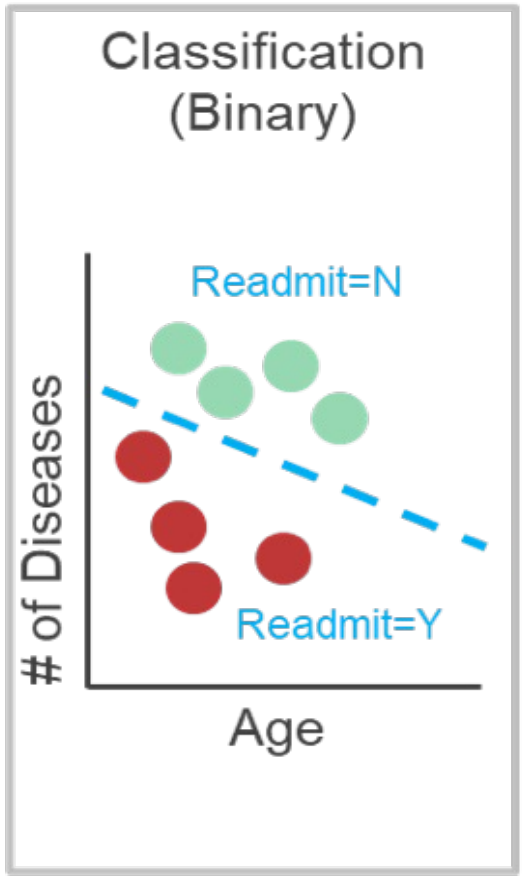
Traditional Maschine Learning



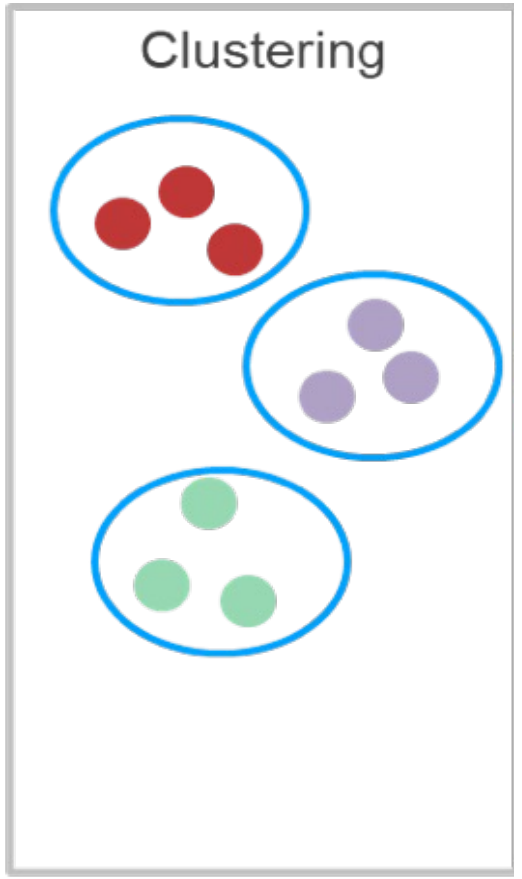
Traditional unsupervised and supervised machine learning



Length of Stay



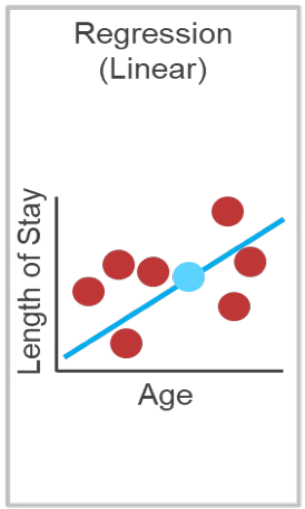
Readmit Yes/No



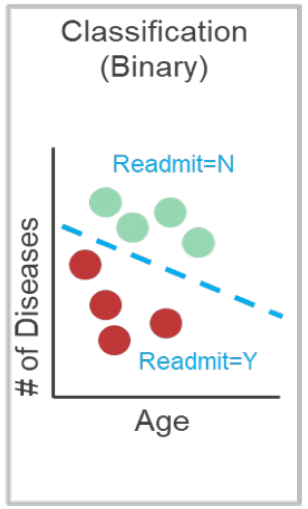
Diabetes Subgroups

Example by https://www.healthcatalyst.com

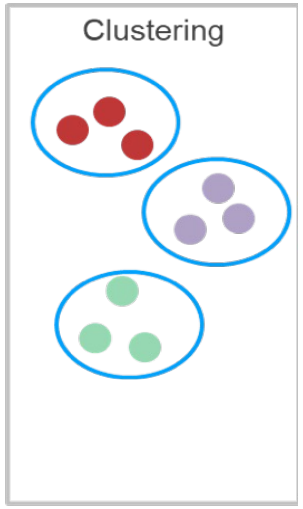
Clustering



Length of Stay

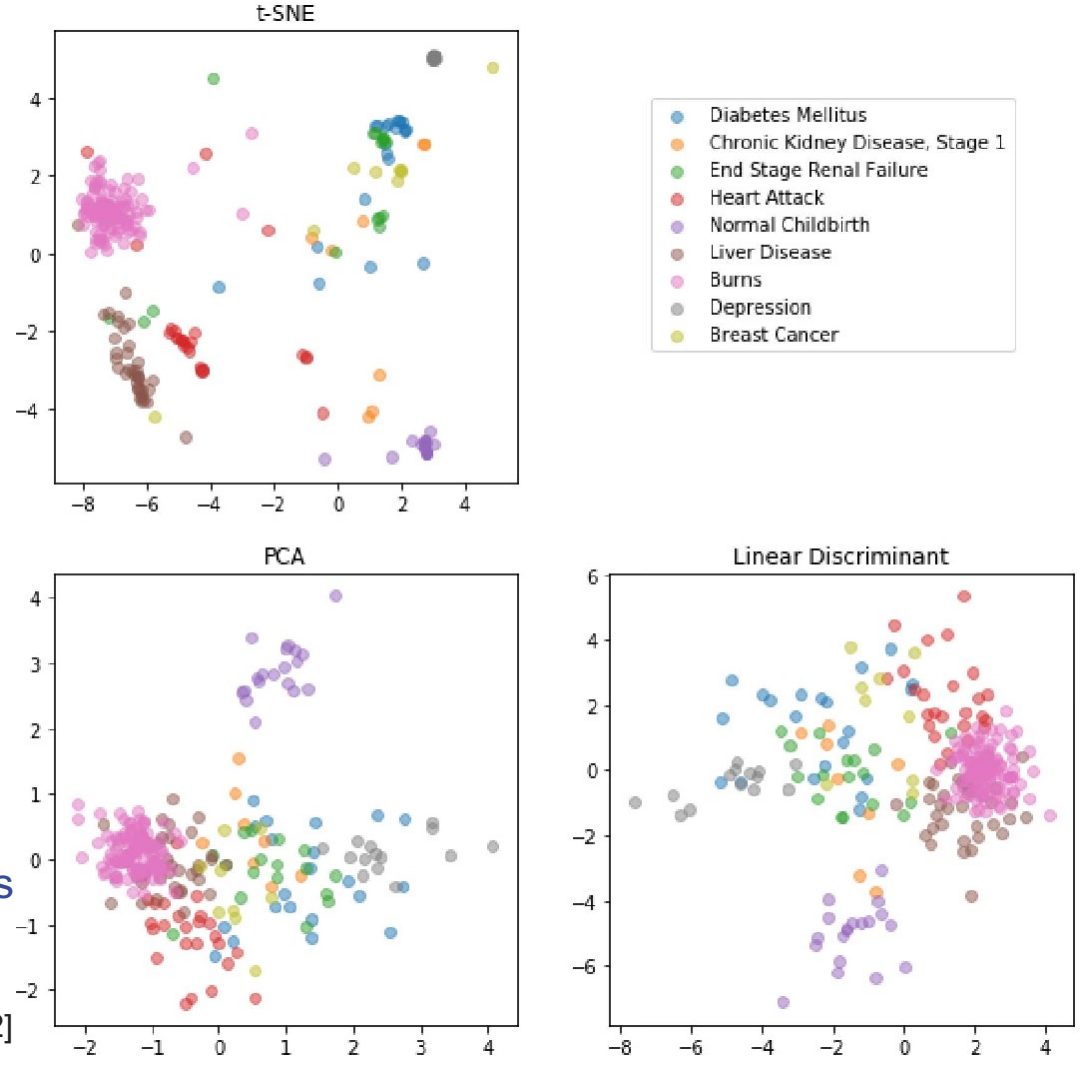


Readmit Yes/No

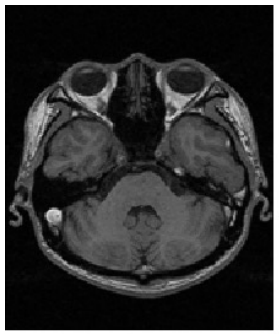


Diabetes Subgroups

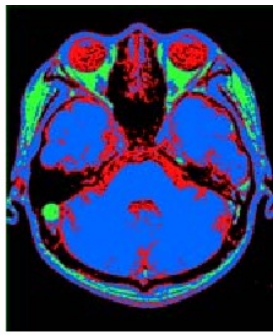
Example by https://www.healthcatalyst.com



[2]



(a)



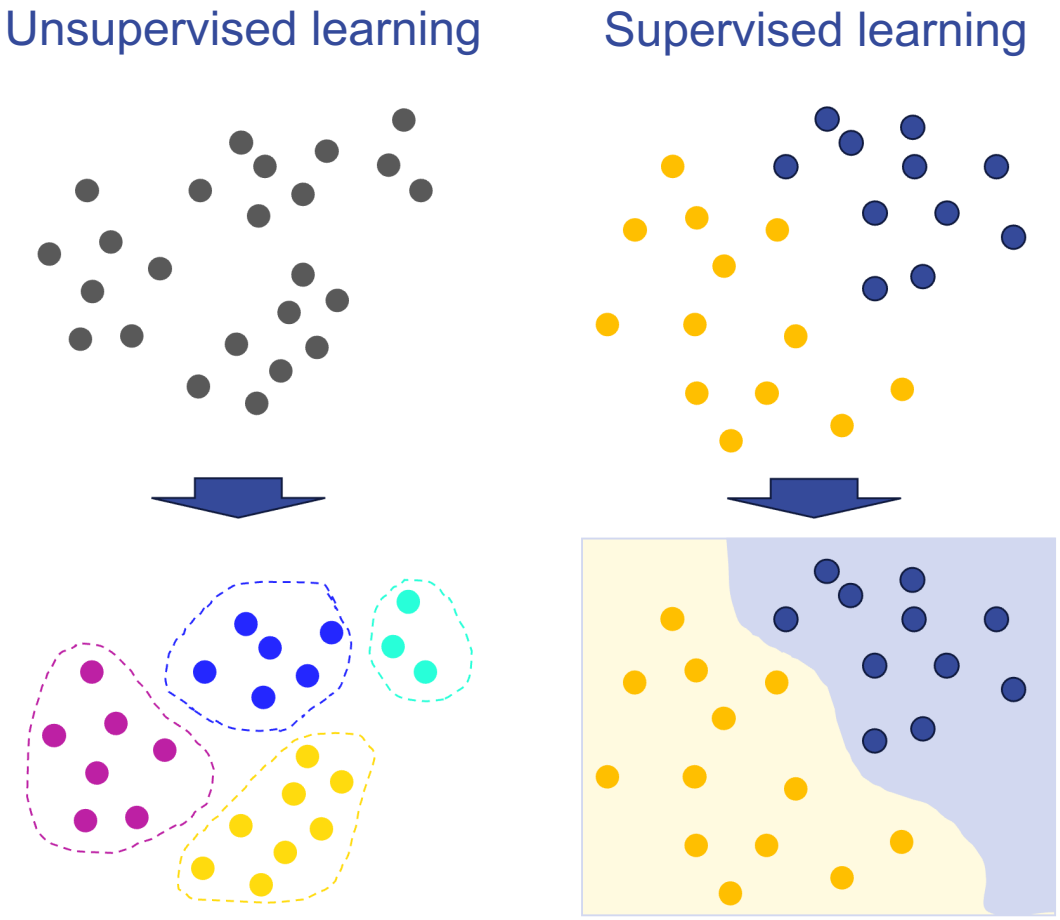
(b)

[1]

see also,
logistic regression
nonlinear regression
Support Vector Machines (SVM)

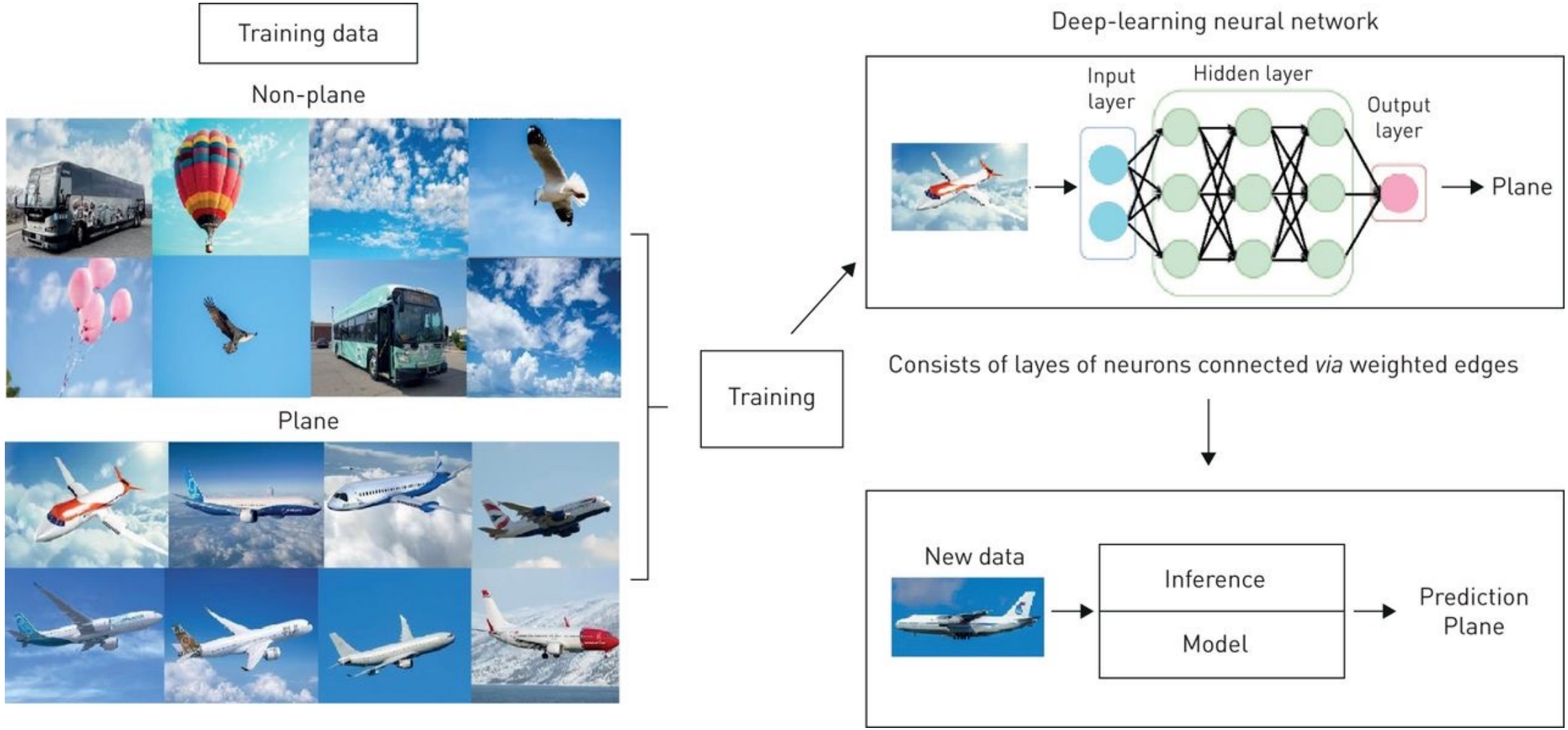
[1] H. P., et al. "Medical image segmentation using k-means clustering and improved watershed algorithm." *2006 IEEE southwest symposium on image analysis and interpretation*. IEEE, 2006.
[2] Kartchner, David, et al. "Code2vec: Embedding and clustering medical diagnosis data." *2017 IEEE International Conference on Healthcare Informatics (ICHI)*. IEEE, 2017.

Traditional unsupervised and supervised machine learning



ANN & Deep Learning

Neural Networks and Deep Learning



Convolution Neural Networks (CNNs)

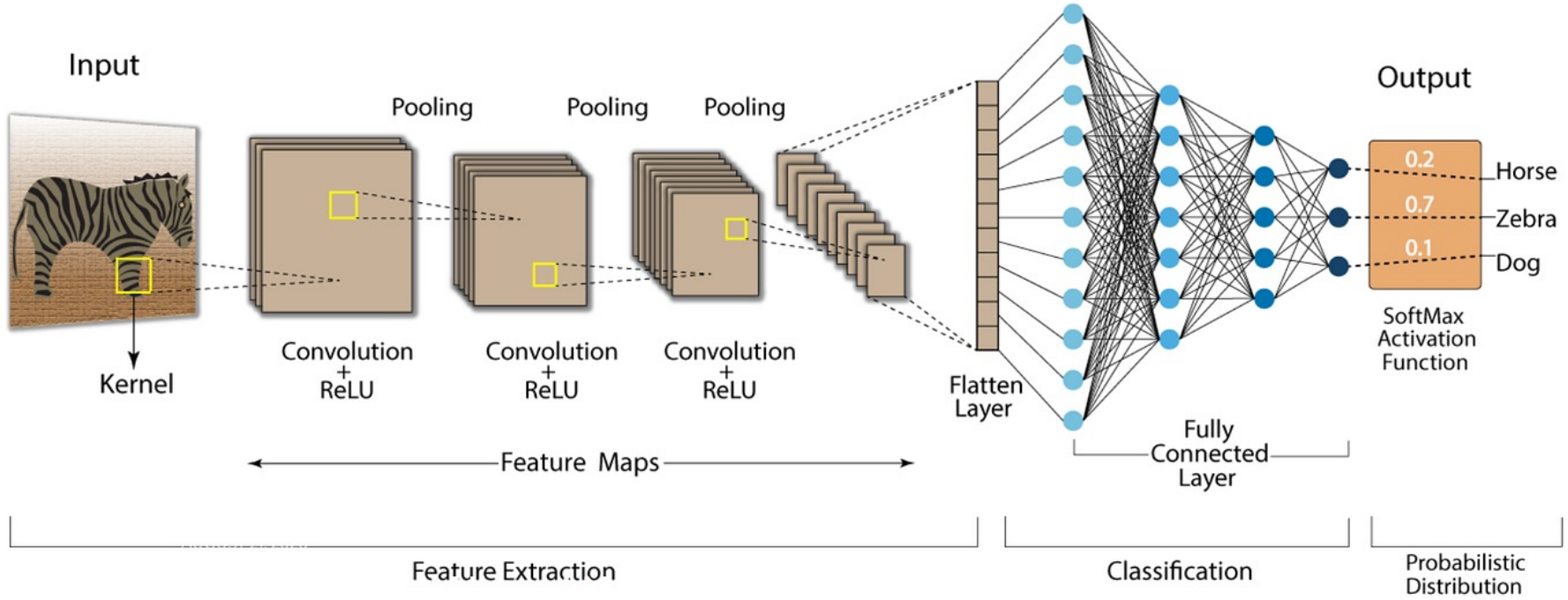
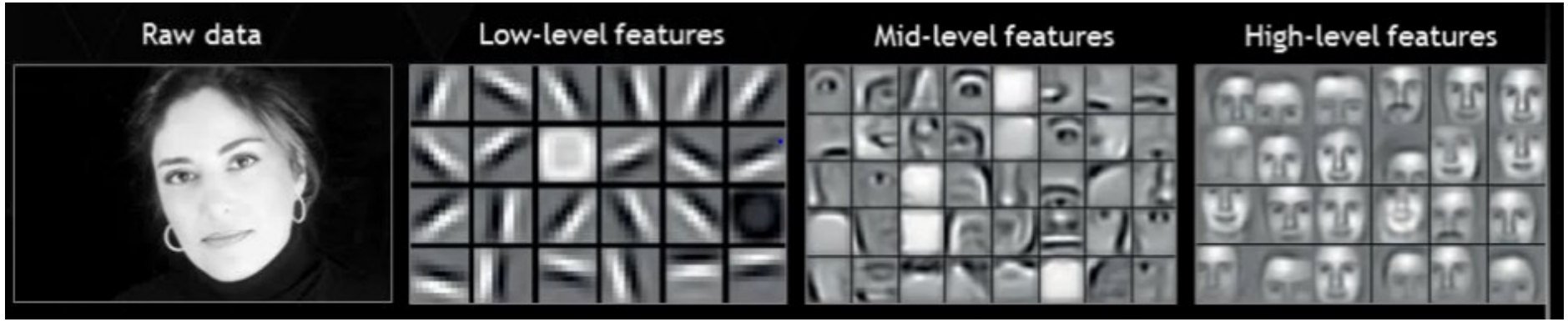
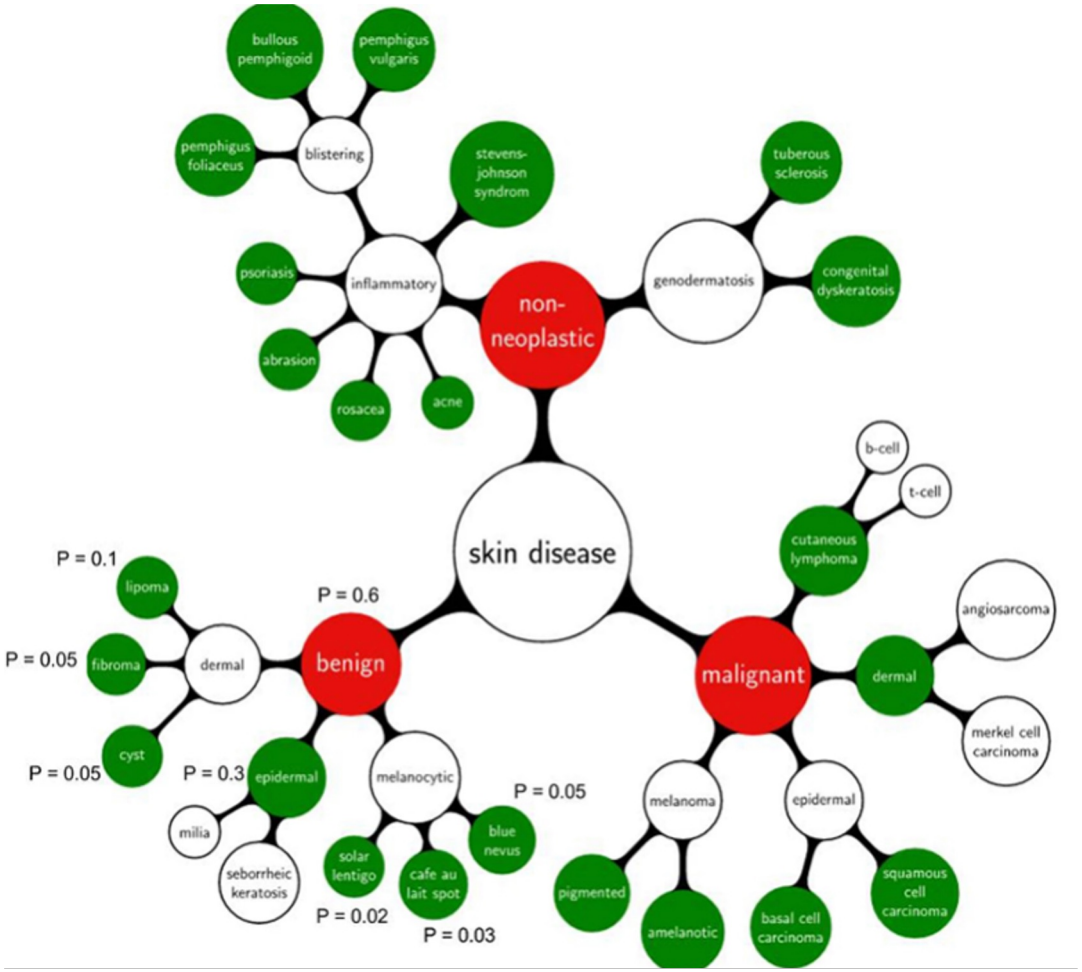
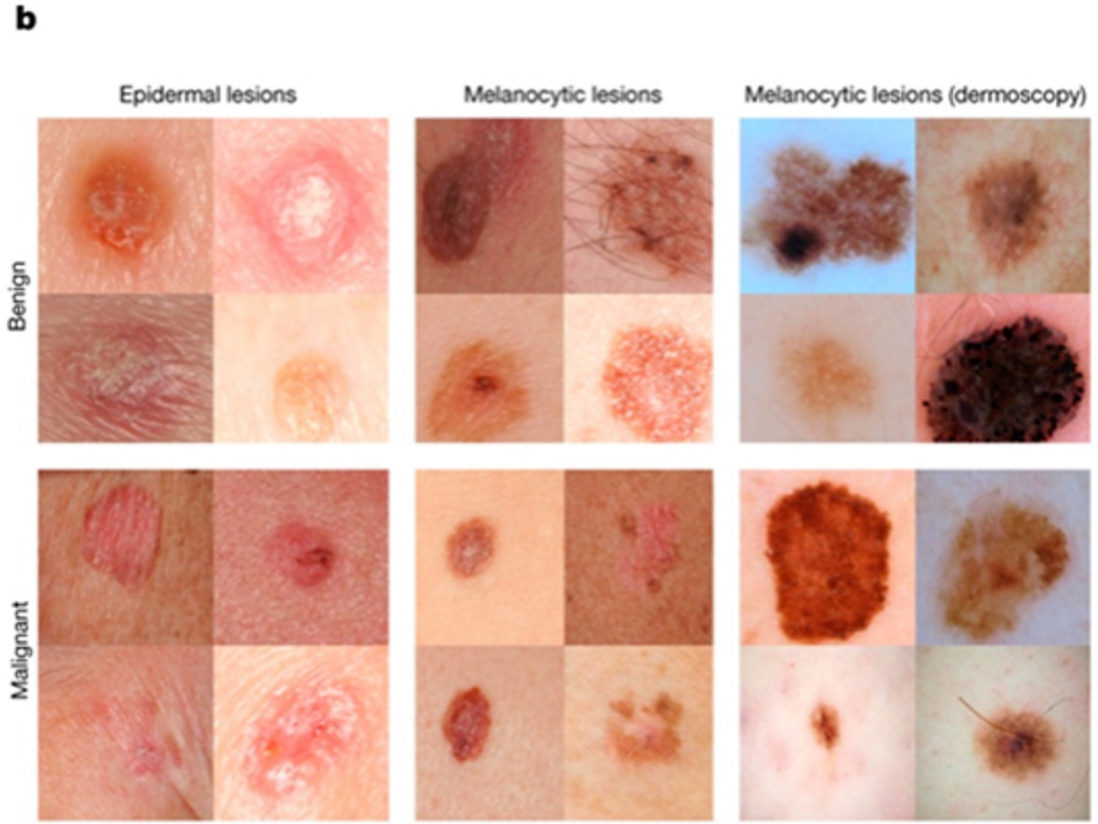


Image by Maurice Peemen.



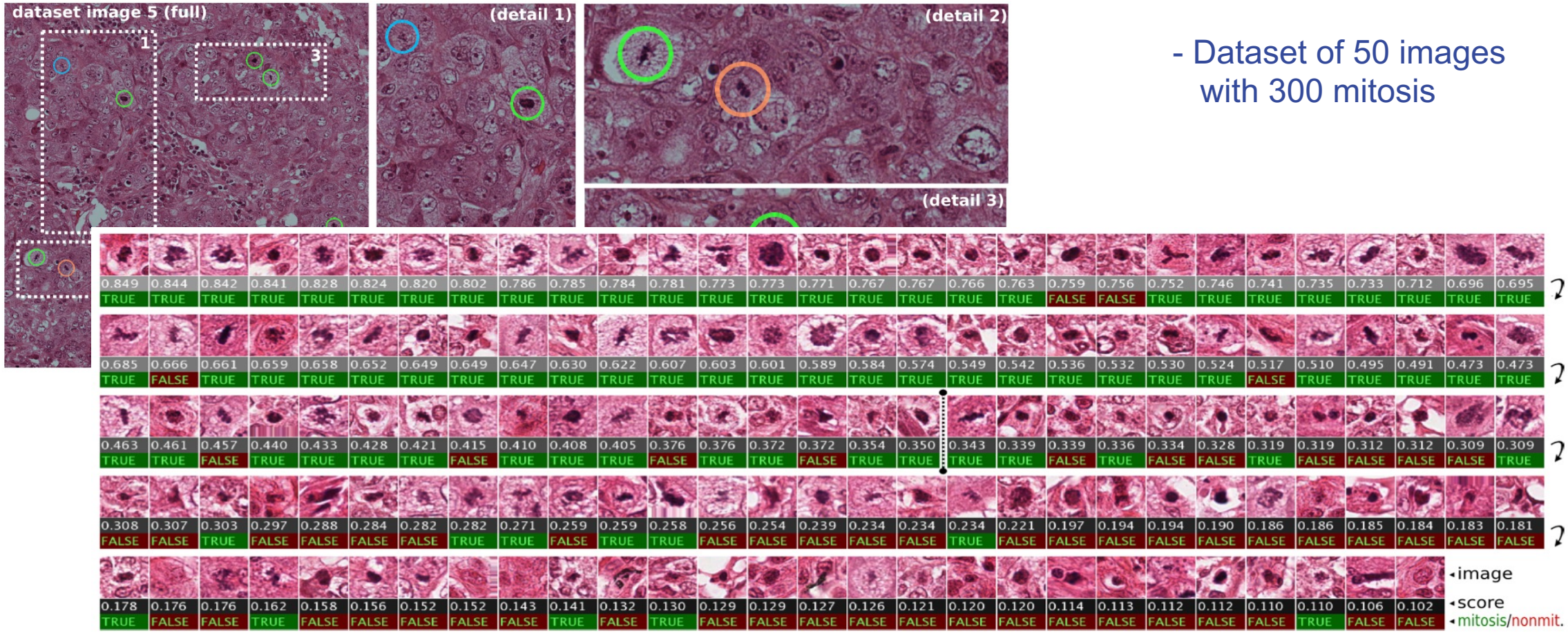
CNN - Derma – Skin cancer detection





Esteva, A. et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542(7639), 115-118.

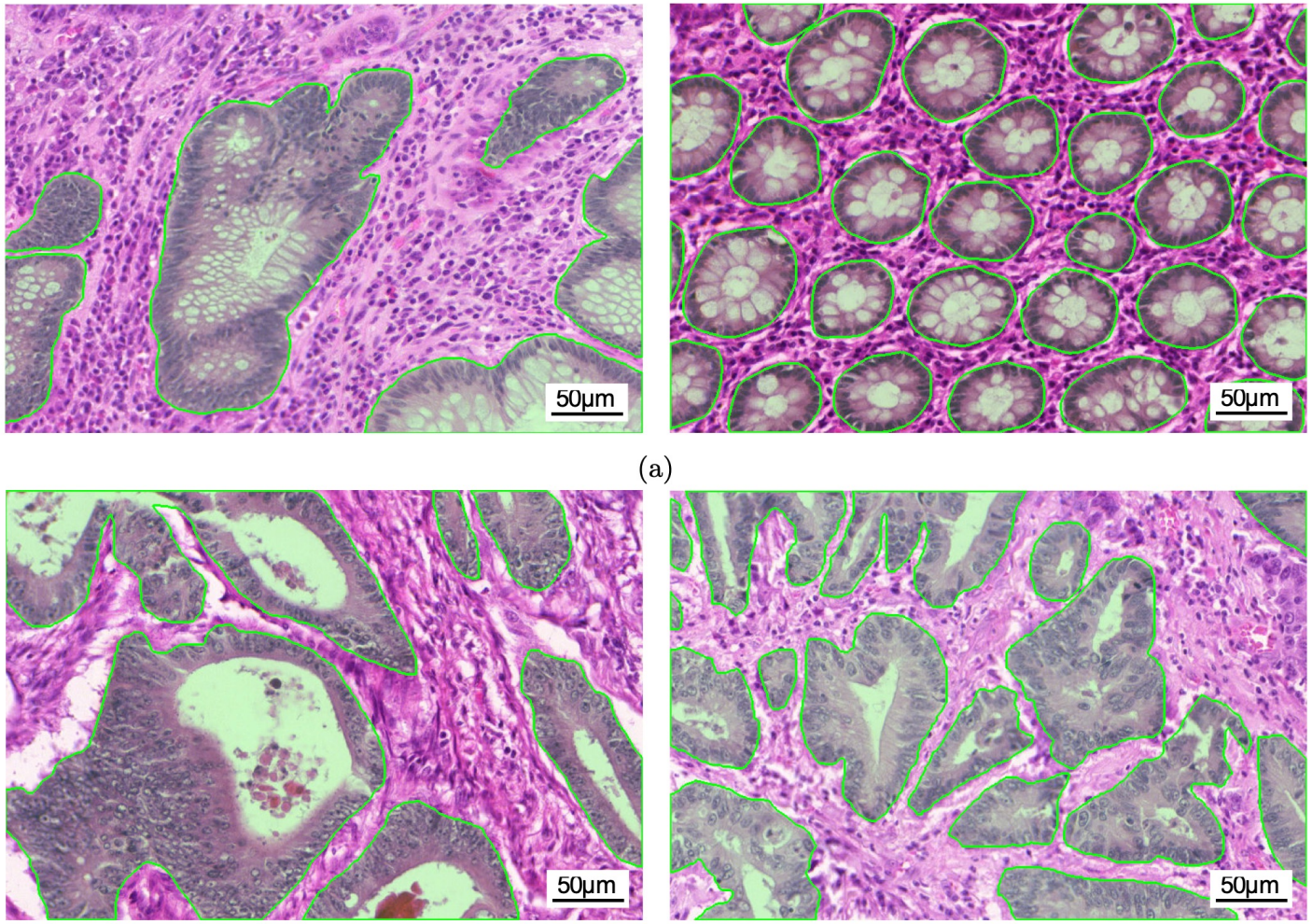
CNN - Pathology - Breast Cancer Histology Images (2012)



- Dataset of 50 images with 300 mitosis

Cireşan, Dan C., et al. "Mitosis detection in breast cancer histology images with deep neural networks." *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2013: 16th International Conference, Nagoya, Japan, September 22-26, 2013, Proceedings, Part II 16*. Springer Berlin Heidelberg, 2013.

CNN - Pathology - Colon Cancer Histology Images (2017)



- Dataset of 165 images
- Gland Segmentation in Colon Histology Images Challenge Contest (GlaS) held at MICCAI'2015

Sirinukunwattana, Korsuk, et al. "Gland segmentation in colon histology images: The glas challenge contest." *Medical image analysis* 35 (2017): 489-502.

CNN - Pathology - Renal Tissue Histology Images (2019)

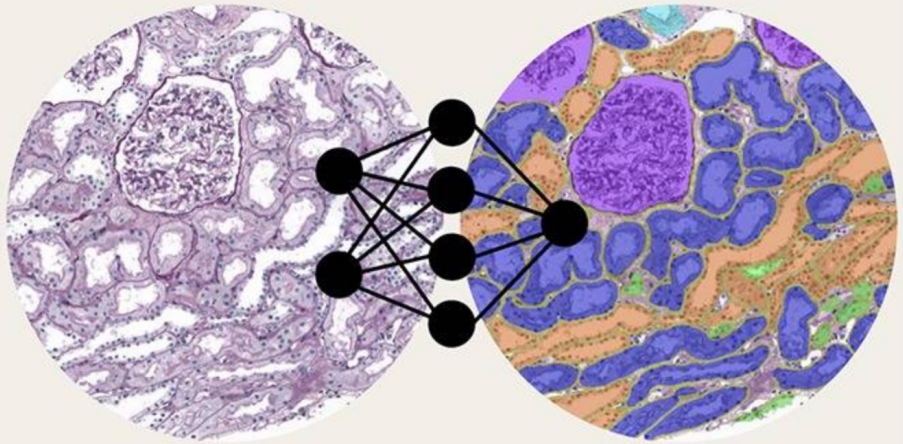
TRAINING
- 40 transplant biopsies
- 10 tissue classes
- 9488 annotations

TEST
- 20 transplant biopsies from two centers
- 15 nephrectomy samples
- 82 transplant biopsies for correlation with visual (Banff) scoring of multiple pathologists

LEGEND
- Border
- Glomeruli
- Undefined tubuli
- Proximal tubuli
- Distal tubuli
- Atrophic tubuli
- Arteries

No fill = interstitium

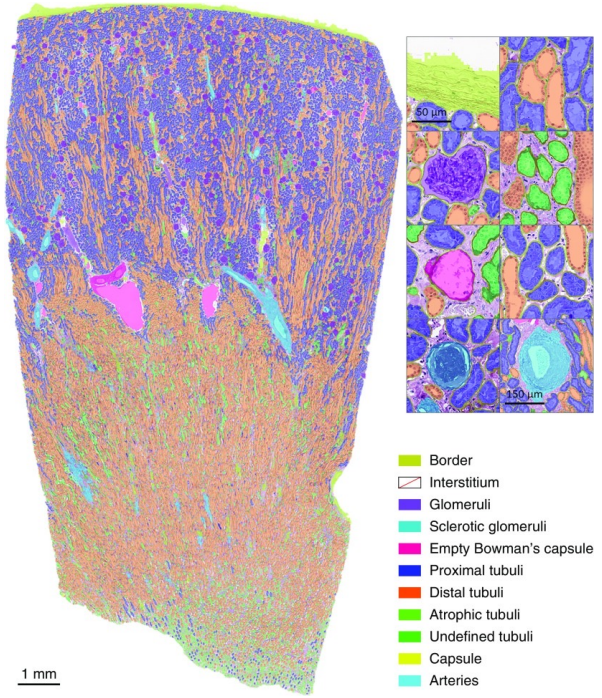
Convolutional Neural Network for segmentation renal tissue



RESULTS
- Highest performance for glomeruli, tubuli and interstitium segmentation
- Average DC[1] 0.88
- Equal performance on images external center
- For analysis of nephrectomy and biopsy samples
- For healthy and pathological tissue
- CNN-based quantifications correlate significantly with components Banff scoring system

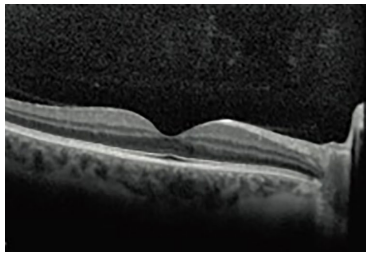
CONCLUSION
This study presents **the first CNN for multi-class segmentation** of periodic acid-Schiff-stained **nephrectomy samples and transplant biopsies.** Our CNN can be of aid for quantitative studies concerning renal histopathology **across centers** and provides opportunities for deep learning applications in routine diagnostics.

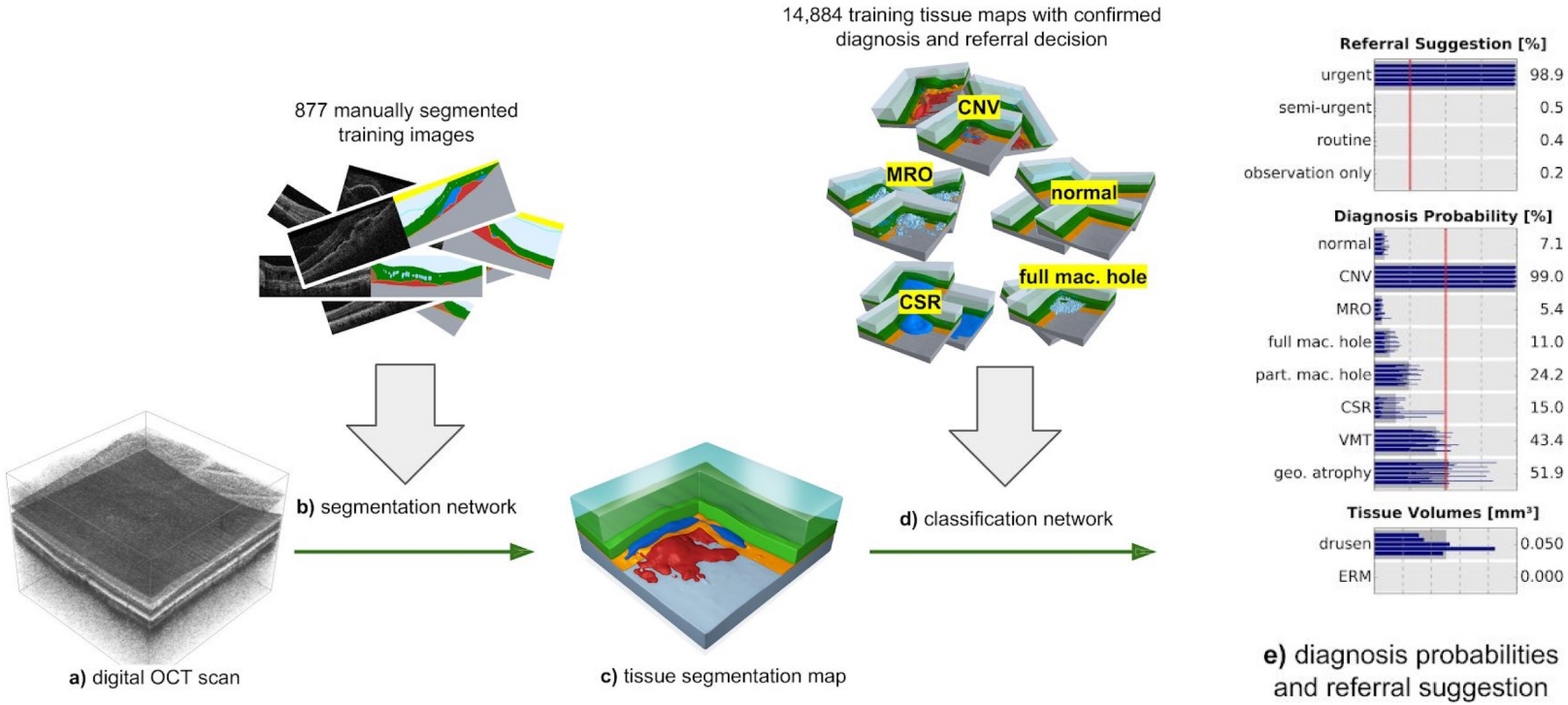
[1]DC= Dice coefficient



Hermsen, Meyke, et al. "Deep learning–based histopathologic assessment of kidney tissue." *Journal of the American Society of Nephrology: JASN* 30.10 (2019): 1968.

CNN - Ophthalmology

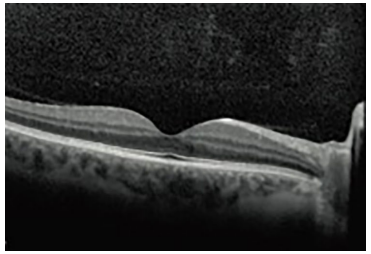


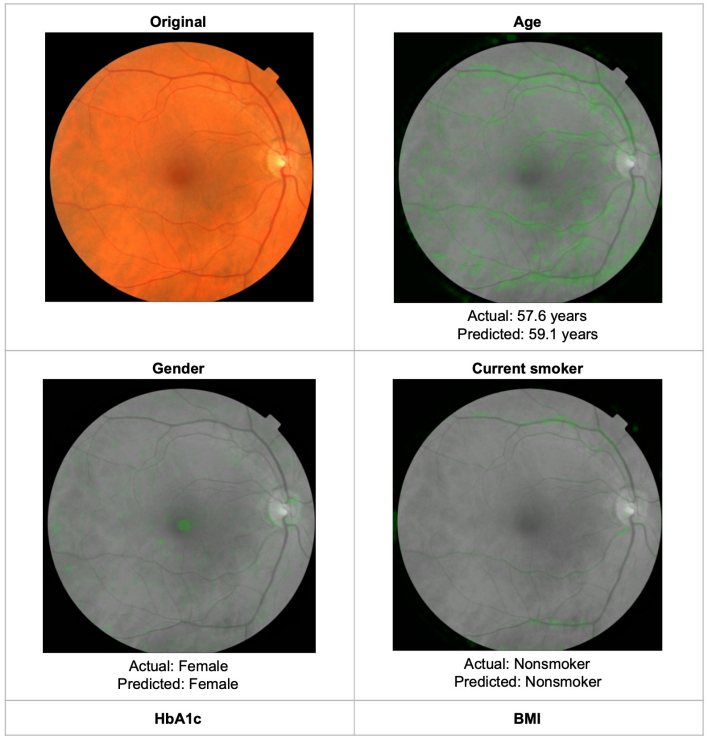


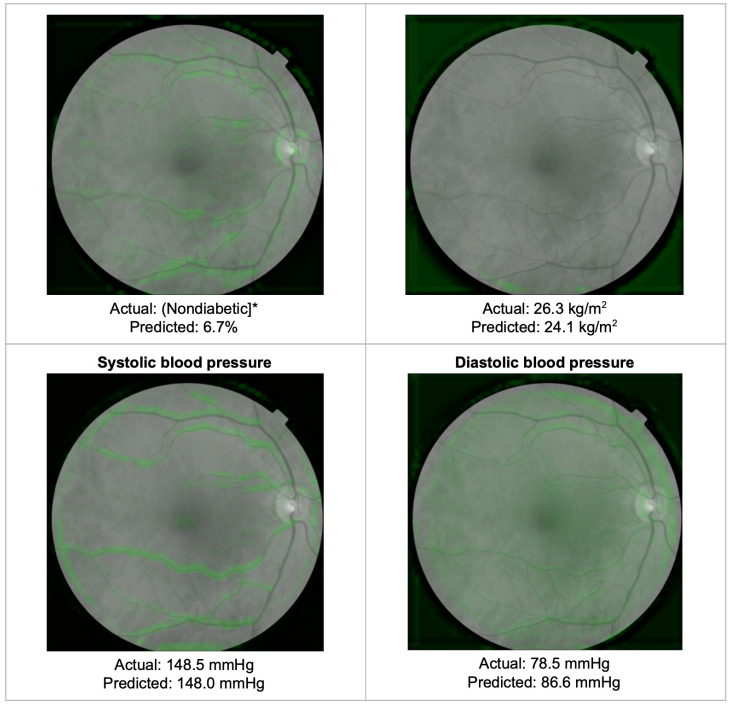
- Prediction of urgency, diganostic probability and tissue volumes
- Over 14.000 cases in dataset

De Fauw, J. et al. (2018). Clinically applicable deep learning for diagnosis and referral in retinal disease. Nature Medicine, 24(9), 1342-1350.

CNN - Genomik – Prediciting Cardiovescular Riskfactors from Retina





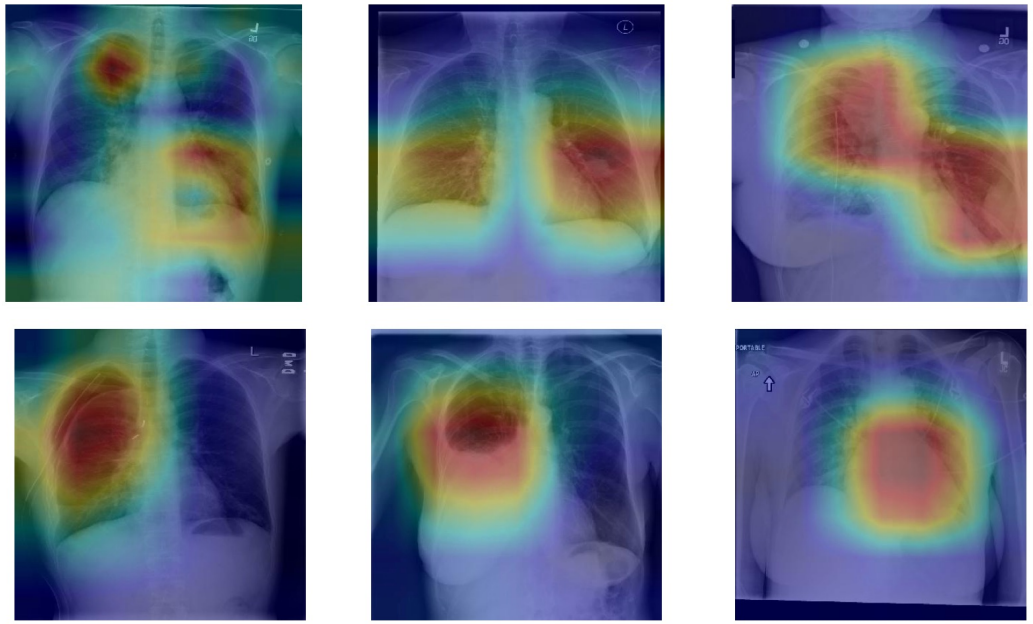


- Prediction of age, gender, smoking, systolic blood pressur, etc.
- Prediction of cardiovascular riskfactors with similar precision as blood tests

Poplin, Ryan, et al. "Prediction of cardiovascular risk factors from retinal fundus photographs via deep learning." *Nature biomedical engineering* 2.3 (2018): 158-164.

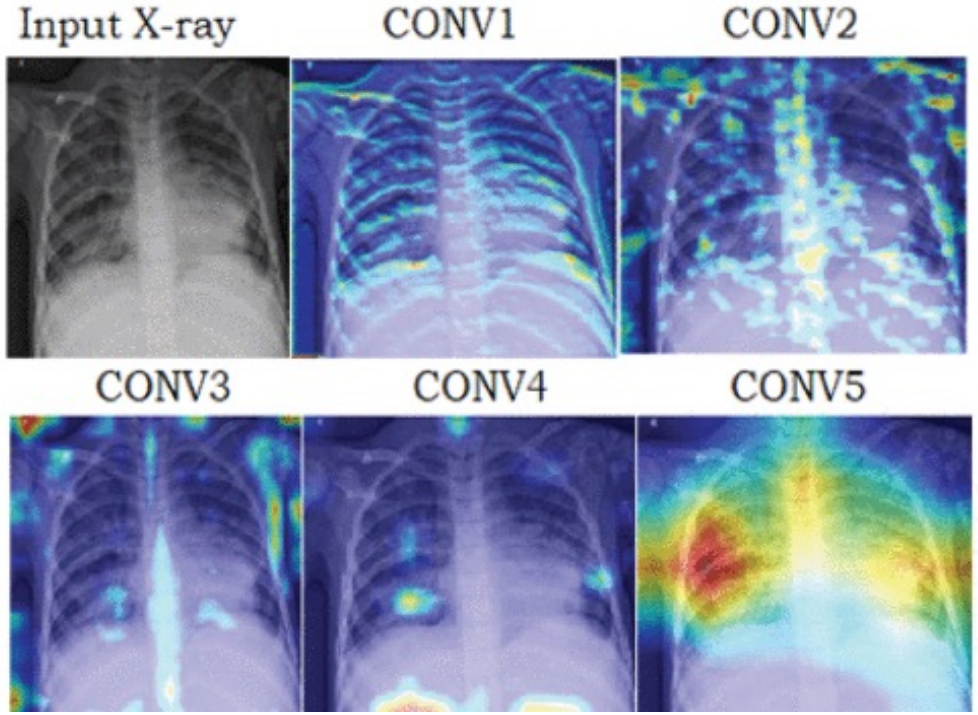
CNN - Radiology – Pneumonia & COVID detection

CheXNet [3]



- 121 Convolution layers
- Over 100.000 images in dataset
- Detection of 14 pathologies

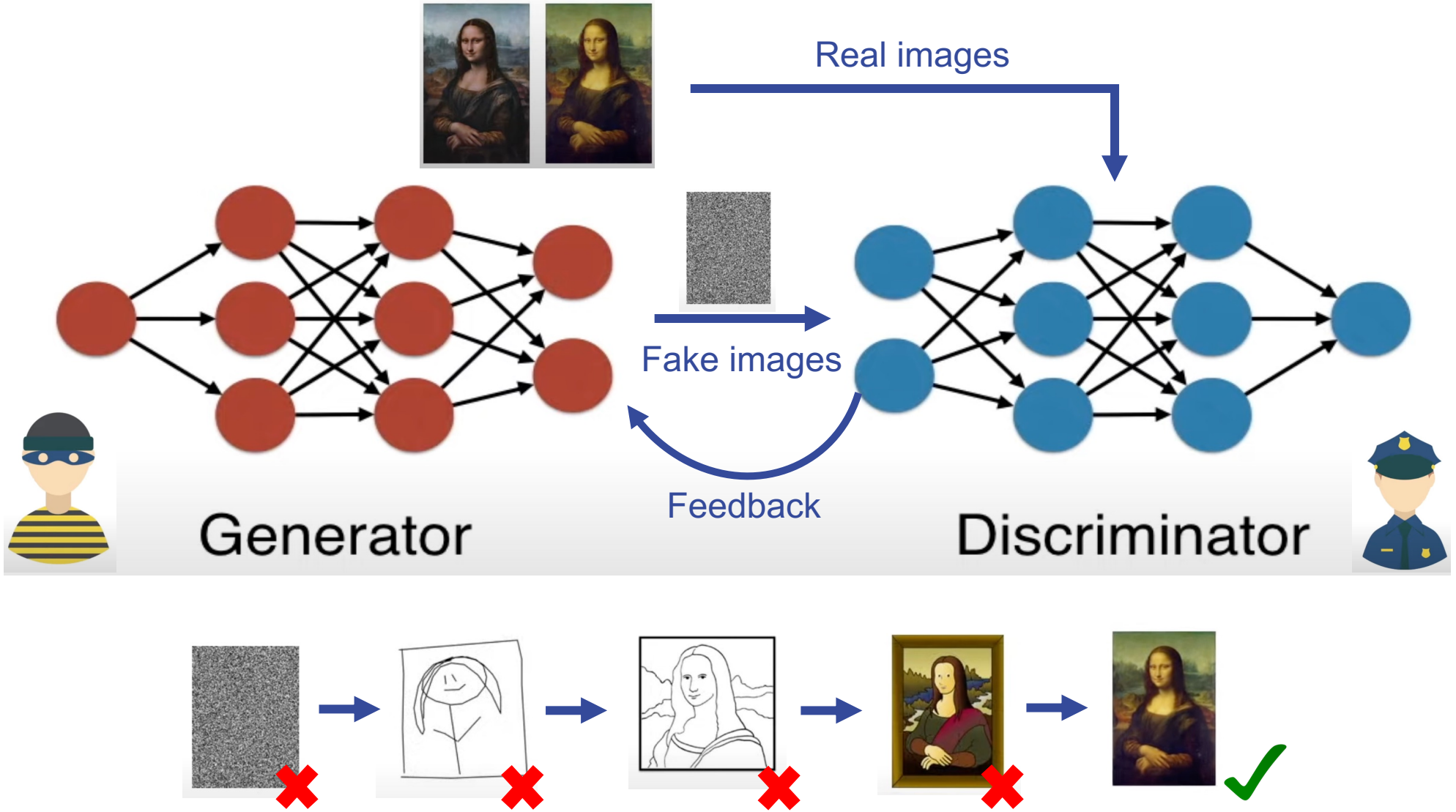
Covid-CheXNet (Transfer Learning) [4]



- Training on 1824 images
- Achieving 99,9% accuracy – AUC = 1*(careful!)*

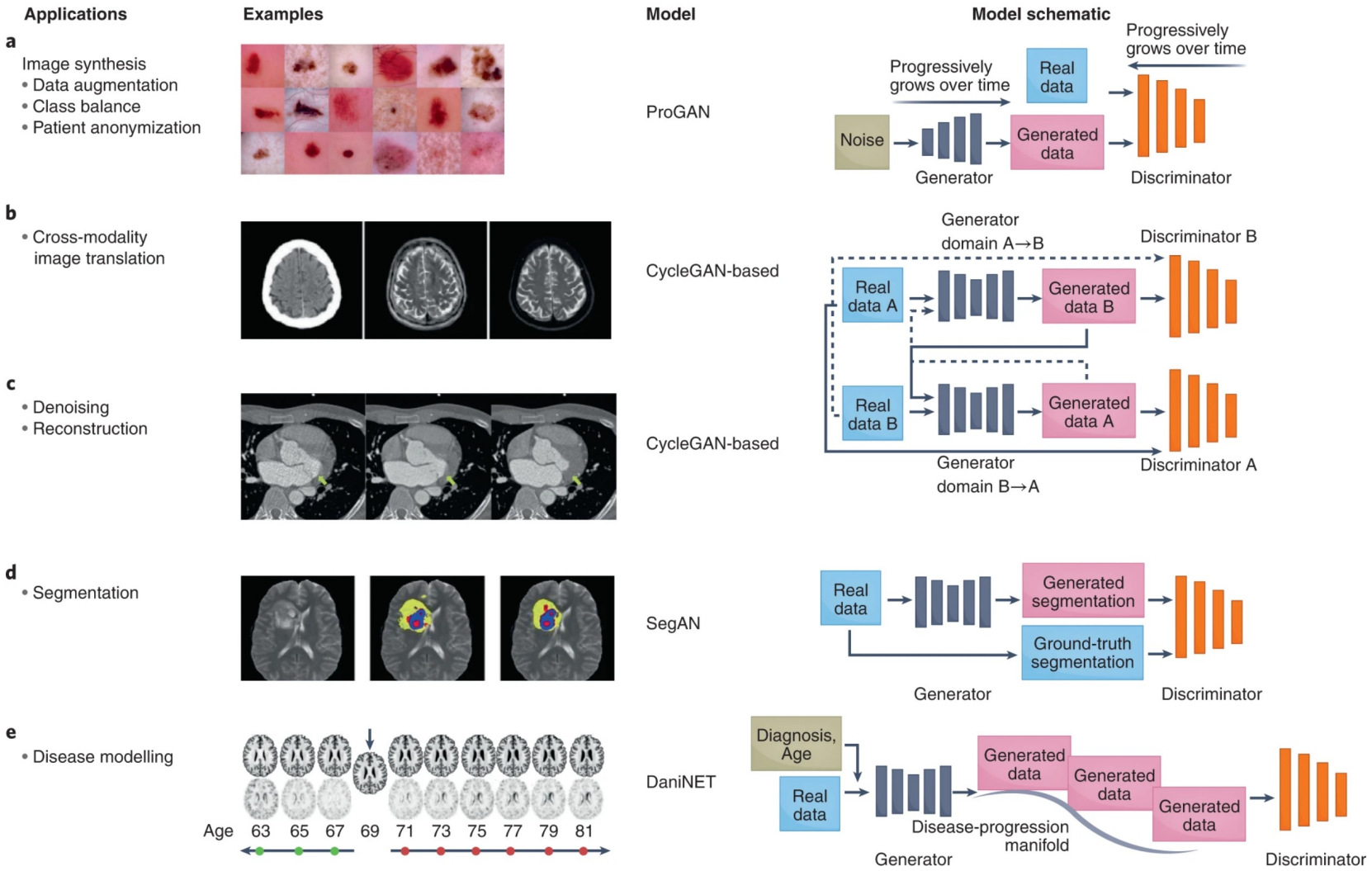
[3] Rajpurkar, Pranav, et al. "Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning." arXiv preprint arXiv:1711.05225 (2017).
]4] Haritha, D., M. Krishna Pranathi, and M. Reethika. "COVID detection from chest X-rays with DeepLearning: CheXNet." 2020 5th international conference on computing, communication and security (ICCCS). IEEE, 2020.

Generative Adversarial Networks (GANs)



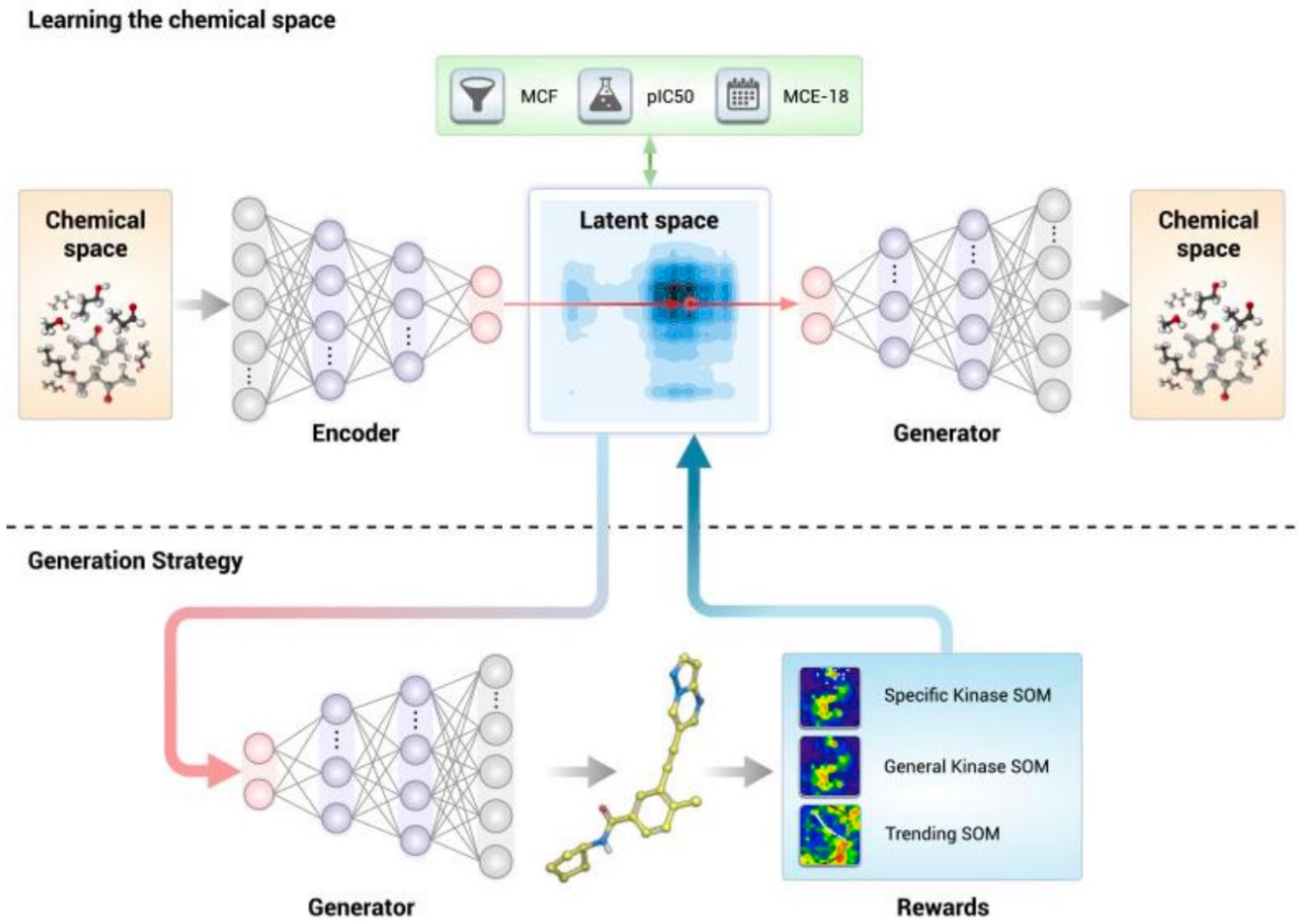
Modified example of Serrano.Academy https://www.youtube.com/watch?v=8L11aMN5KY8 (2020)

GANs – Data augmentation

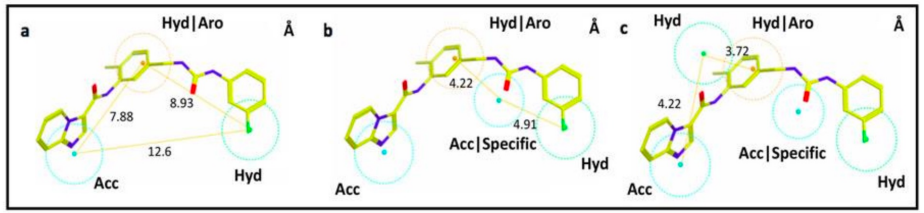


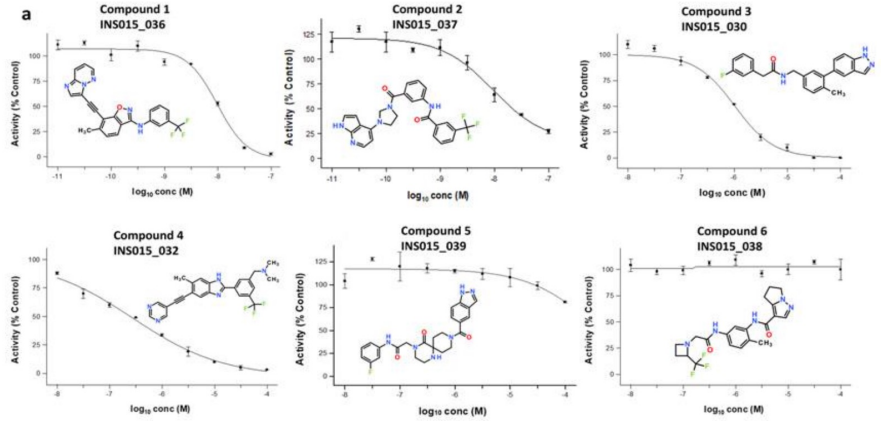
Zhang, Angela, et al. "Shifting machine learning for healthcare from development to deployment and from models to data." *Nature Biomedical Engineering* 6.12 (2022): 1330-1345.

GANs - Molecular analysis for drug discovery – DDR1 blocker



- Dataset includes over 1.000.000 molecules
- After 21 days they have idetified promising DDR1 blocker → Usually it may take years





Zhavoronkov, Alex, et al. "Deep learning enables rapid identification of potent DDR1 kinase inhibitors." *Nature biotechnology* 37.9 (2019): 1038-1040.

Generative Adversarial Networks (GANs)

thispersondoesnotexist.com

Video posted on Instagram using LLMs

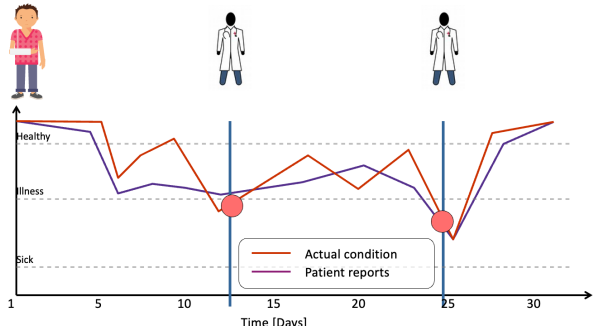
Modified example of Serrano.Academy https://www.youtube.com/watch?v=8L11aMN5KY8 (2020)

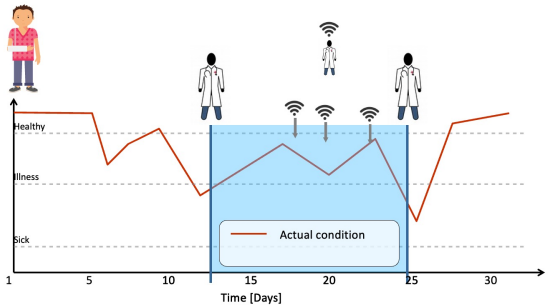
RNNs & LSTMs

Recurrent Neural Networks (RNNs) & Long-Short Term Memory

Motivation

So far, in those neural networks variation one had to predefine the number of input (or kernal size)
--> In sequential data we may be interested in





mart ring (Oura)

Smart watch (~2011)

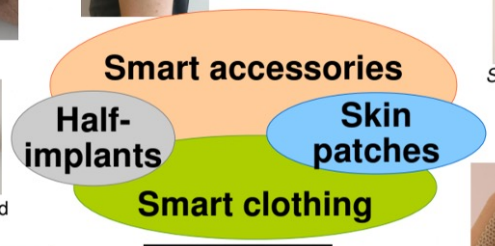
BodyMedia arm band (~2010)

Smart eyeglasses (2015-17)

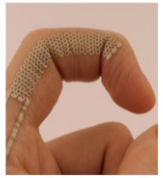
QBIC: Q-Belt Integrated Computer (2004)

Smart phoes

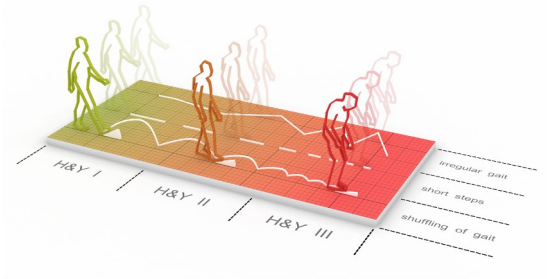
SimpleSkin: Textile sensor building blocks (2017)

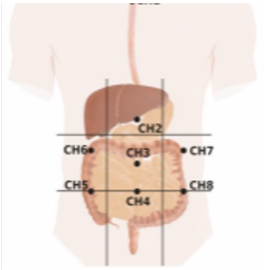


Widex in-ear hearing aid (2015)


Printed interactive skin electronics (2017, Courtesy Prof. Steimle).

Amft & v Learhoven, IEEE Perv Comput 16(4), '17

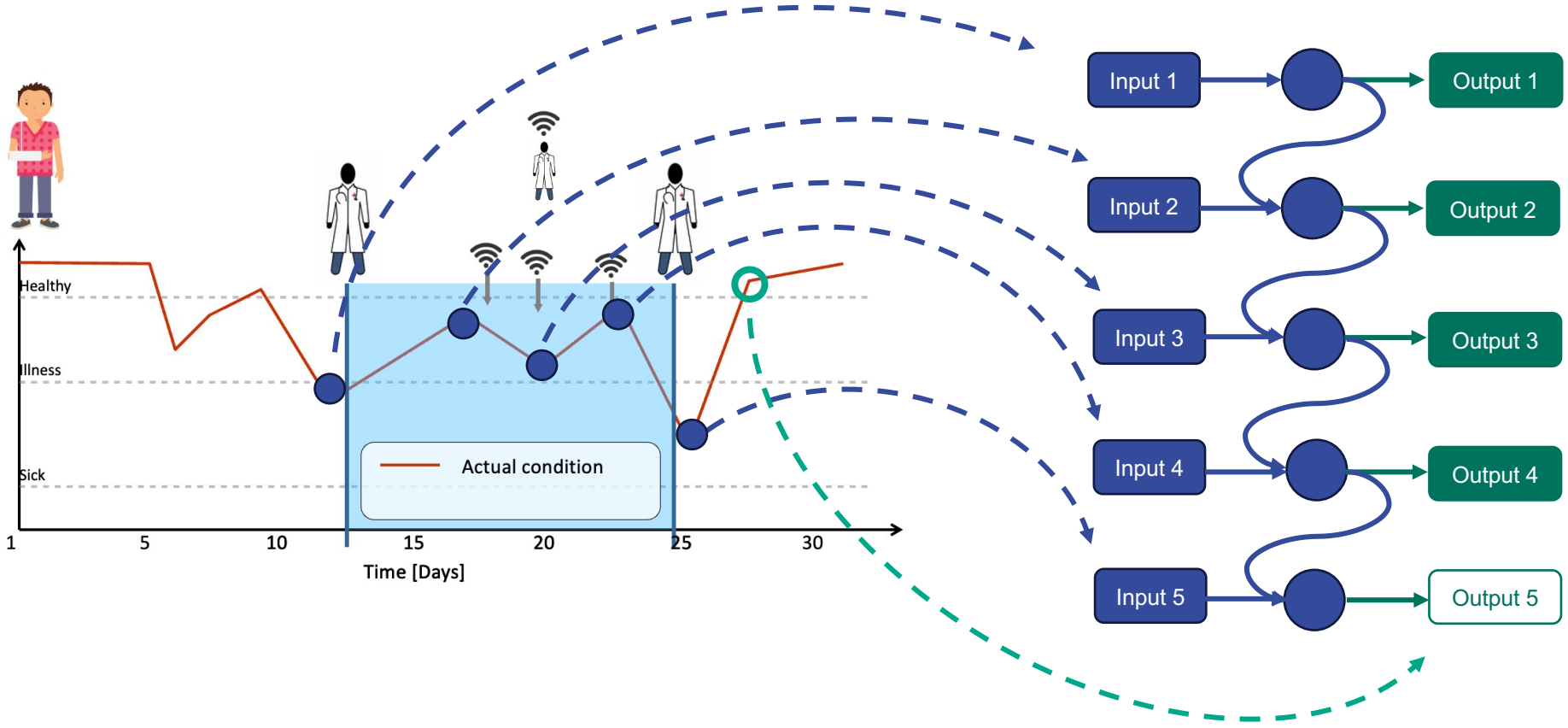




Recurrent Neural Networks (RNNs)

Motivation

So far, in those neural networks variation one had to predefine the number of input (or kernal size)
--> In sequential data we may be interested in



Recurrent Neural Networks (RNNs) – ICU example

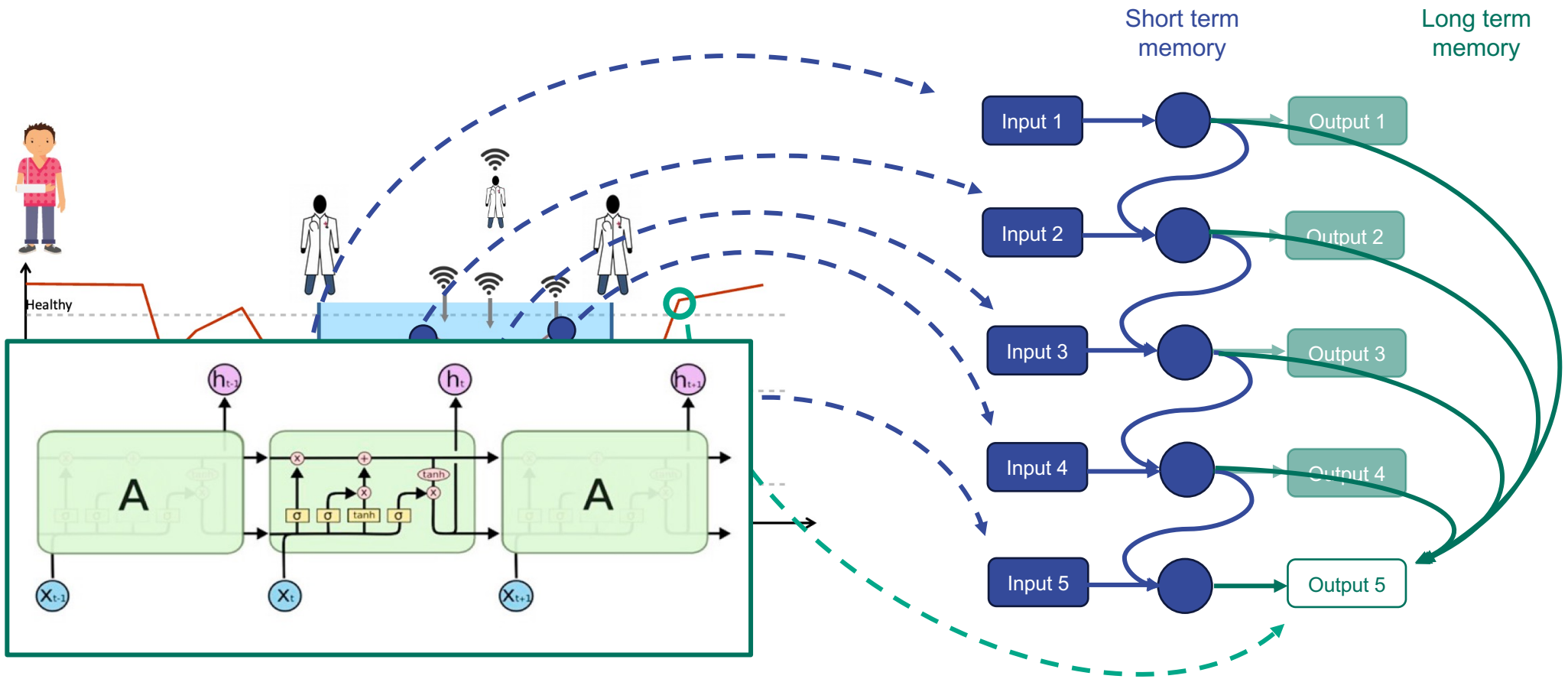
Prediction of several complications incl.
- Mortality
- Renal failure
- Postoperative bleeding

Model based on
Over 9.000 patient cases (from init. 40k)

Meyer, Alexander, et al. "Machine learning for real-time prediction of complications in critical care: a retrospective study." *The Lancet Respiratory Medicine* 6.12 (2018): 905-914.

Long-Short Term Memories (LSTMs)

Problem with RNN for large sequences: Vanishing/Exploding Gradient



Solves the RNN problem but its activation unit (for each input) is initially more complicated, because of comparing and updating of short and long-term memories.

Moghar, Adil, and Mhamed Hamiche. "Stock market prediction using LSTM recurrent neural network." *Procedia Computer Science* 170 (2020): 1168-1173.

LSTM – Forecasting stress level of students

- Data from 142 participants for 7 days
- Comparing LSTMs with SVMs and LRs
- LSTMs reach 81.4% only with objective data

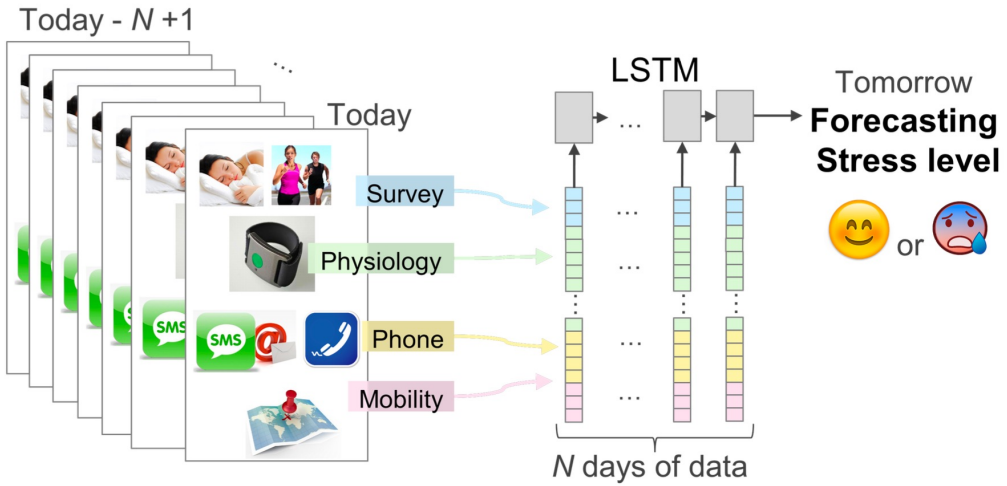


TABLE I
CONFUSION MATRIX FOR THE 7-DAY ALL FEATURE MODELS

		Predicted Label					
		LSTM		SVM		LR	
		Stress	Calm	Stress	Calm	Stress	Calm
Actual Label	Stress	438	127	334	231	152	413
	Calm	75	591	55	611	32	634

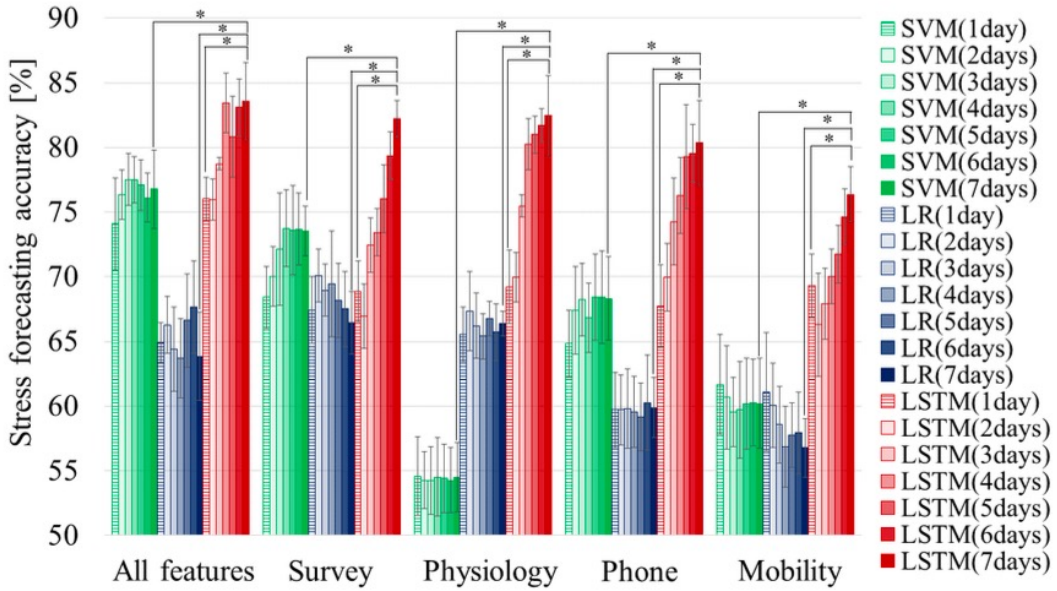


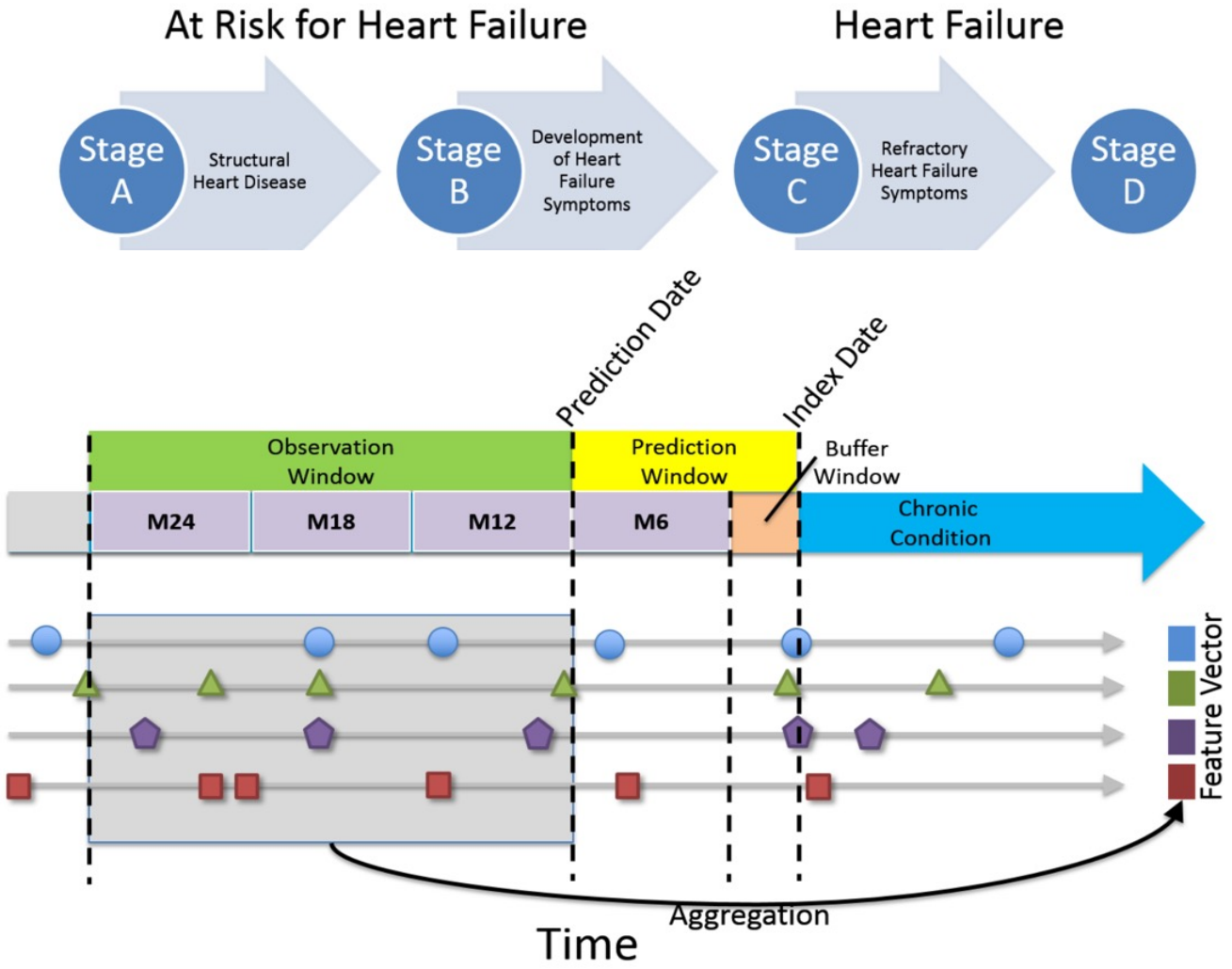
Fig. 2. Stress forecast accuracy for support vector machine (SVM), logistic regression (LR), and long short-term memory neural network models (LSTM) (* : One way ANOVA, Tukey's HSD test, p <0.05)

Umematsu, Terumi, et al. "Improving students' daily life stress forecasting using LSTM neural networks." *2019 IEEE EMBS international conference on biomedical & health informatics (BHI)*.

LSTM – Predicting heart failure

- Comparing LSTMs with several models including Random Forest, 1-D CNN, Logistic Regression

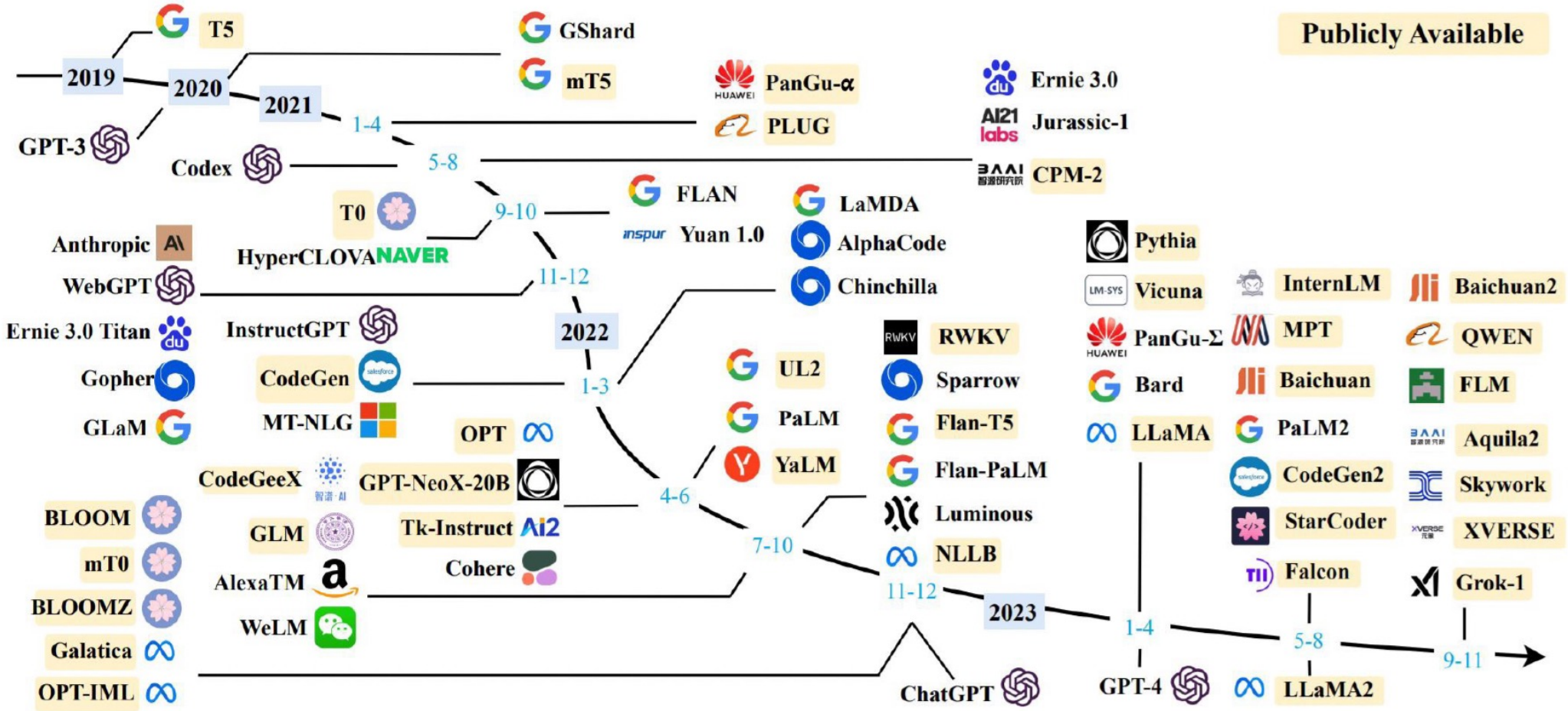
- LSTM achieves high AUC of 0.92, followed by Random Forrest with 0.91.



Mallya, Sunil, et al. "Effectiveness of lstms in predicting congestive heart failure onset." *arXiv preprint arXiv:1902.02443*(2019).

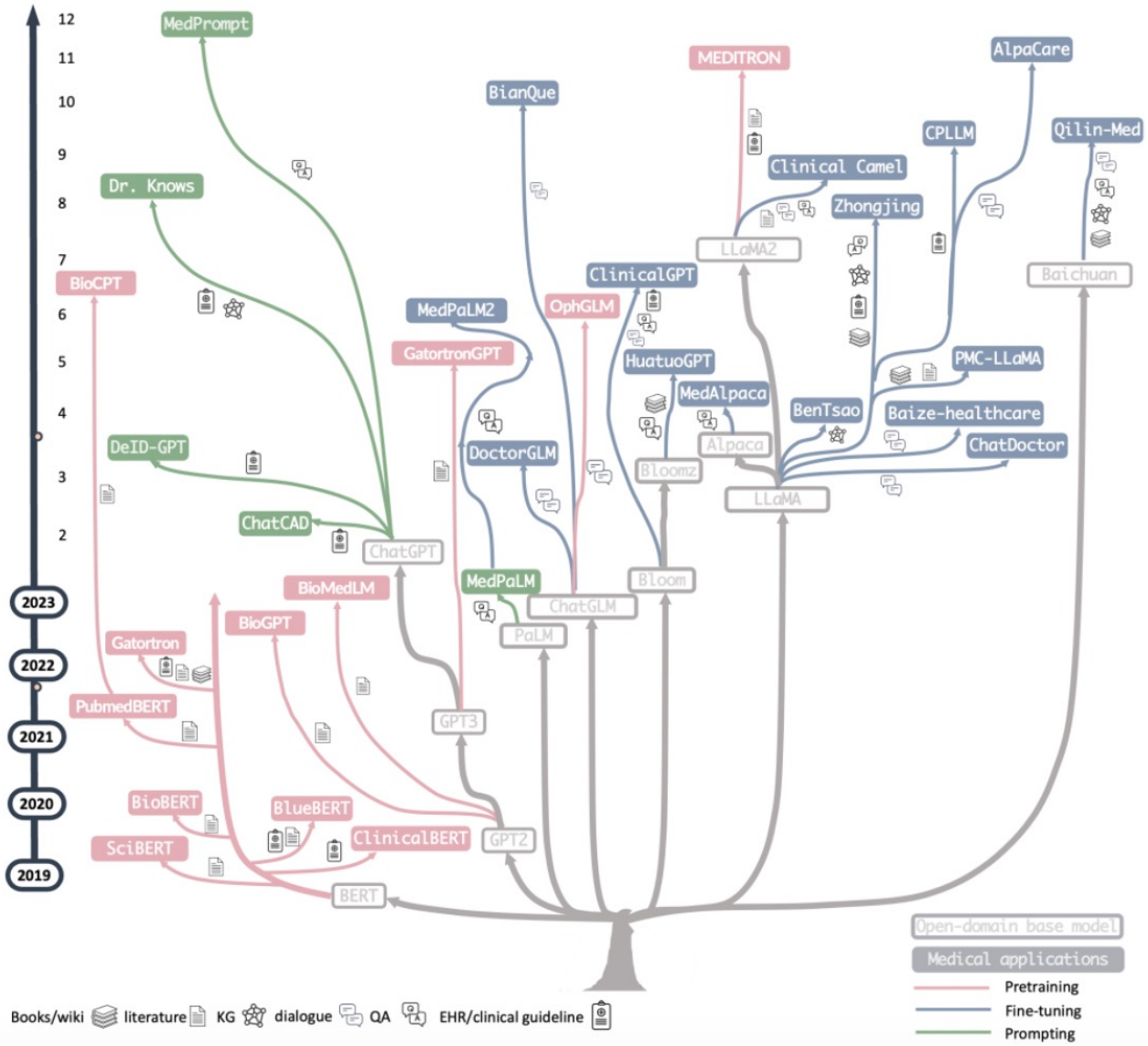
LLMs & VLMs

LLMs and VLMs – Rapid Evolution of Foundation Models



Zhao et al. 2023

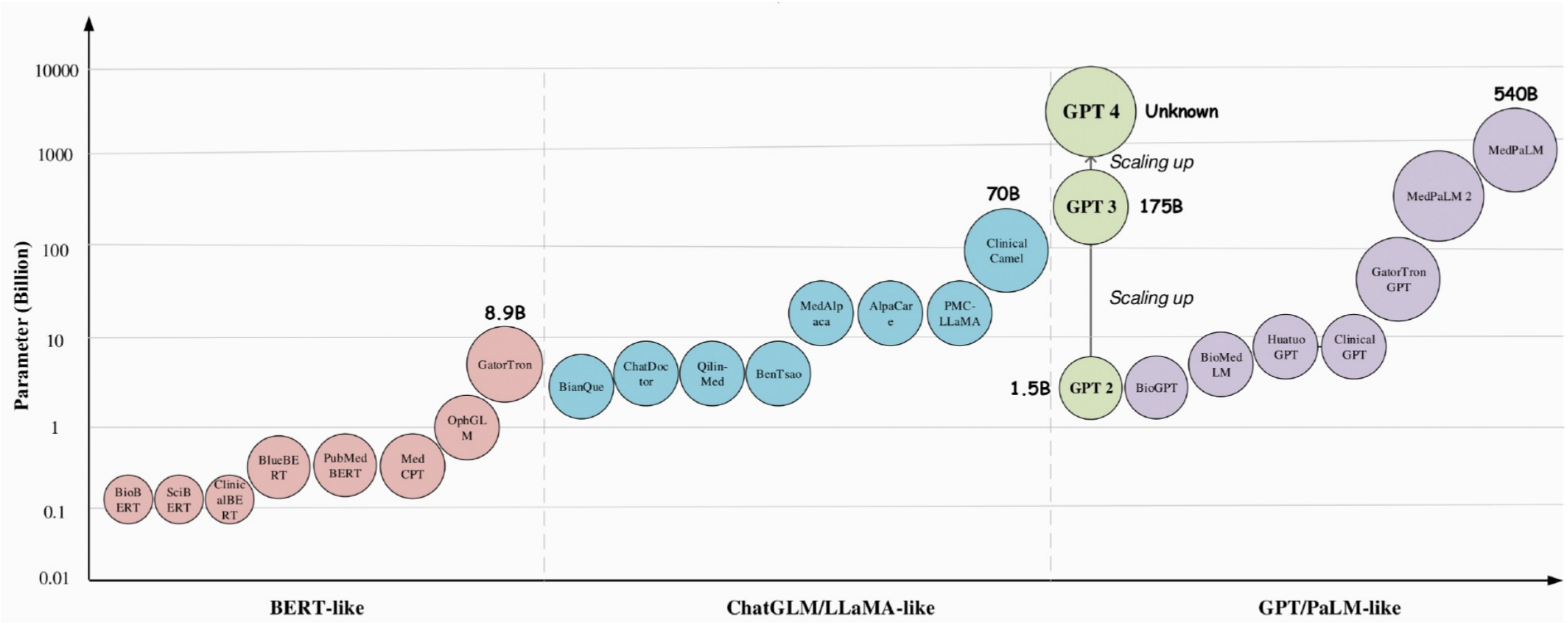
LLMs in medicine



Roots
- Bert (Google)
- GPT (OpenAI)
- PaLM (Google)
- LLaMA (Meta)
- Bloom (Hugging Face)
- ChatGLM, Baichuan (Tsinghua Univ.)

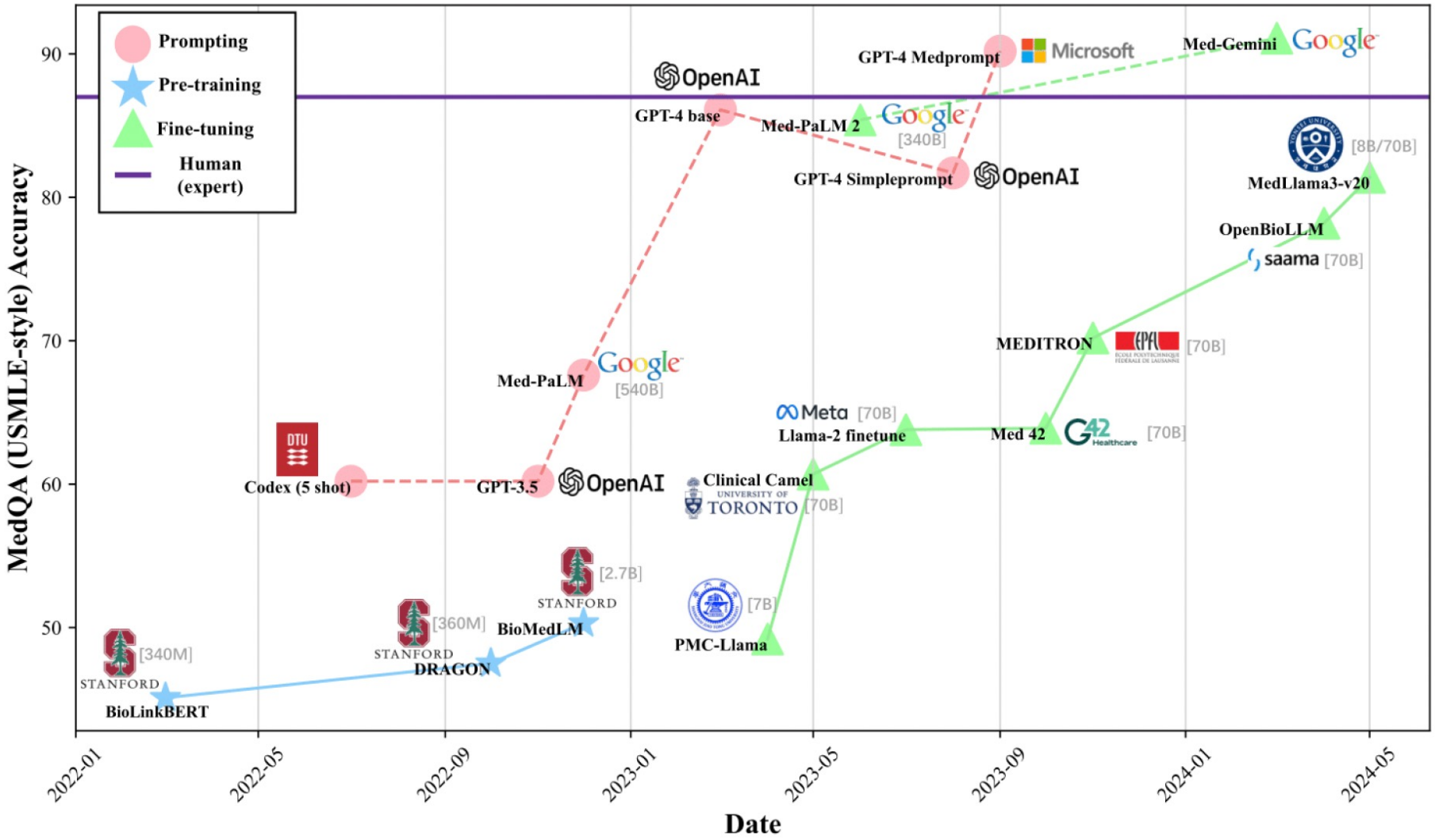
https://github.com/AI-in-Health/MedLLMsPracticalGuide/tree/main

LLMs in medicine



https://github.com/AI-in-Health/MedLLMsPracticalGuide/tree/main

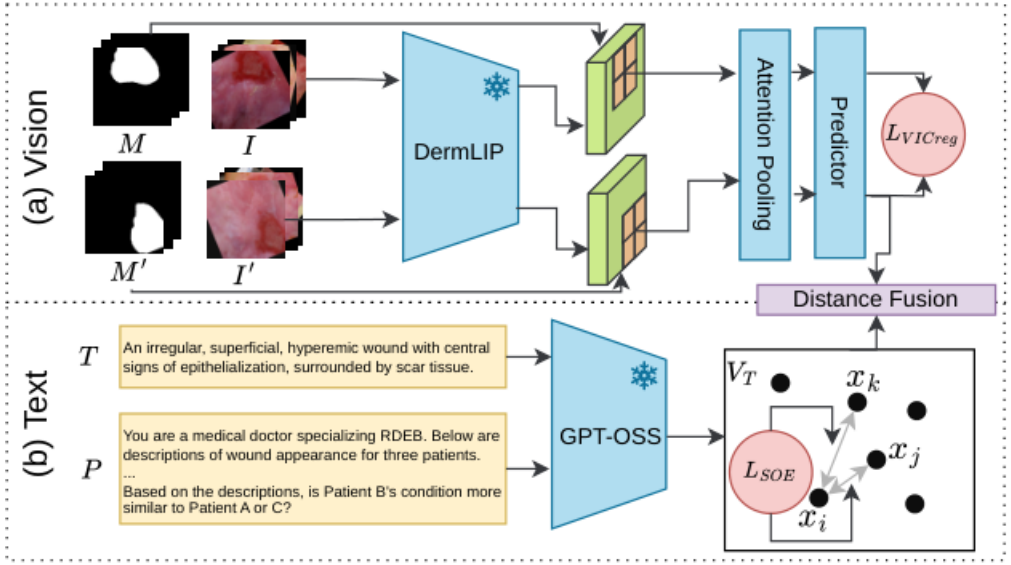
Potential applications of LLM in medicine



https://github.com/AI-in-Health/MedLLMsPracticalGuide/tree/main

LLM as Knowledge Source for Statistical Modelling

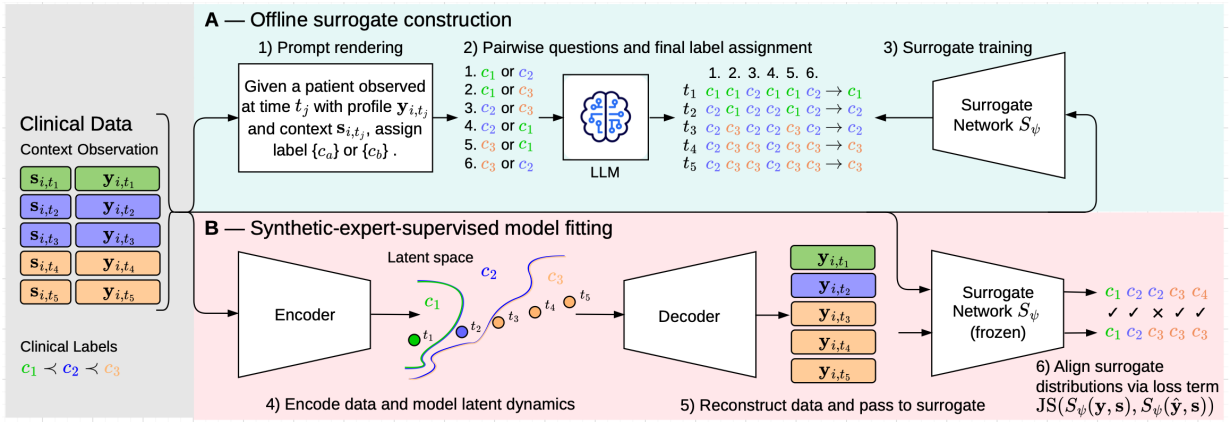
- LLMs supply knowledge for statistical modelling
- Wounds: combine images and clinical descriptions
- Comparisons reveal clinically meaningful similarities
- COPD/MS: elicit associations between clinical variables
- SMA: guide longitudinal disease representation learning
- Expert validation remains essential



[Ka26]

[Ka26] Kabus, F., Hindel, J., Bratulić, J., Karakioulaki, M., Gupta, A., Has, C., ... & Binder, H. (2026). Assessing Multimodal Chronic Wound Embeddings with Expert Triplet Agreement. arXiv preprint arXiv:2603.29376.

[Sc26] Schächter, C., Pechmann, A., Kirschner, J., Hasenauer, J., & Binder, H. (2026). Large language models as synthetic clinical experts to inform longitudinal rare-disease modeling. arXiv preprint arXiv:2608.16507.



[Sc26]

LLMs and VLMs – Rapid Evolution of Foundation Models

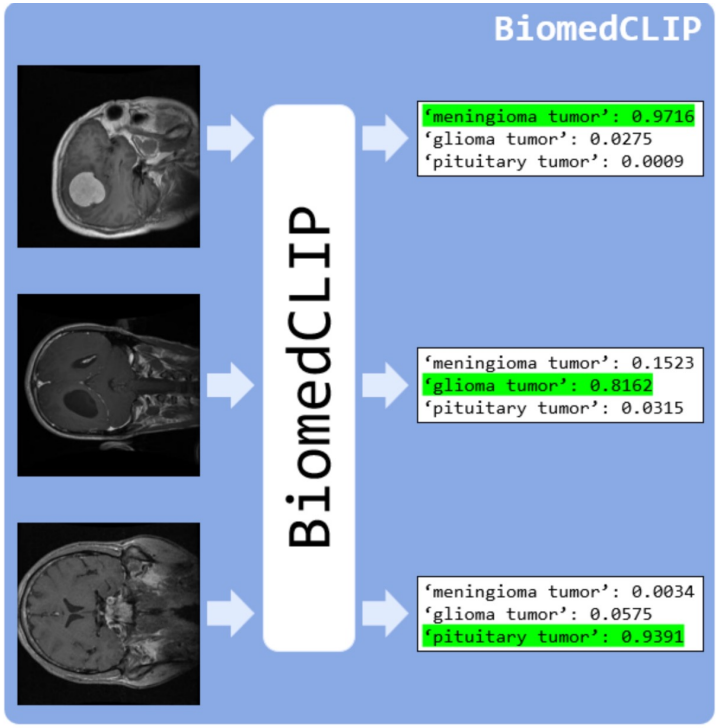
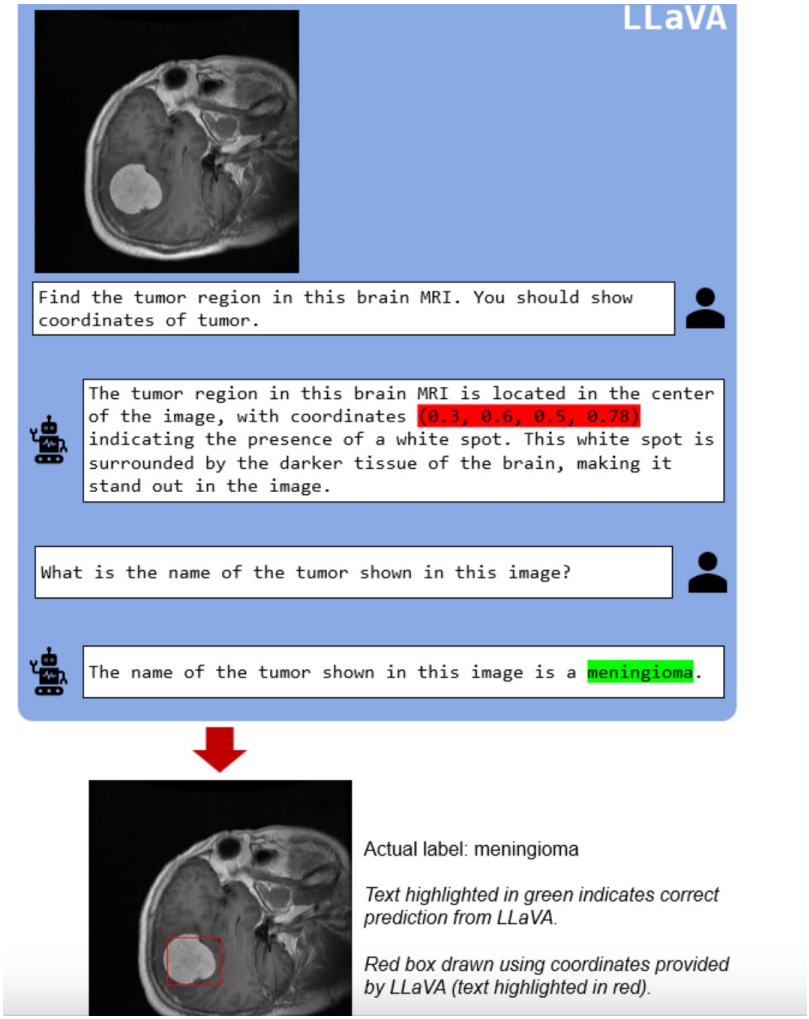


Figure 1: An example of BiomedCLIP in predicting brain tumor type from MRIs. Green highlighted text indicates correct prediction by the method.



Van, Minh–Hao, Prateek Verma, and Xintao Wu. "On large visual language models for medical imaging analysis: An empirical study." *2024 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*. IEEE, 2024.

LLMs and VLMs – Rapid Evolution of Foundation Models

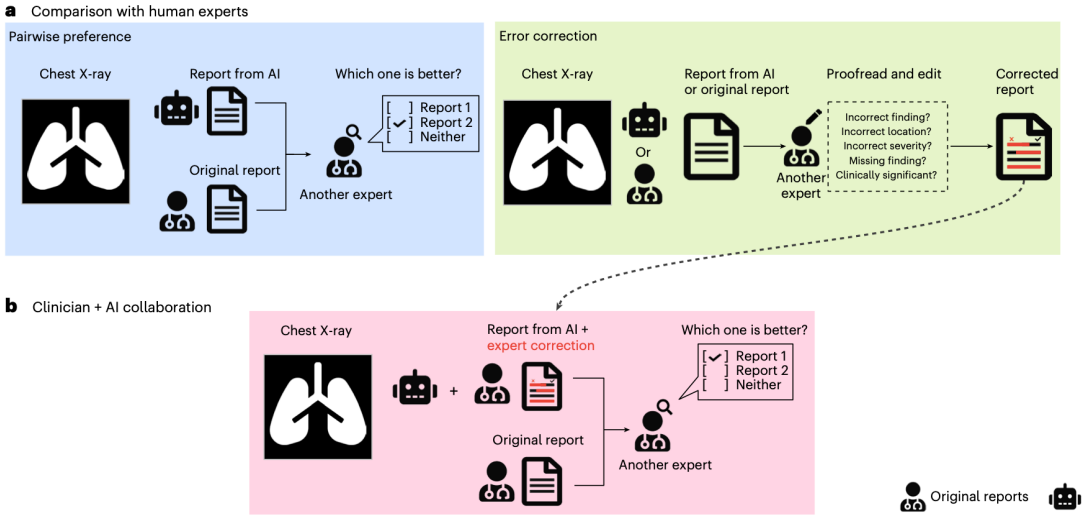
- LLaVA-1.5-7b vs. Melanoma
  - Image + question -> answer

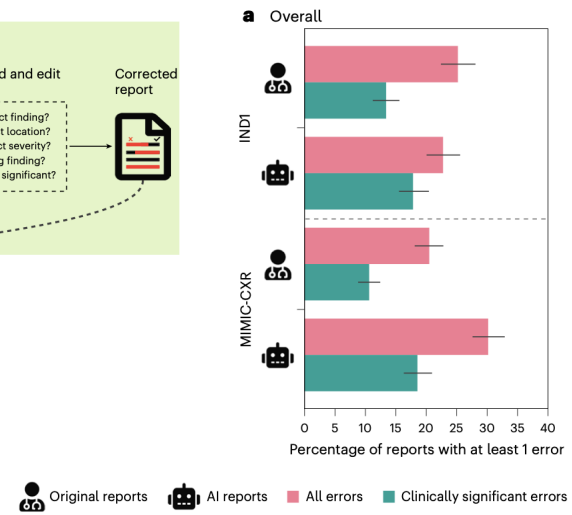
USER: Please describe this skin condition in dermoscopy.

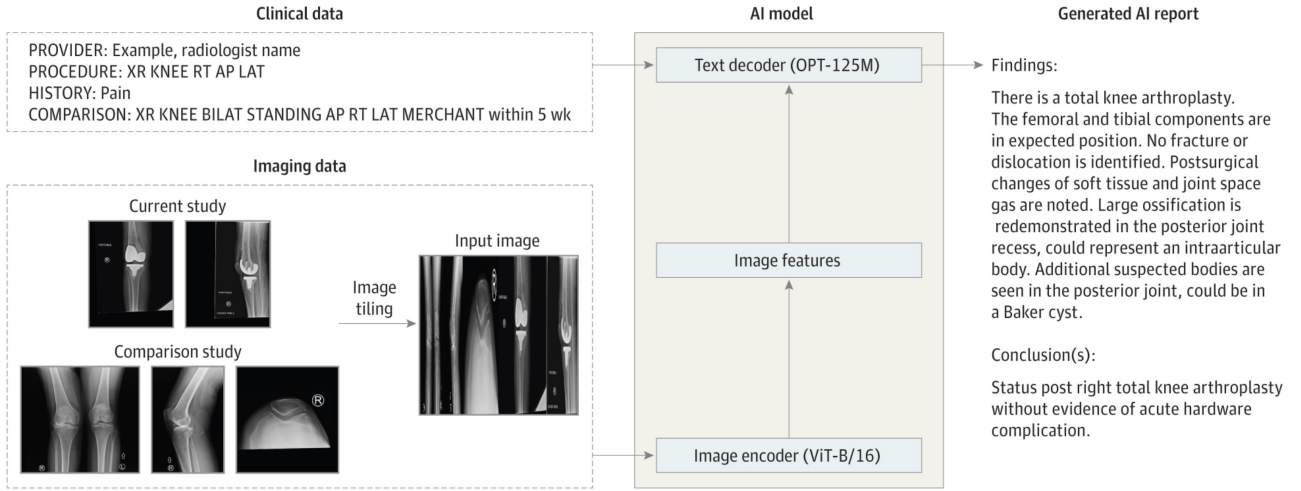
ASSISTANT: The skin condition in the image is characterized by a brown spot with a black center, surrounded by a white circle. This pattern is commonly associated with melanoma, a type of skin cancer. The presence of a black spot in the center of the brown spot suggests that it could be a **melanoma**, which requires further examination by a dermatologist to confirm the diagnosis. It is essential to monitor and follow up on such skin conditions to ensure early detection and treatment.

Vision–Language Models in Radiology

- Combine radiographs with clinical context
- Generate draft radiology reports
- Radiologists review and correct drafts
- Flamingo-CXR: expert evaluation across clinical settings [Ta25]
- Clinical deployment (~12K cases): 15.5% shorter documentation time [Hu25]
- Human oversight remains essential







[Ta25] Tanno, Ryutaro, et al. "Collaboration between clinicians and vision–language models in radiology report generation." Nature Medicine 31.2 (2025): 599-608.

[Hu25] Huang, Jonathan, et al. "Efficiency and quality of generative AI–assisted radiograph reporting." JAMA Network Open 8.6 (2025): e2513921.

Potential applications of LLM in medicine

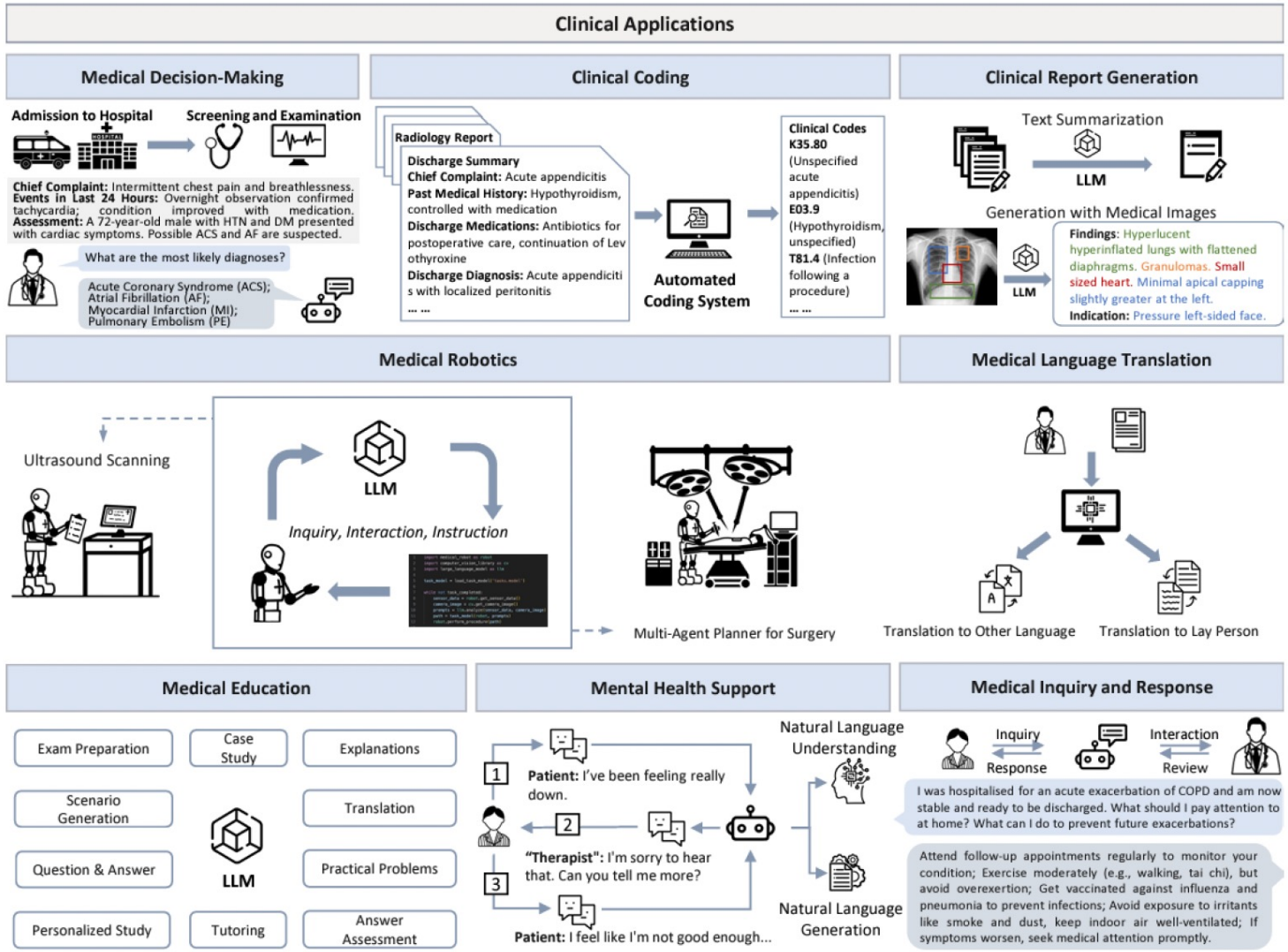


Figure 5. Integrated overview of potential applications [114,150,151,152,153] of large language models in medicine.

https://github.com/AI-in-Health/MedLLMsPracticalGuide/tree/main

Activation!

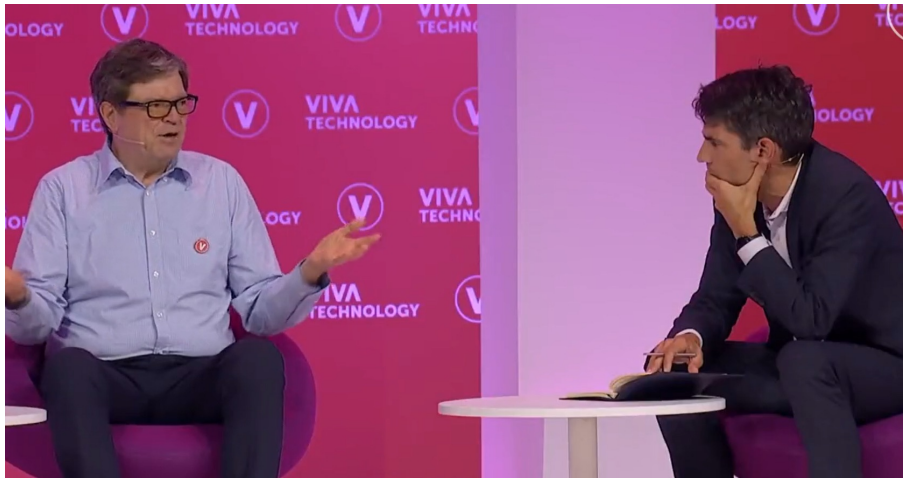
10 . How much do you use LLMs (aka chatbots, copilot, etc.)?

11. What LLM limitations do you know from your own experience?

12. What are your counter measures against hallucination?

pingo.coactum.de/815896

Limitations of LLM



Meta’s AI chief Yann LeCun
World Economic Forum in Davos

“[illegible] three to five years because of the limitations of current systems,”

Source: https://www.hpcwire.com

Limitations of LLM

Limitations	Description
Recency	- GPT training datasets do not include content created after September 2021. - All pretraining datasets necessarily 'cut off' at an arbitrary date.
Accuracy	- GPT-3 limited to 570 gigabytes data. - Models not trained to 'understand'; instead limited to learning probabilistic associations between words. - Training data sourced from unverified and unvalidated websites and books.
Coherence	- Model outputs based on learned associations between words rather than understanding input queries or information utilised in outputs. - Fabricated facts presented as if they were true.
Transparency and interpretability	- Unclear how models generate answers from input queries and architectural data and algorithms: 'black box' issues. - Unclear which parts of the training dataset are leveraged in generated responses.
Ethical concerns	- Responses may be dangerous, discriminatory, or offensive. - Risk of privacy and security breaches. - No established accountability for consequences of model outputs.

Thirunavukarasu et al., Nature Medicine, 2023

Limitations of LLM

Limitations	Mitigating strategies
Recency	- Gathering training data from more recent sources. - Real-time internet access (*e.g.* Bing AI, Sparrow, BlenderBot 3).
Accuracy	- Validation of training data. - Uncertainty indicator. - Finetuning to optimise medical accuracy. - Self-improvement through intelligent prompts (*e.g.* chain-of-thought).
Coherence	- Redeveloping model architecture and training strategies to develop true semantic knowledge. - Finetuning to eliminate presentation of inaccurate information.
Transparency and interpretability	- Outputs required to cite parts of the dataset which contributed to the model's answers. - Explainable artificial intelligence research and development.
Ethical concerns	- Finetuning to reduce the incidence of undesirable outputs. - Establishment of governance systems and overseeing authorities. - Installation of a reporting system for users to flag dangerous responses.

Thirunavukarasu et al., Nature Medicine, 2023

Limitations of LLM

The Foundation Model Transparency Index v1.1
May 2024

Rishi Bommasani* Stanford University
Kevin Klyman* Stanford University
Sayash Kapoor Princeton University
Shayne Longpre Massachusetts Institute of Technology
Betty Xiong Stanford University
Nestor Maslej Stanford University
Percy Liang Stanford University

Abstract

Foundation models are increasingly consequential in society, yet they are extremely opaque. To characterize the status quo, the Foundation Model Transparency Index was launched in October 2023 to measure the transparency of leading foundation model developers. The October 2023 Index (v1.0) assessed 10 major foundation model developers (e.g. OpenAI, Google) on 100 transparency indicators (e.g. does the developer disclose the wages it pays for data labor?). At the time, developers publicly disclosed very limited information with the average score being 37 out of

2024 Foundation Model Transparency Index Indicators

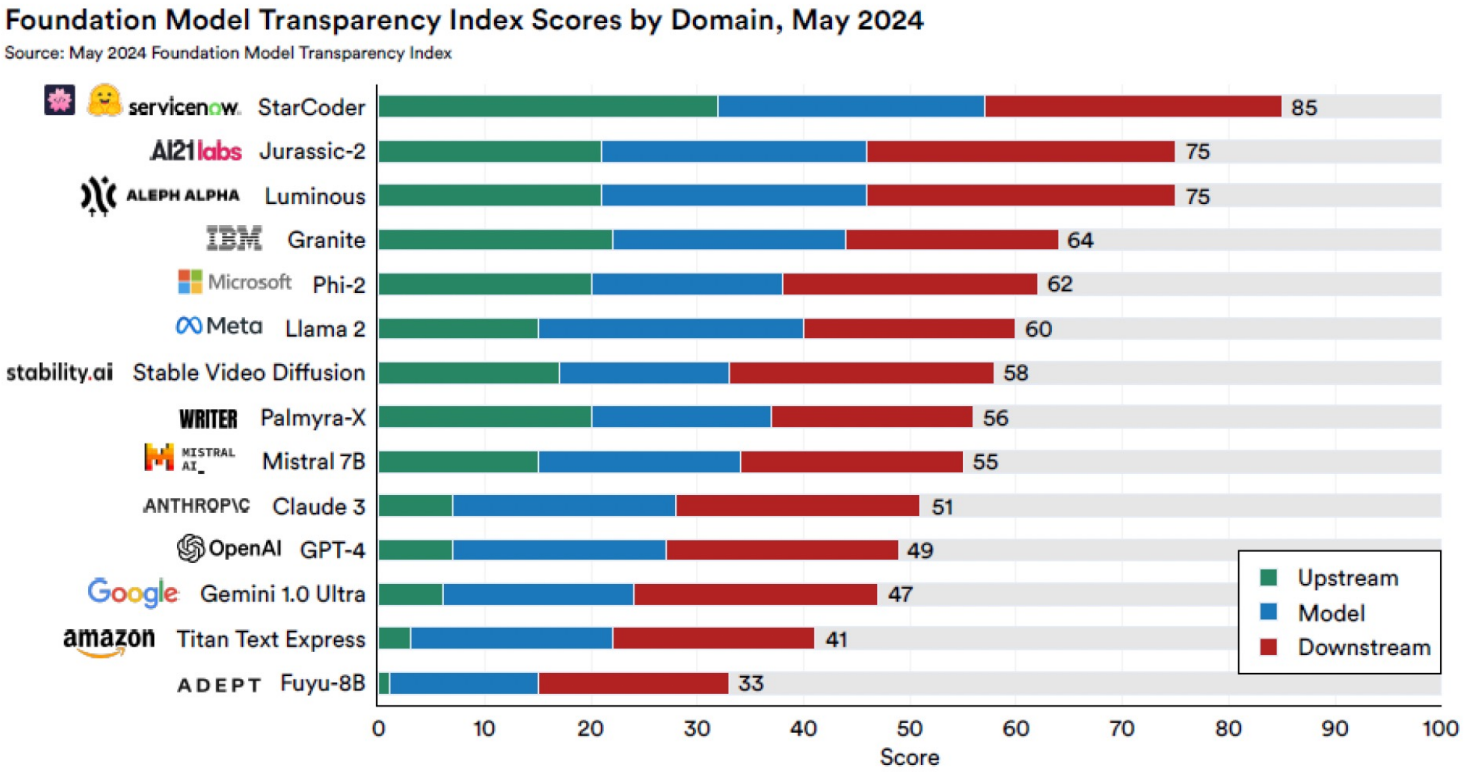


Figure 1: **Scores by Domain.** The overall scores disaggregated into the three domains: upstream, model, and downstream.

Explainable AI

In Healthcare intepretation is necessary:
Intrepetable, Explainable and Trustworthy predictions

Explainable AI

Transparency: Users need to comprehend how models make predicitons

Risk understanding: Users need to understand, quantify and manage risk

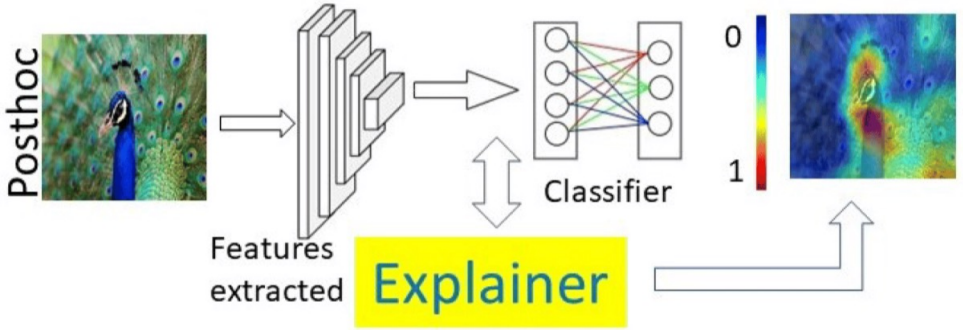
Avoid implicit bias: Users need to be able to check whether the model does not learn biases

Discovery: Users need to distil insights and new knowledge from the learned model

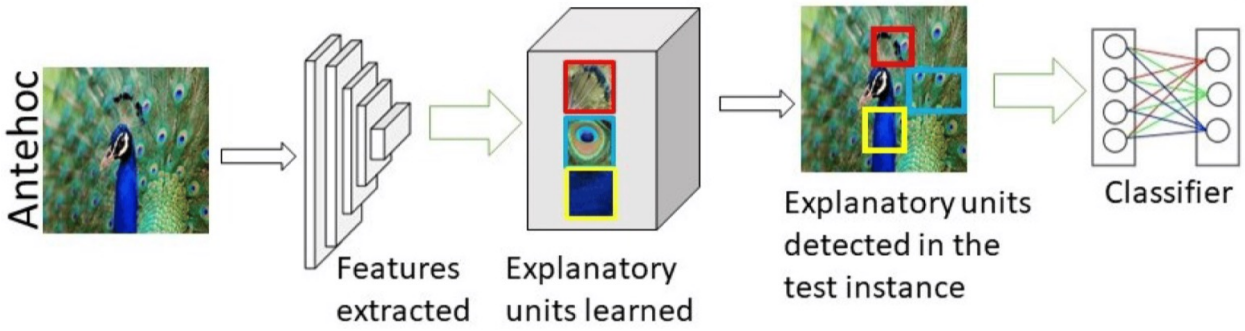
Know what we do not know: Users need to have a quantifictation of the model's prediction unceirtainty

Explainable AI

Explainable AI for Modellers and users



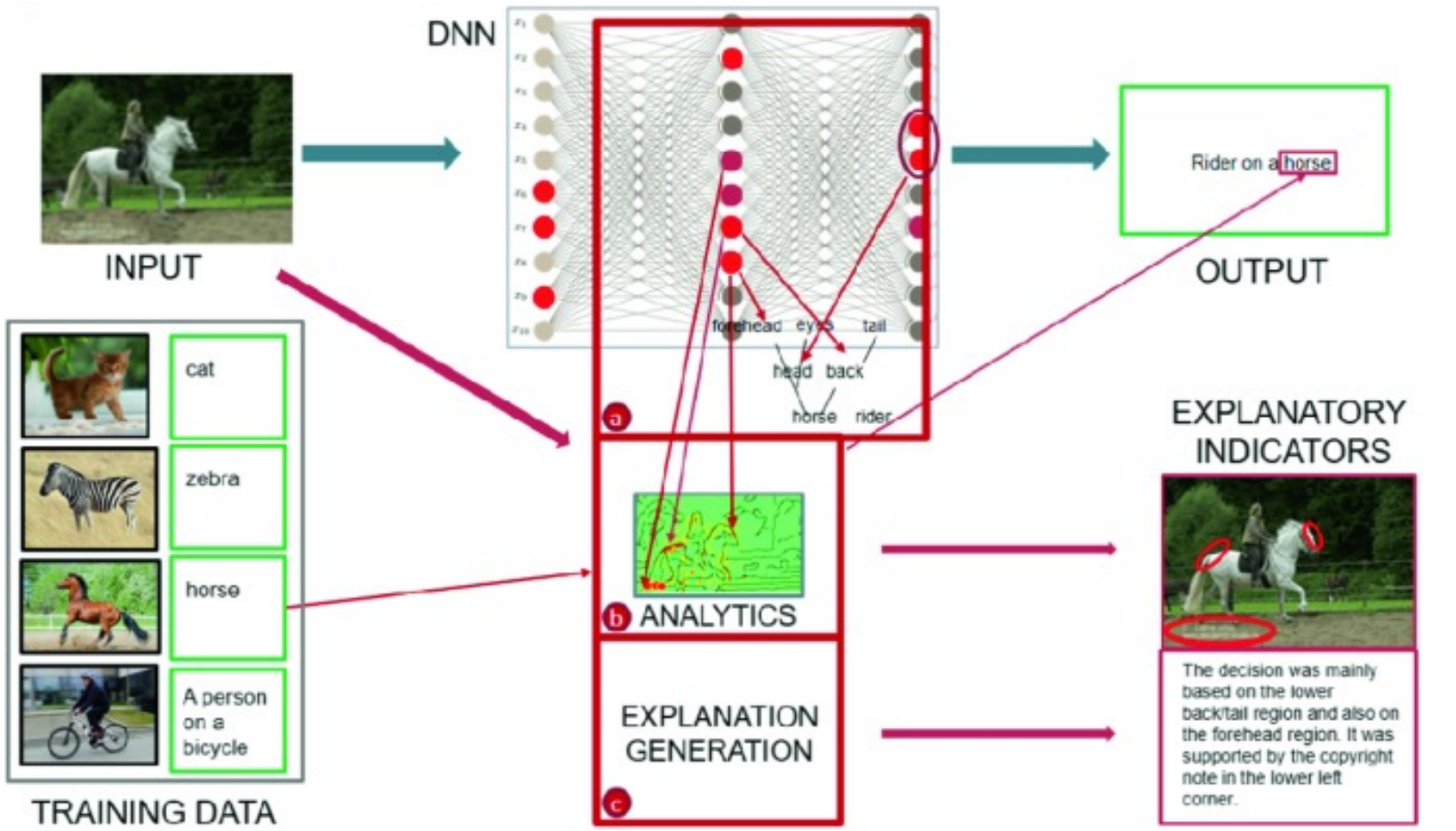
Aspect	Posthoc	Antehoc
Modifications to black-box architecture	Not Necessary	Necessary
Training/ Retraining	Not Necessary	Necessary
Faithfulness	Challenging	Guaranteed
Retaining black-box accuracy	Guaranteed	Challenging

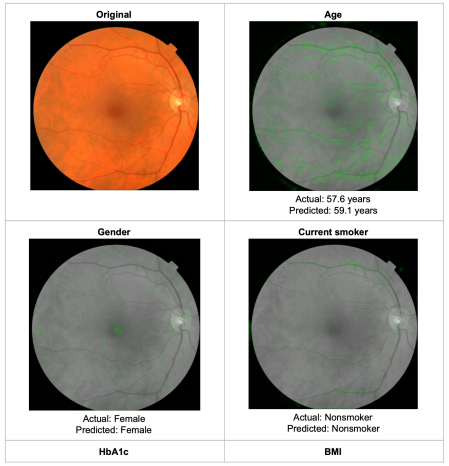


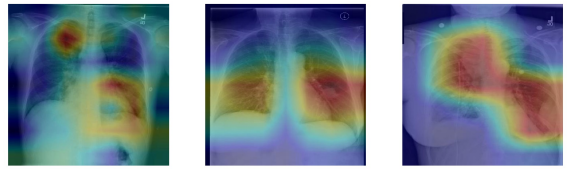
Kamakshi, Vidhya, and Narayanan C. Krishnan. "Explainable Image Classification: The Journey So Far and the Road Ahead." *AI* 4.3 (2023): 620-651.

Explainable AI

Explainable AI for Modellers and users



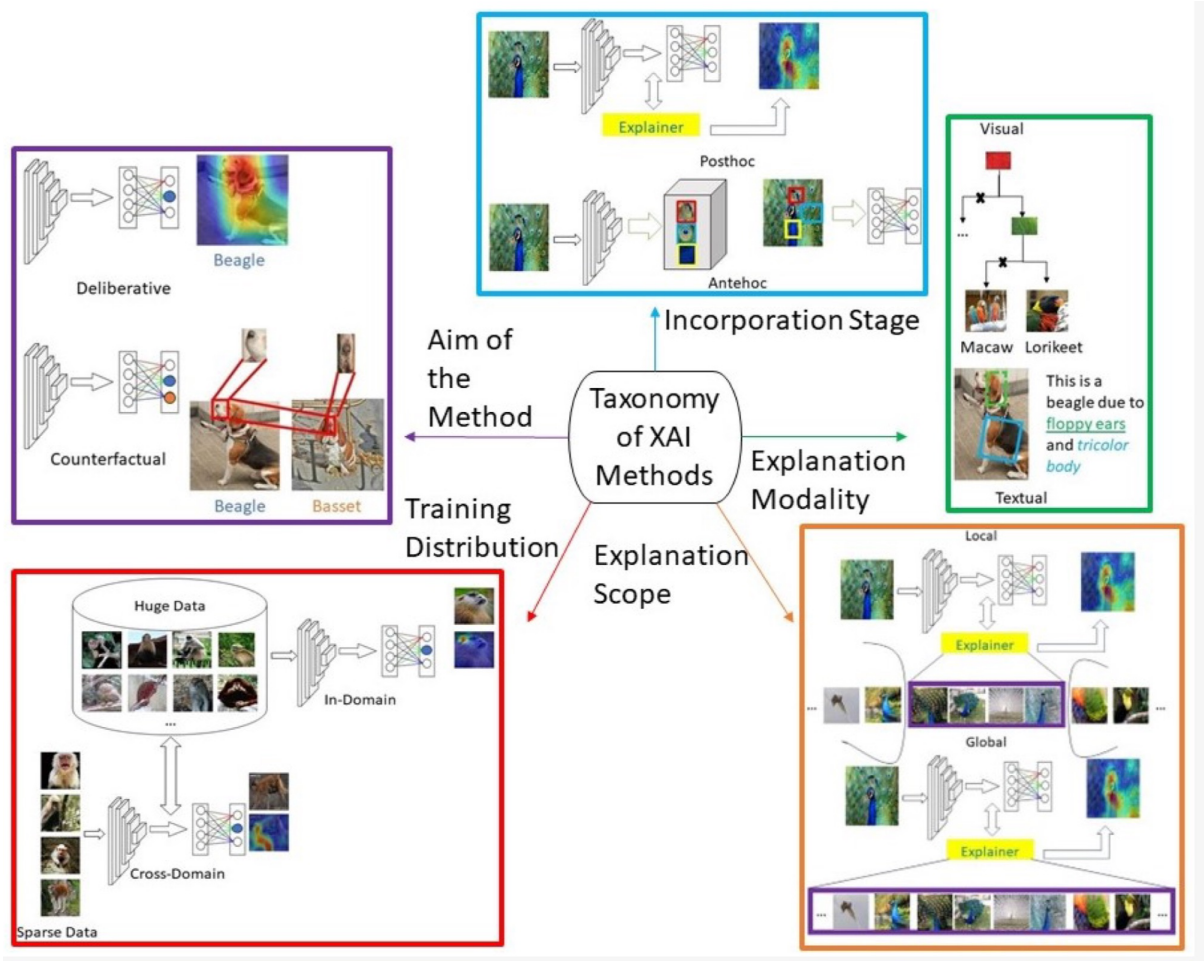




Xu, Feiyu, et al. "Explainable AI: A brief survey on history, research areas, approaches and challenges." *Natural Language Processing and Chinese Computing: 8th CCF International Conference, NLPCC 2019, Dunhuang, China, October 9–14, 2019, Proceedings, Part II 8*. Springer International Publishing, 2019.

Explainable AI

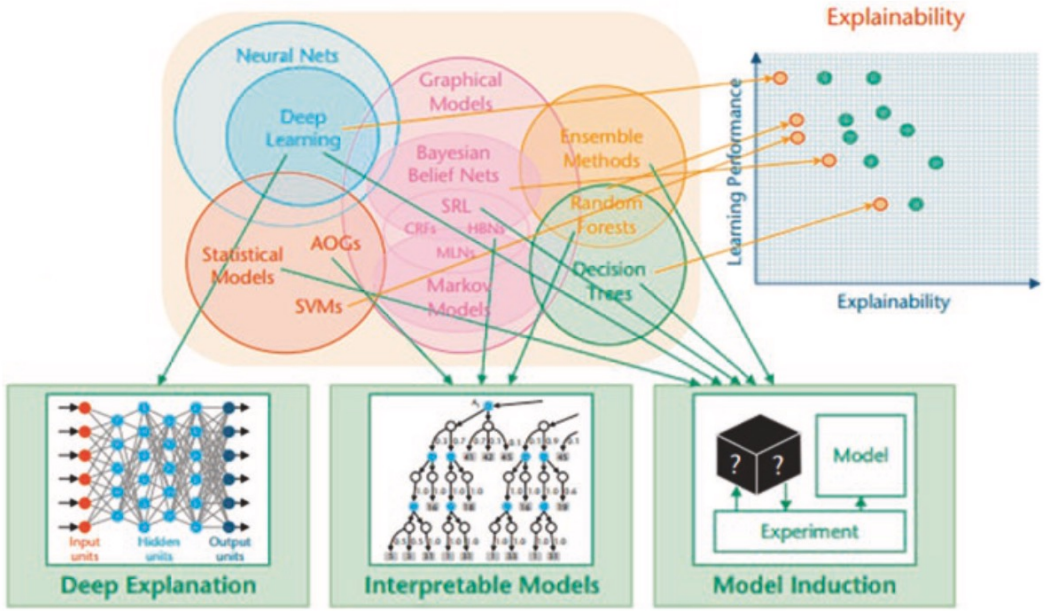
Explainable AI for Modellers and users

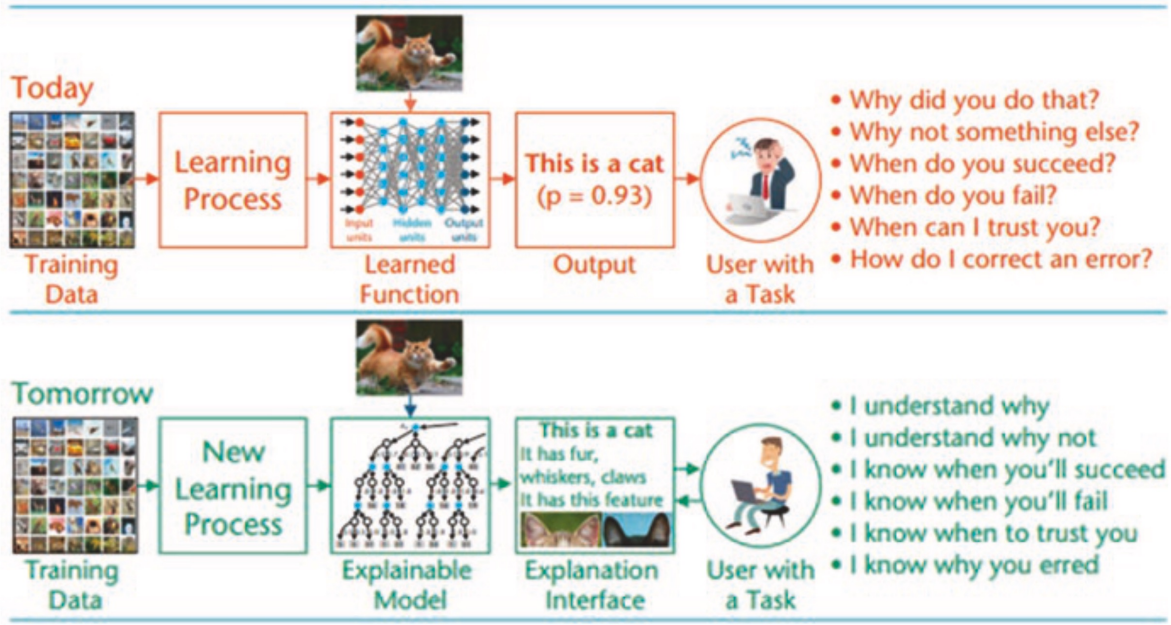


Kamakshi, Vidhya, and Narayanan C. Krishnan. "Explainable Image Classification: The Journey So Far and the Road Ahead." *AI* 4.3 (2023): 620-651.

Explainable AI

Using transparent, interpretablel model





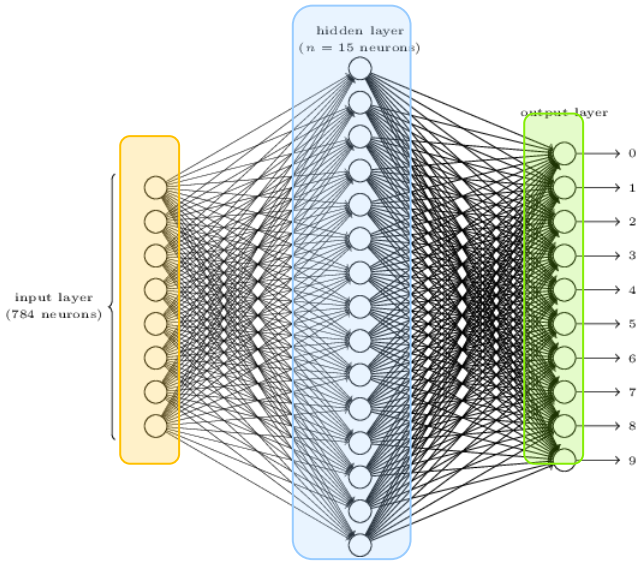
Gaur, Loveleen, and Biswa Mohan Sahoo. "Introduction to Explainable AI and Intelligent Transportation." *Explainable Artificial Intelligence for Intelligent Transportation Systems: Ethics and Applications*. Cham: Springer International Publishing, 2022. 1-25.

Supervised ML and knowledge-based AI

Supervised machine learning

Nonknowledge-based systems (data driven)

- Artificial neural networks
- Genetic networks
- Deep Learning



http://neuralnetworksanddeeplearning.com

Knowledge-based systems (data driven, modeled or both)

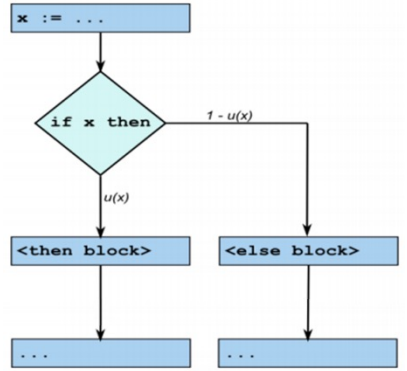
- Rule-based systems
- Decision trees
- Markov random fields
- Bayesian networks

```
fever_limit      :=

FUZZY SET (37.5,0), (38,1);
temperature   := 37.9;

message := “patient has no fever”;
IF temperature > fever_limited
THEN
    message := “patient has fever”;
END IF
```

➔ Result message: “patient has fever” (with an applicability of 0.8)



Arden Syntax.
Medexter Healthcare GmbH, 2014

Supervised machine learning

Constrains for building a Therapy Decision Support Systems

- Limited data (availability and accessibility)
- Distinguish multi-disciplinary viewpoints
- Transparency
- Reproducibility
- Extended model analyses

Diagnose

One disease = **one diagnose**

Differential diagnoses:

More examinations ➔ more certainty

Therapy

One disease = **one or more therapy options**

More examinations ➔ more certainty?

Decision Trees

Table 8
Error analysis results for the developed CART model (using 5 features) and the literature models while conducting CAD diagnosis.

Algorithm	Parameter		
	ACC%	**TNV%**	**TPR%**
Bagging SMO [51]	93.40	87.36	95.83
Naïve Bayes [51]	75.51	**95.40**	67.59
SMO [51]	**94.08**	88.51	96.30
ANN [51]	88.11	80.46	91.20
ANN [53]	84.62	83.00	86.00
ANN-GA [53]	93.85	92.00	97.00
N2GC-nuSVM [79]	93.08	*	*
SMO [75]	92.09	79.31	97.22
SVM [75]	89.11	83.91	91.20
C4.5 [75]	83.85	55.17	95.37
Naïve Bayes [75]	80.15	94.25	74.54
KNN ($k = 1$) [75]	74.61	28.74	93.06
KNN ($k = 2$) [75]	74.94	17.24	98.15
KNN ($k = 10$) [75]	72.62	4.60	**100**
Naïve Bayes (average) [76]	60.61	47.56	73.02
C.45 (average) [76]	68.76	67.59	65.80
KNN (average) [76]	60.05	56.62	57.85
Naïve Bayes [80]	87.22	76.50	91.50
SMO [80]	86.95	79.00	90.11
Ensemble [80]	88.52	82.05	91.12
C4.5 (average) [81]	68.30	60.14	71.04
Bagging (average) [81]	69.64	61.32	71.38
J48 + MFA (average) [82]	91.09	*	91.10
BF tree + MFA (average) [82]	87.70	*	87.70
REP tree + MFA (average) [82]	84.28	*	84.27
NB tree + MFA (average) [82]	93.77	*	93.77
CART (5 features)	92.41	77.01	98.61

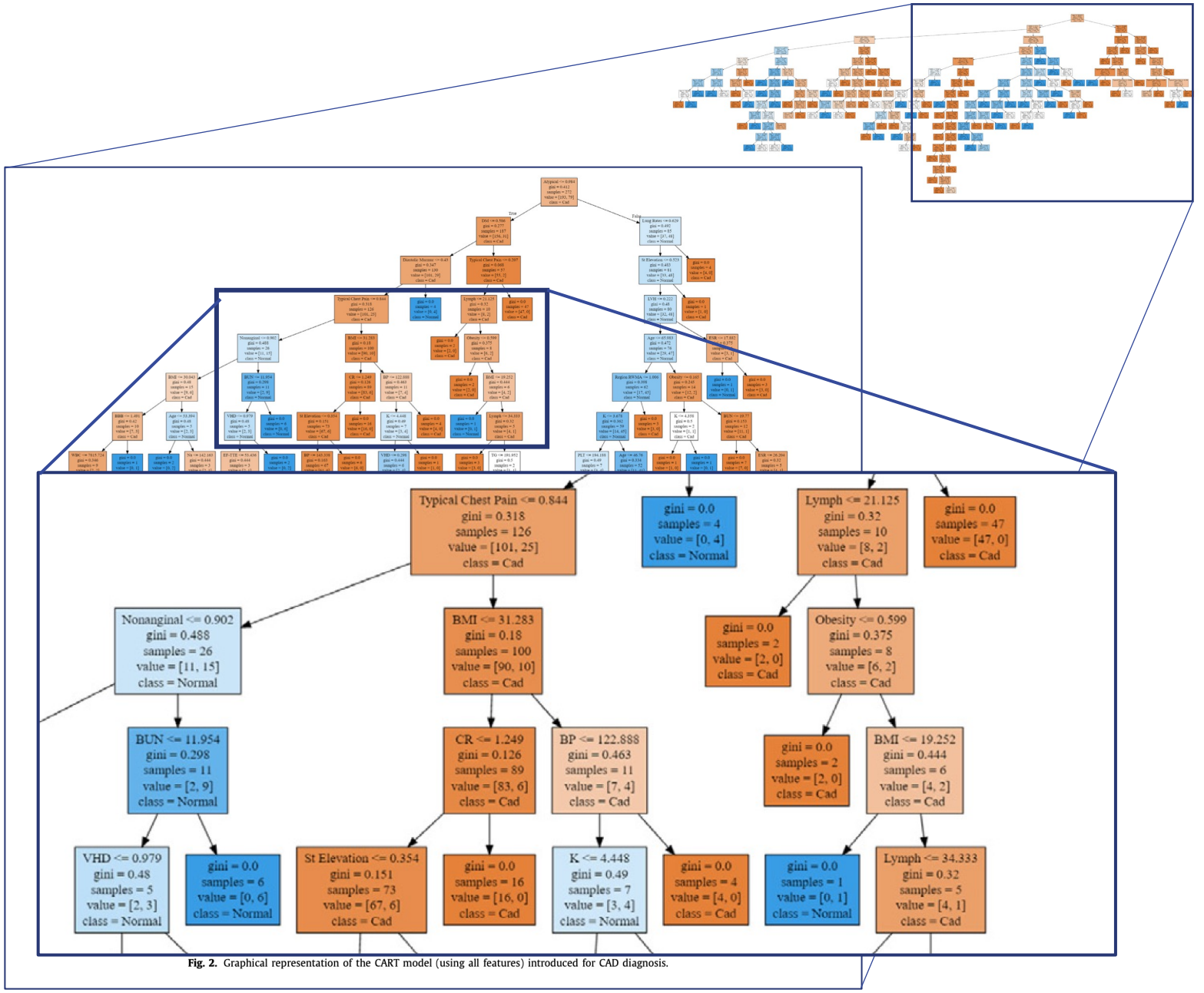
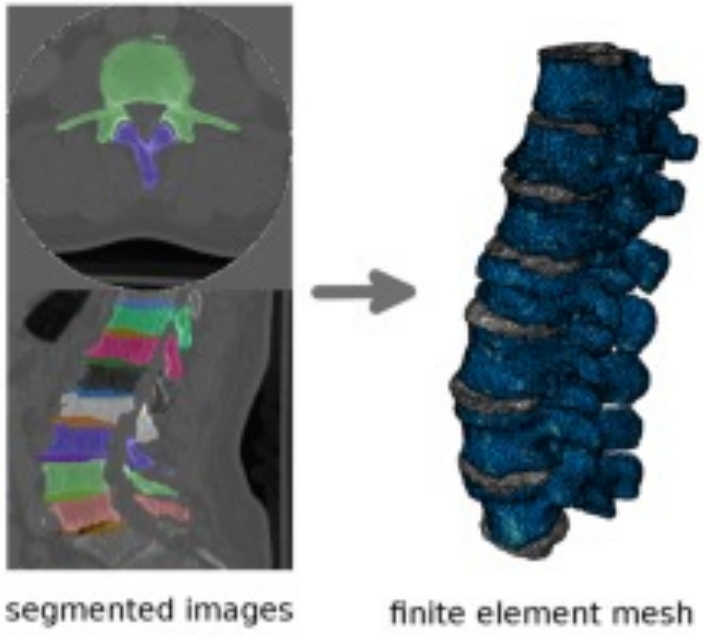
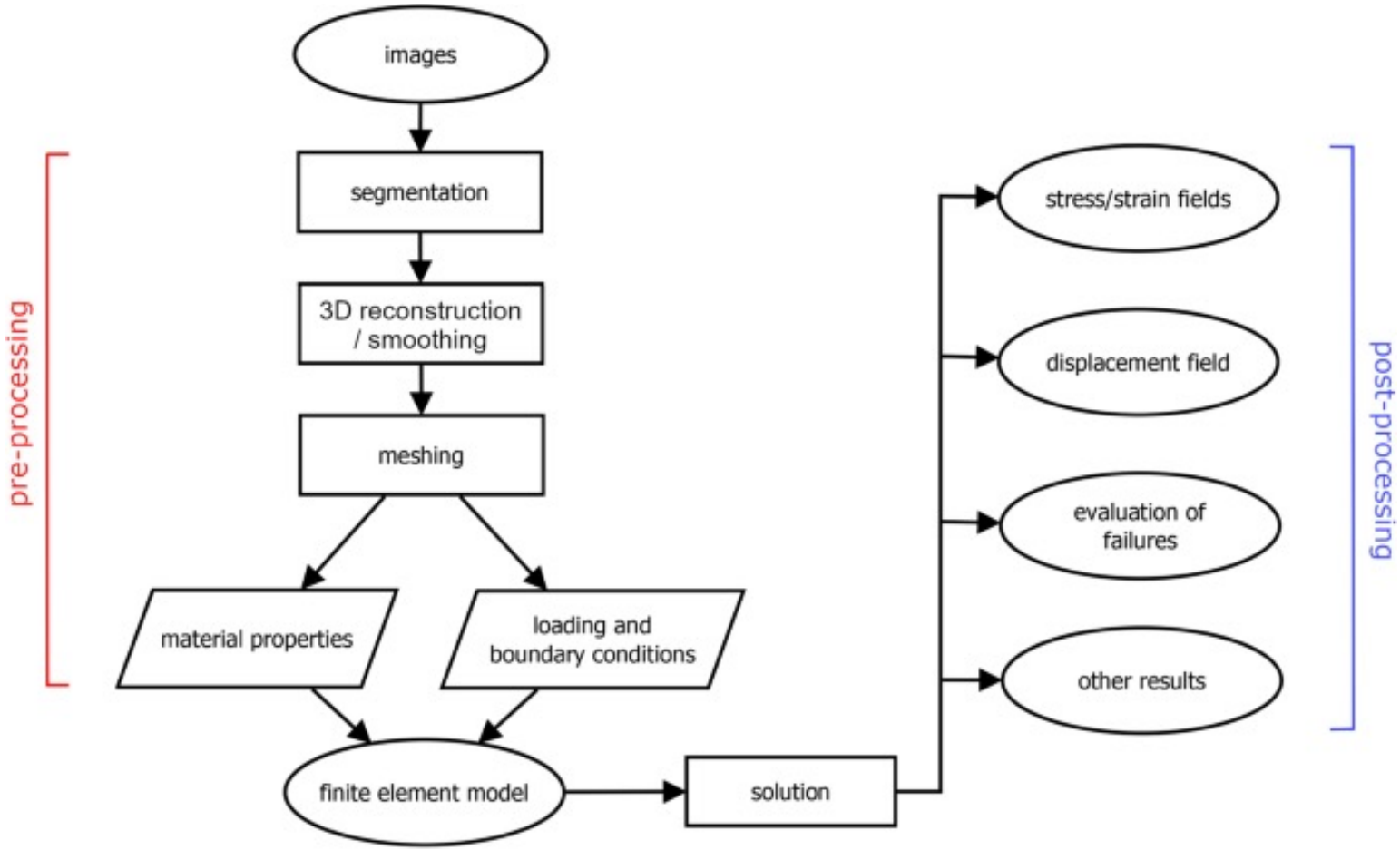


Fig. 2. Graphical representation of the CART model (using all features) introduced for CAD diagnosis.

Ghiasi, M. M., et al. "Decision tree-based diagnosis of coronary artery disease: CART model." *Computer methods and programs in biomedicine* 192 (2020): 105400.

Finite Models

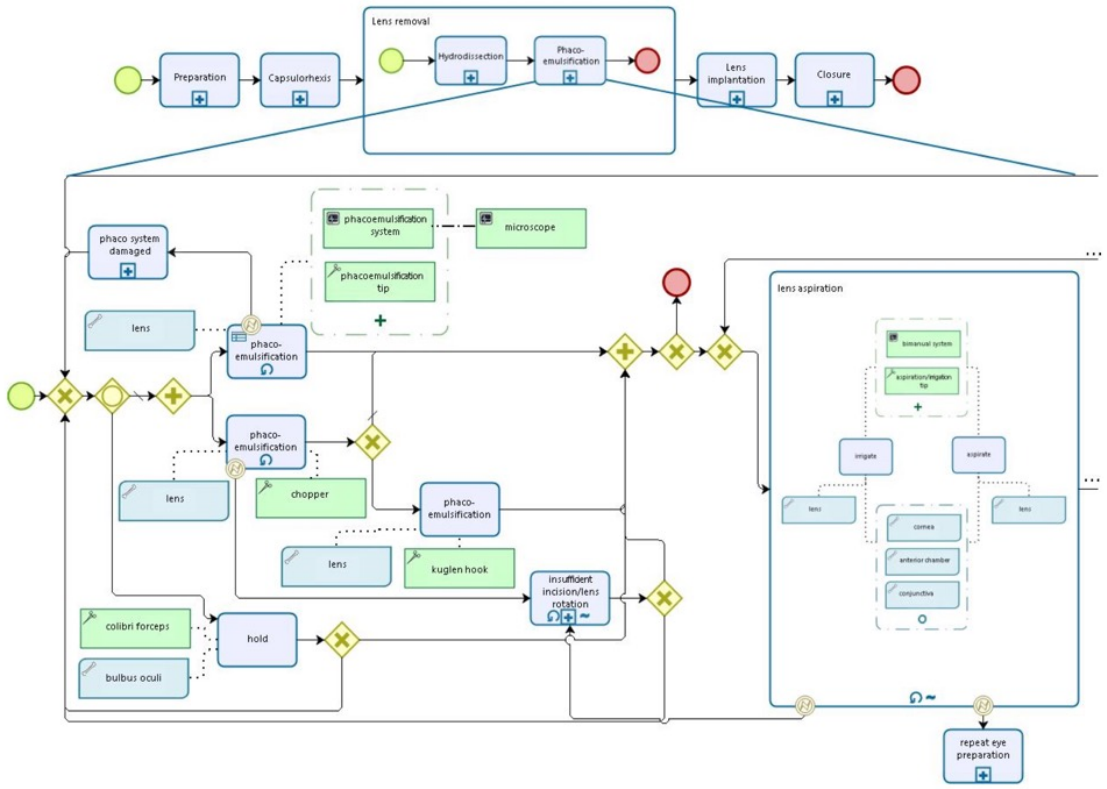




Galbusera, Fabio, et al. "Image-based biomechanical models of the musculoskeletal system." *European radiology experimental* 4.1 (2020): 49.

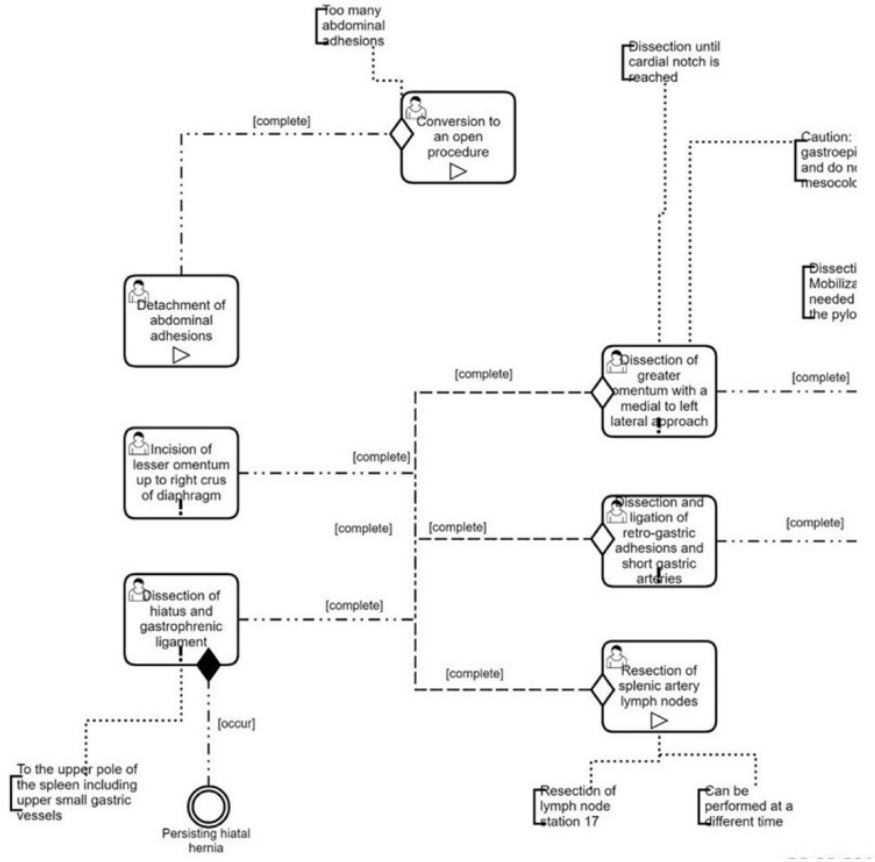
Process Models | BPMN (SPMs) and CMMN

Business Process Model and Notation



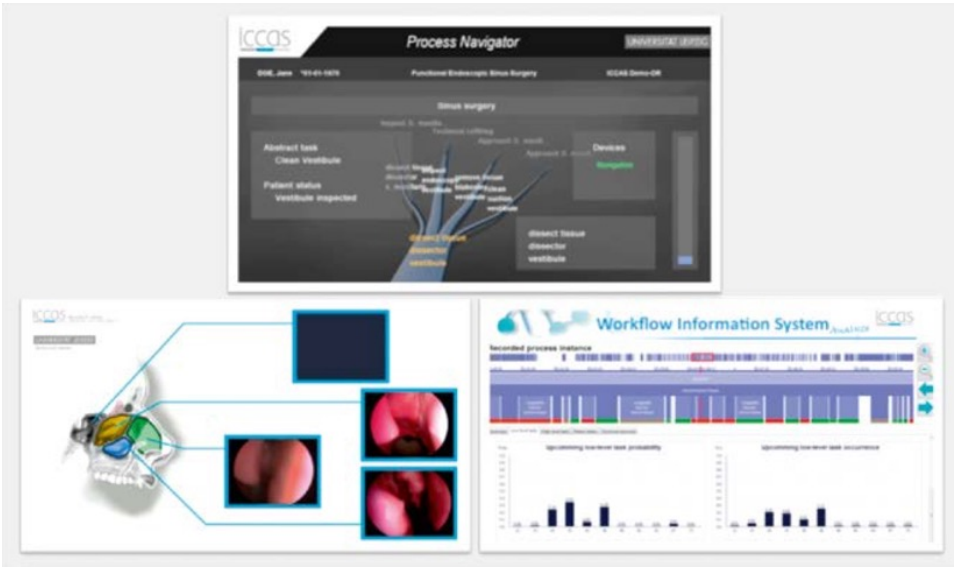
Neumann, Juliane, et al. "Application of activity semantics and BPMN 2.0 in the generation and modeling of generic surgical process models." *IJCARS* 12 (2017): 48-49.

Case Management Model and Notation

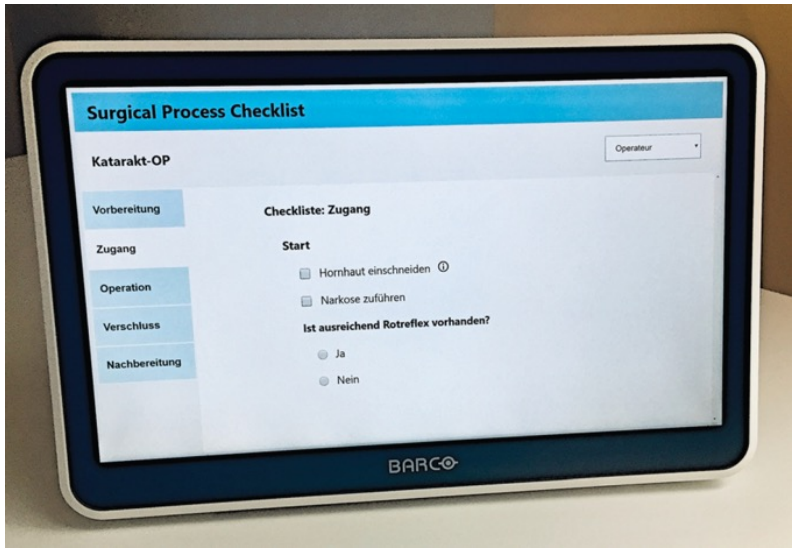


Junger, Denise, et al. "Toward an interoperable, intraoperative situation recognition system via process modeling, execution, and control using the standards BPMN and CMMN." *International journal of computer assisted radiology and surgery* 19.1 (2024): 69-82.

Process Models | Examples



Surgical process recognition
ICCAS, Leipzig (Pirlich, Stöhr, Neumuth & Dietz. 2019)

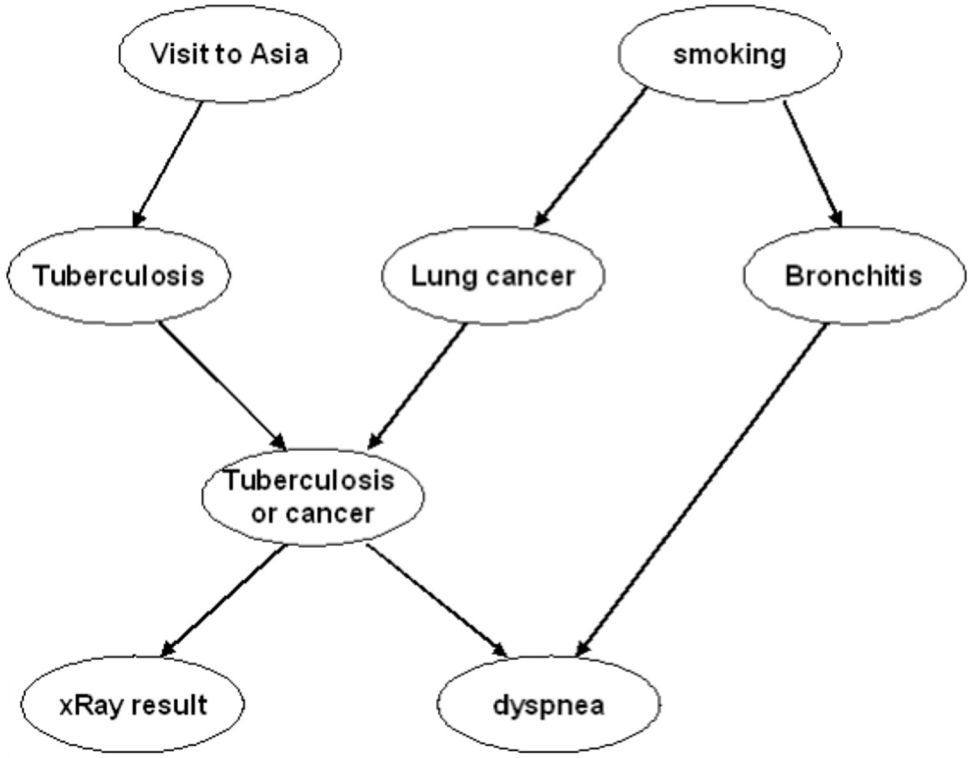


Automatically generated surgical checklists
HS Reutlingen (Ryniak, Burgert. 2020)

IT-assisted process management in healthcare.
UMIT, Austria (Elske, Hackl. 2020)

Bayesian networks

Bayesian networks (aka causal networks)



Example of a BN - https://www.cs.cmu.edu/afs/cs/academic/class/15381-s07/www/hw5/hw5Sols.pdf

Activation!

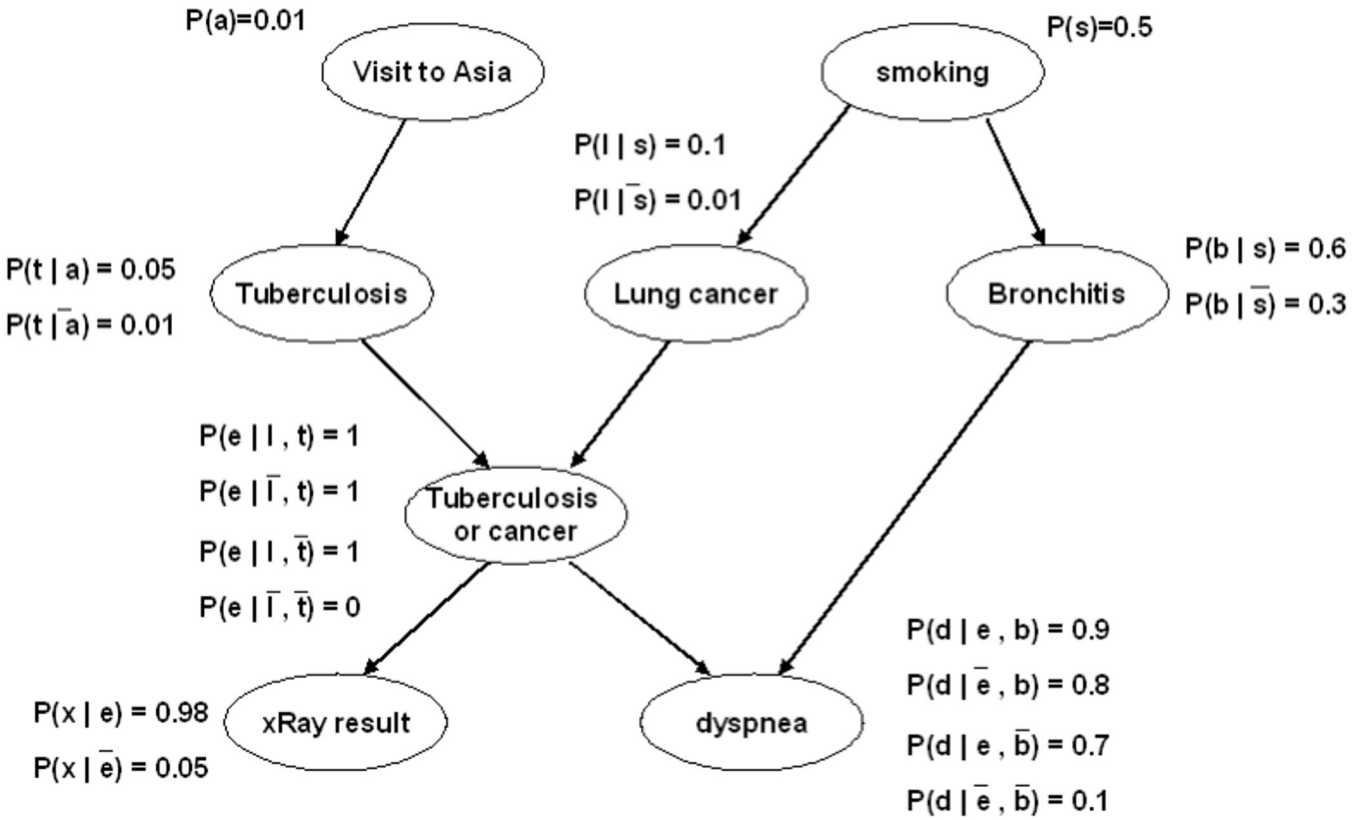
13. How are the information entities *Cancer* and *CT* causally related?
    Cancer → *CT* or *CT* → *Cancer*?

14. How are the information entities *Cancer* and *Alcohol Abuse* causally related?
    Cancer → *Alcohol Abuse* or *Alcohol Abuse* → *Cancer*?

pingo.coactum.de/815896

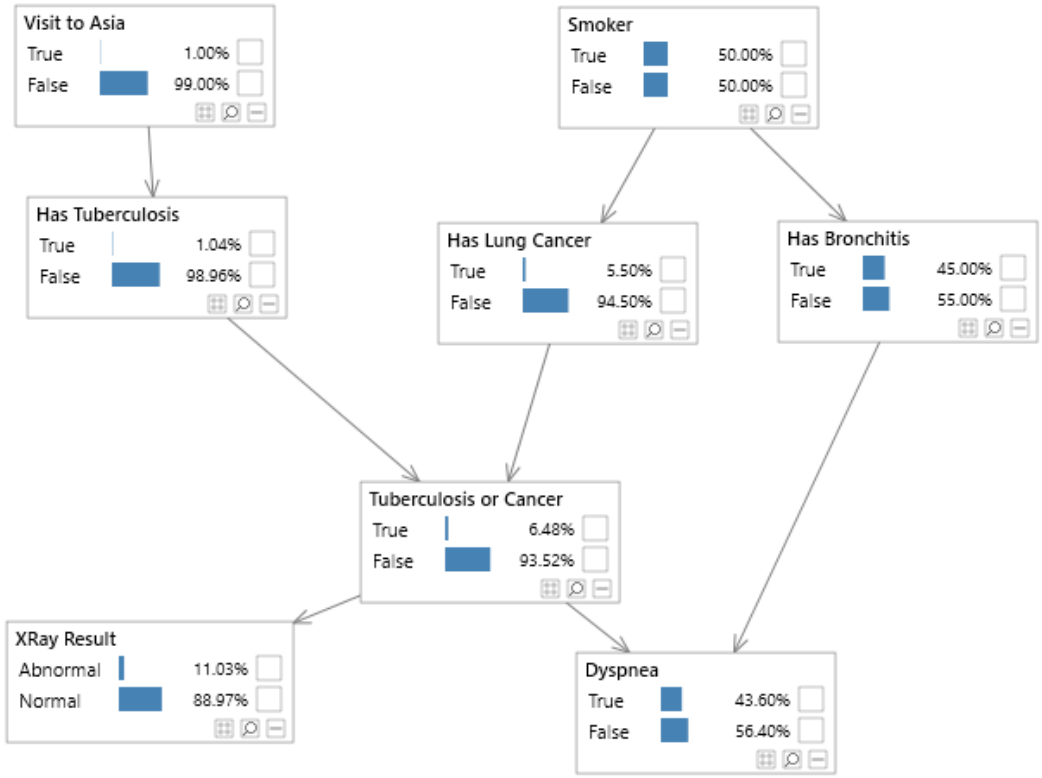
15. How are the information entities *CT* and *Alcohol Abuse* causally related?
    CT → *Alcohol Abuse* or *Alcohol Abuse* → *CT*?

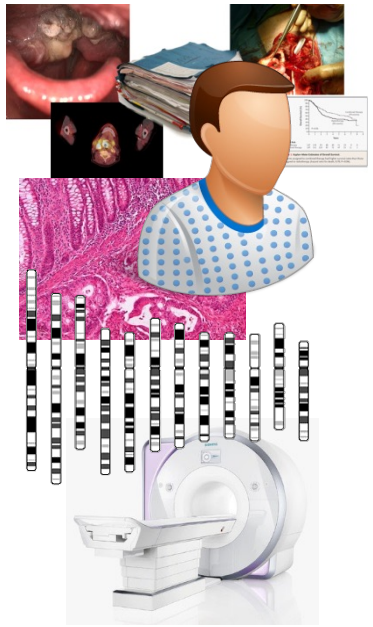
Bayesian networks



Example of a BN - https://www.cs.cmu.edu/afs/cs/academic/class/15381-s07/www/hw5/hw5Sols.pdf

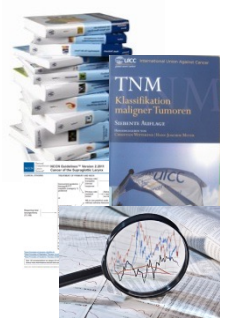
Bayesian networks

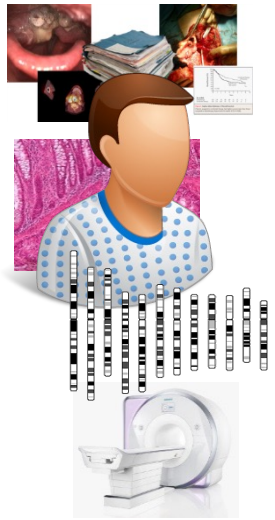


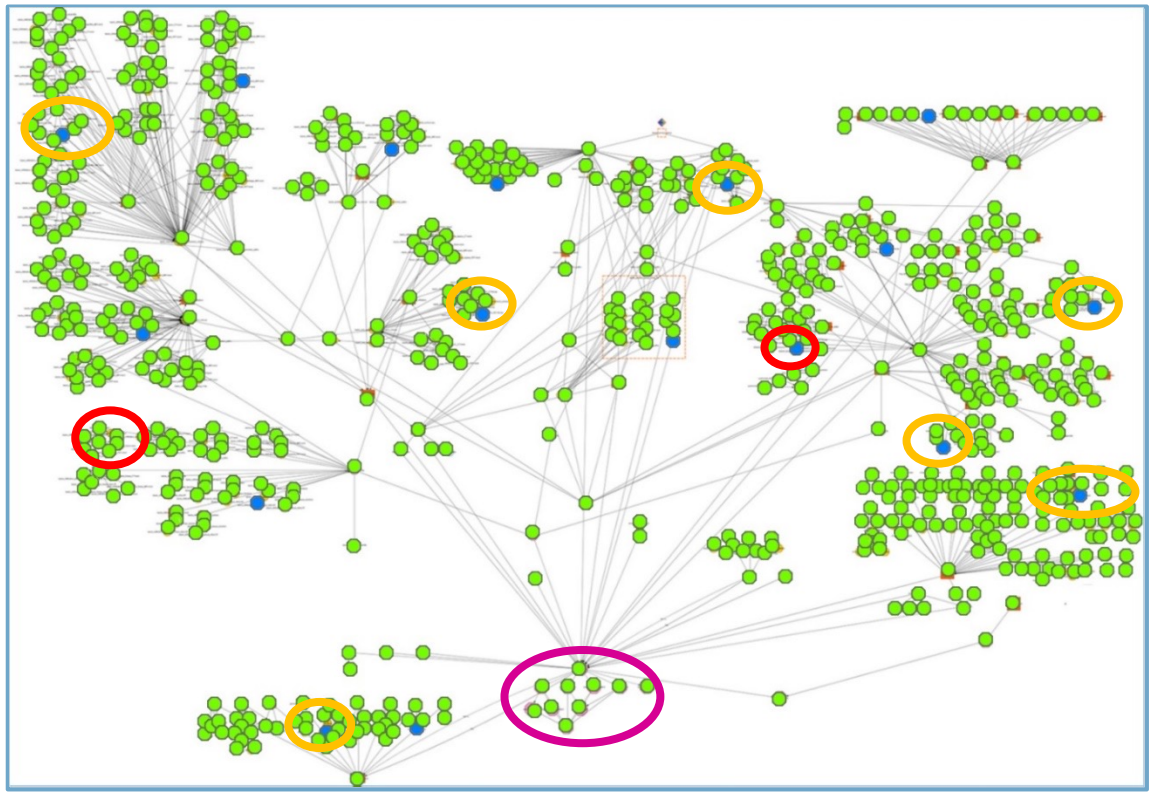


Example of a BN - https://www.bayesserver.com/docs/introduction/bayesian-networks/

Bayesian networks







Methods to analyze patient-specific BN:

- Diagnostic value
- Sensitive analysis

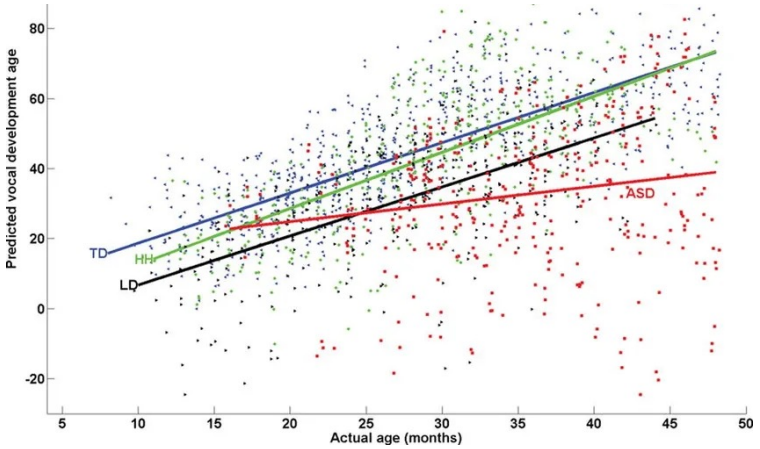
Bayesian networks | Example German Heart Center

Czerny, Martin, et al. "Prediction of mortality rate in acute type A dissection: the German Registry for Acute Type A Aortic Dissection score." *European Journal of Cardio-Thoracic Surgery* 58.4 (2020): 700-706.

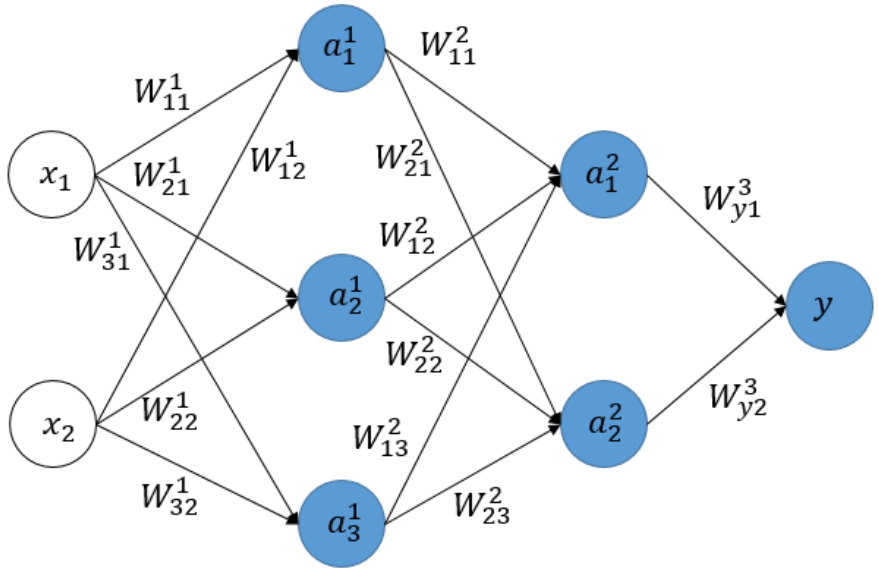
Mortality prediction (2537 patients)

- Linear regression (AUC = 0.712)
- Elastic net regularization
- Gradient boosted models

Google: Visual example of Multi Linear Regression Model



Google: Visual example of Deep Neural Network



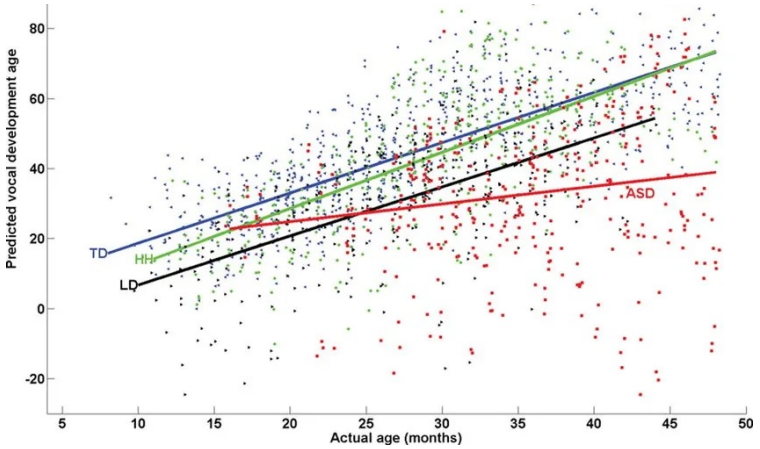
Bayesian networks | Example German Heart Center

Czerny, Martin, et al. "Prediction of mortality rate in acute type A dissection: the German Registry for Acute Type A Aortic Dissection score." *European Journal of Cardio-Thoracic Surgery* 58.4 (2020): 700-706.

Mortality prediction (2537 patients)

- Linear regression (AUC = 0.712)
- Elastic net regularization
- Gradient boosted models

Google: Visual example of Multi Linear Regression Model

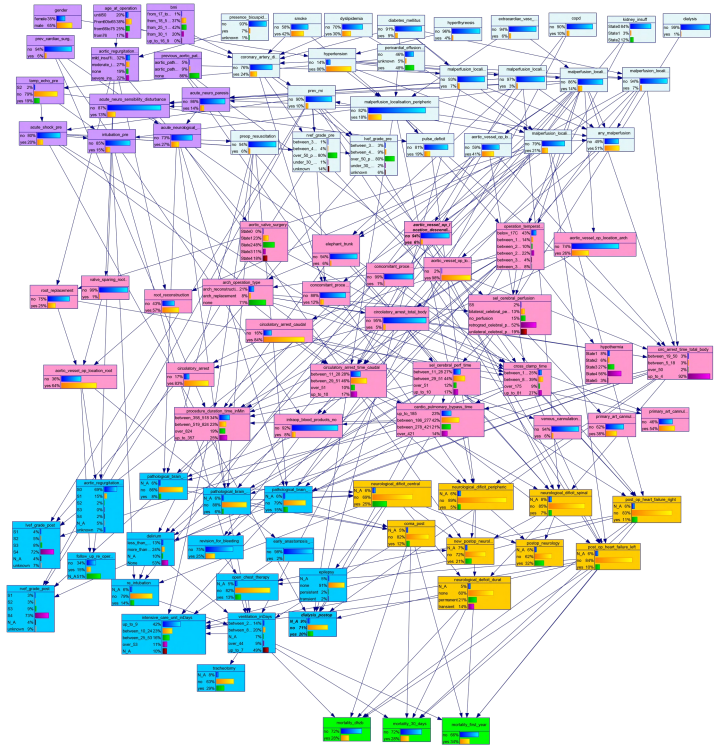


Cypko, Mario et al. „From Emergency to Recovery: A Multipurpose Bayesian Model for Type-A Patients in Cardiac Surgery“ (unpublished)

Pre-OP

Intra-OP

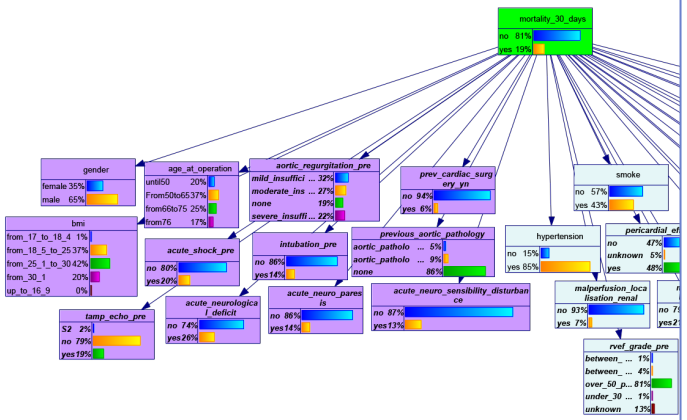
Post-OP



Legend: Pre-OP = Checklist questionary + In-hospital examinations
Intra-OP = Surgrical treatment and durations
Post-OP = MACCE + examinations beyond MACCE

Bayesian networks | Example German Heart Center

Naive Bayesian Networ



ROC 1-Year-Moratlity: Yes/No
Models: V1 + Naive BN

AUC of NBN = 0.77

Hahn Schickard

Conditions

If the following causes occur to 100%...

History of hospitalization — absent
Surgery in the past — present
Choledocholithotomy — absent

History of transfusion

ca. 1 min.

How likely is it, that **History of transfusion** is: **present**?

71%

(almost) impossible | improbable | uncertain | fifty-fifty | expected | probable | (almost) certain

How likely is it, that **History of transfusion** is: **absent**?

34%

(almost) impossible | improbable | uncertain | fifty-fifty | expected | probable | (almost) certain

How certain are you?

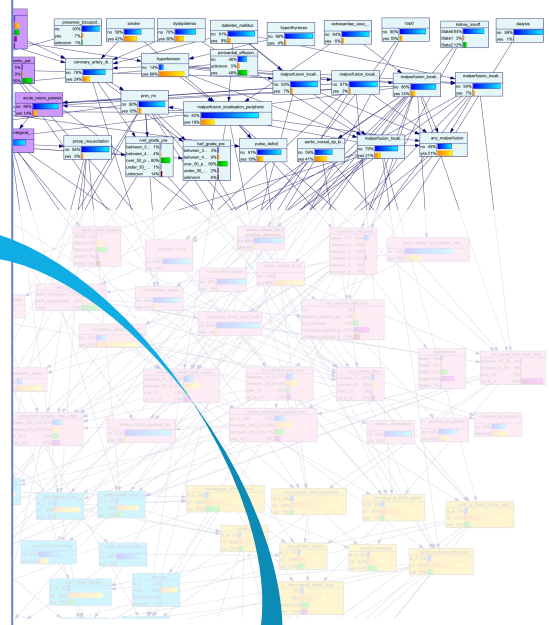
uncertain | rather uncertain | rather certain | certain

← Back | Continue →

Overview

Your progress is saved.

ry: A Multipurpose Bayesian
npublished)

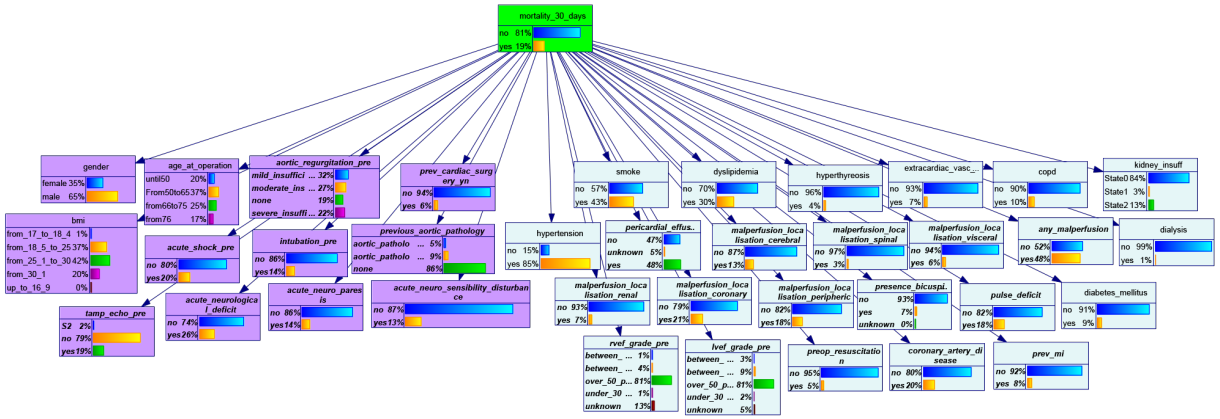


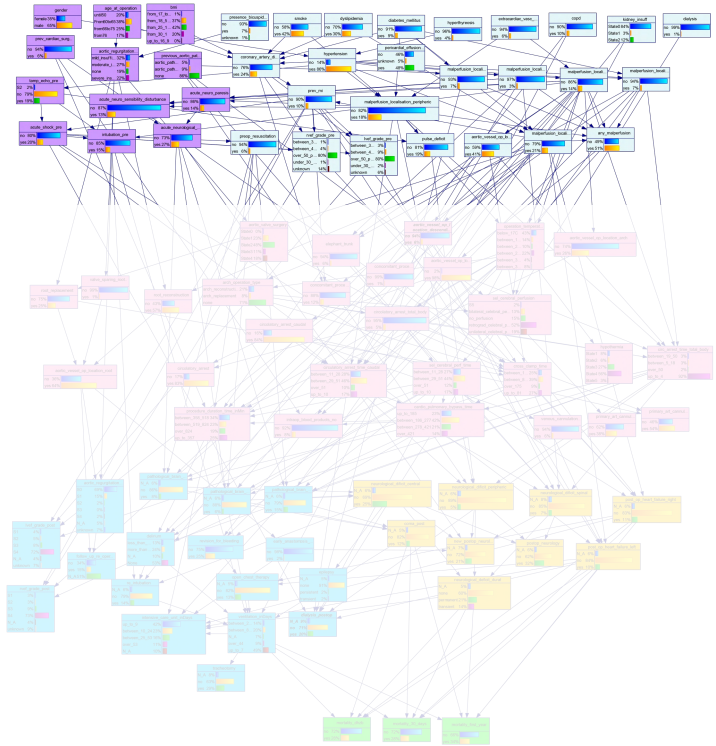
bmi		from_25_1_to_30				from_3_1				up_to_16			
smoke	es	no		yes		no		yes		no		yes	
prev_mi	yes	no	yes	no	yes	no	yes	no	yes	no	yes	no	yes
State0	0.77777778	0.7969697	0.7979798	0.91262136	0.78947368	0.82894737	0.63636364	0.8178913	0.85185185	0.66666667	0.33333333	0.83333333	0.33333333
State1	0.055555556	0.039393939	0.01010101	0.0097087...	0.052631579	0.032894737	0.090909091	0.060869565	0.12962963	0.16666667	0.33333333	0.083333333	0.33333333
State2	0.16666667	0.16363636	0.19191919	0.077669903	0.15789474	0.13815789	0.27272727	0.12173913	0.018518519	0.16666667	0.33333333	0.083333333	0.33333333

Bayesian networks | Example German Heart Center

Naive Bayesian Network

Cypko, Mario et al. „From Emergency to Recovery: A Multipurpose Bayesian Model for Type-A Patients in Cardiac Surgery“ (unpublished)

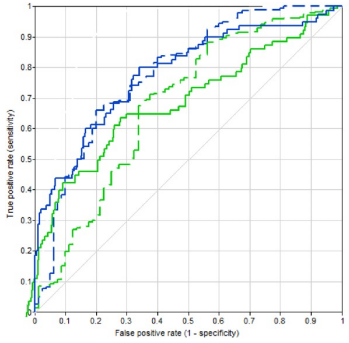




ROC 1-Year-Moratlity: Yes/No

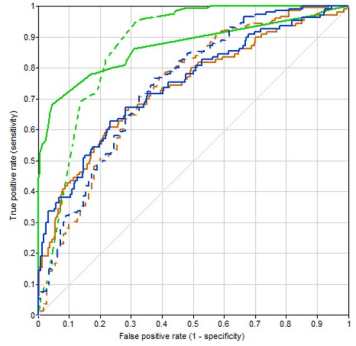
Models: Naive BN

AUC of NBN = 0.77

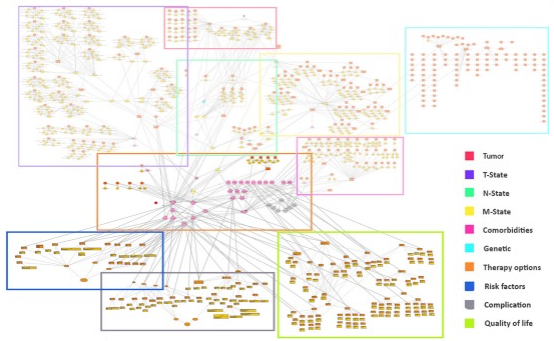


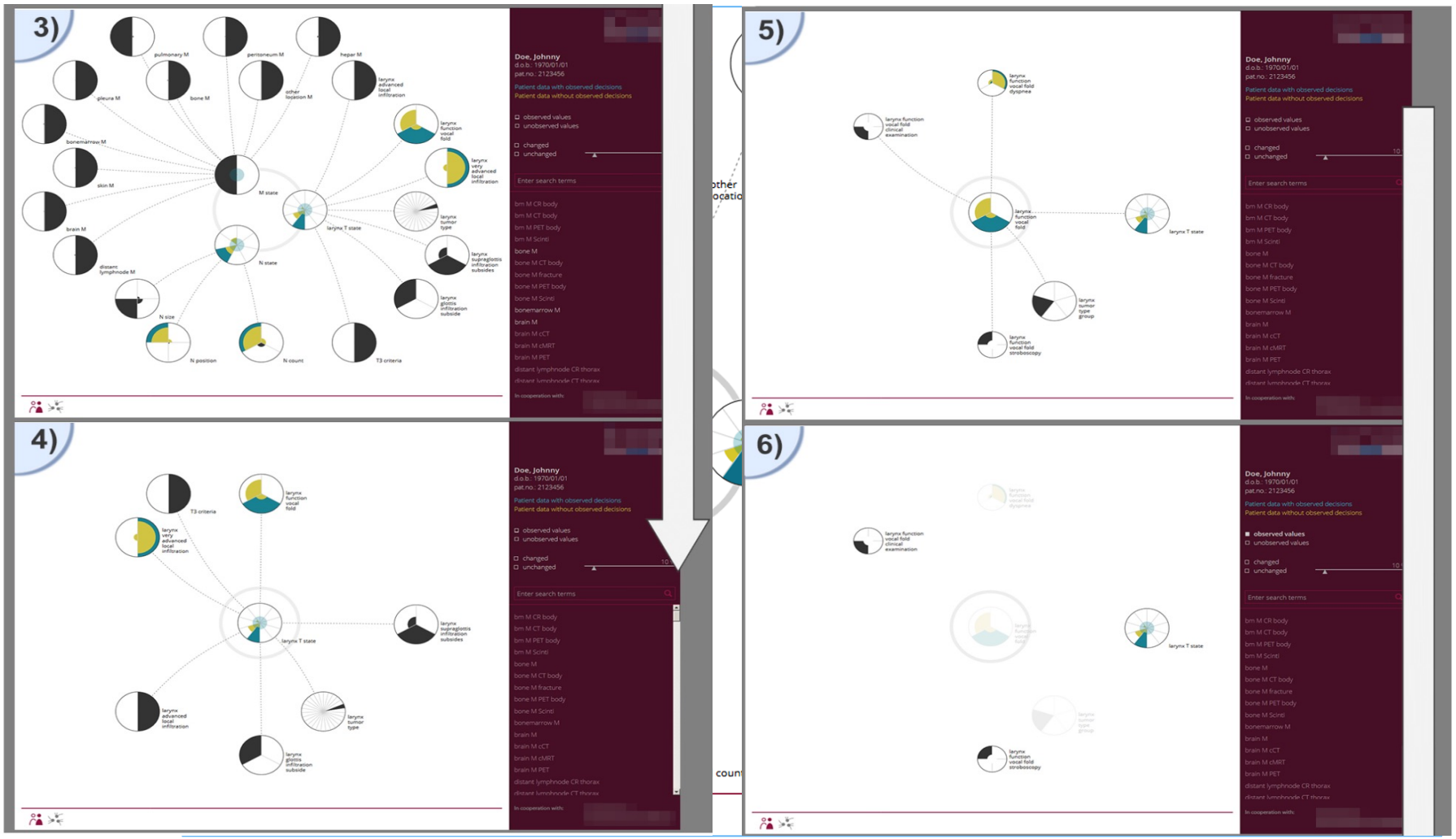
Models: Updated Naive BN

AUC of NBN = 0.87



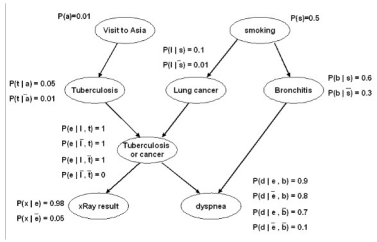
Bayesian network | Example of GUI

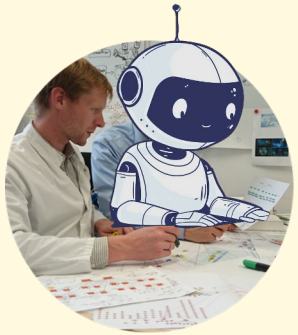




Cypko MA et al. Visual Verification of Cancer Staging for Therapy Decision Support. *Computer Graphics Forum*, 2017.

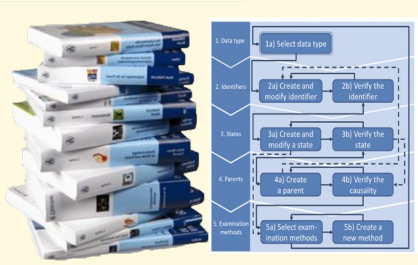
Bayesian network | Example LLM-supported Expert Modelling



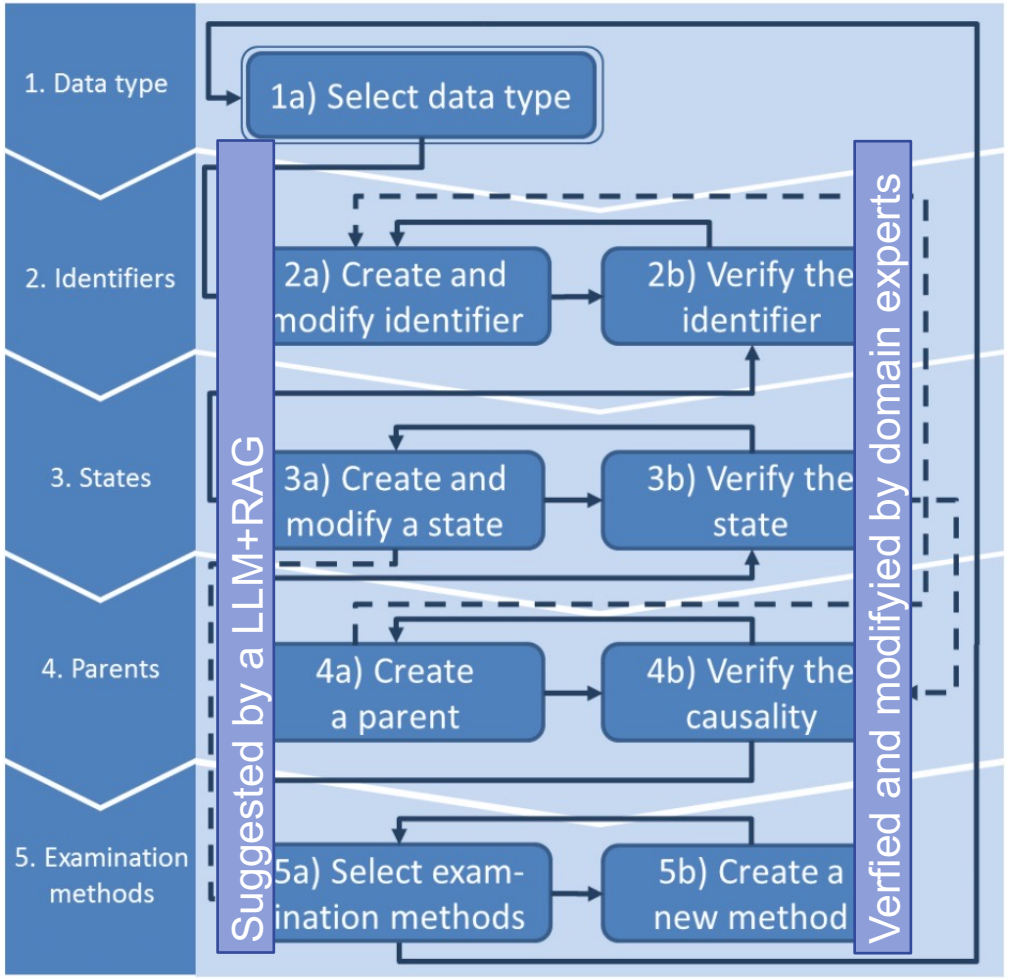


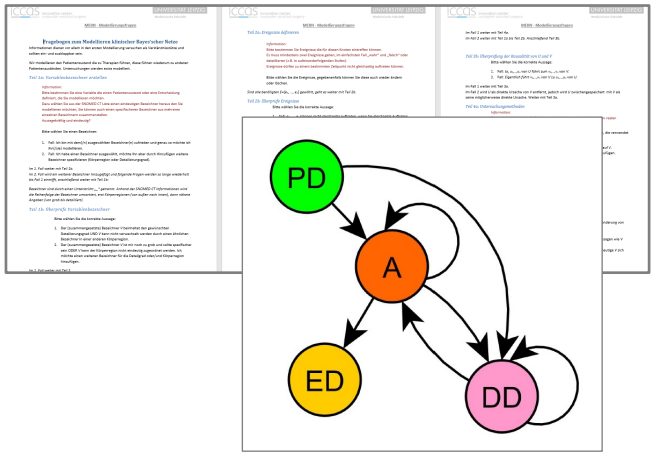
Knowledge Engineer

Domain Expert

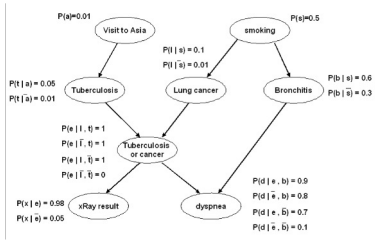


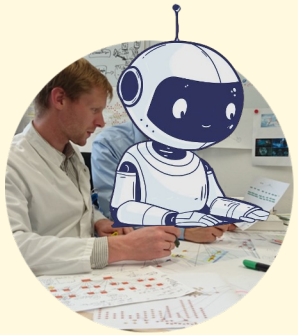
Traditional Process





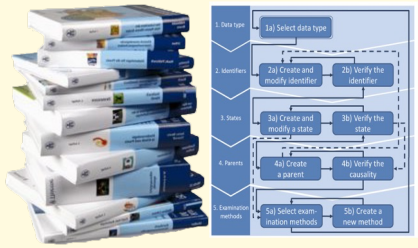
Bayesian network | Example LLM-supported Expert Modelling



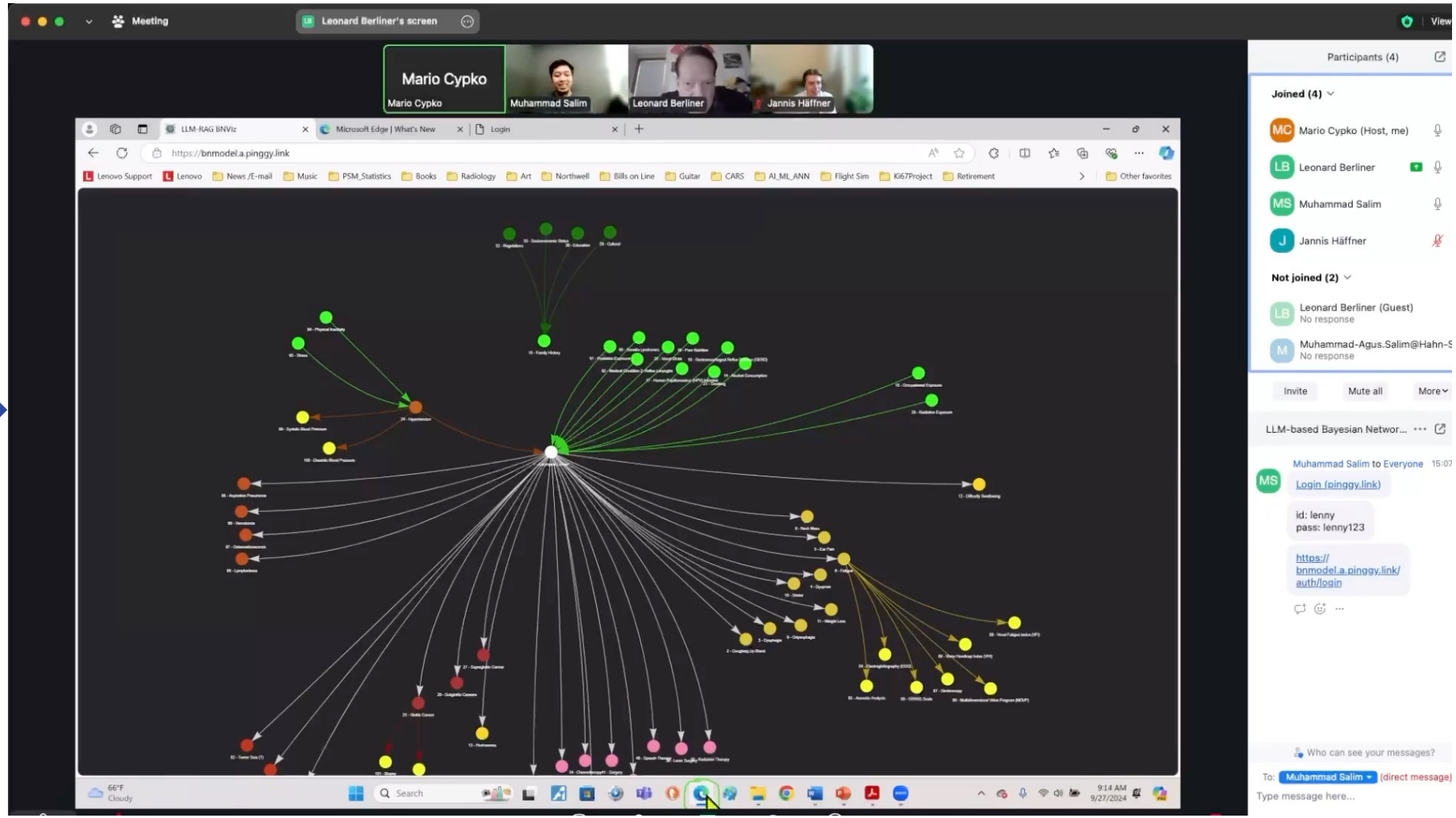


Knowledge Engineer

Domain Expert



Traditional Process

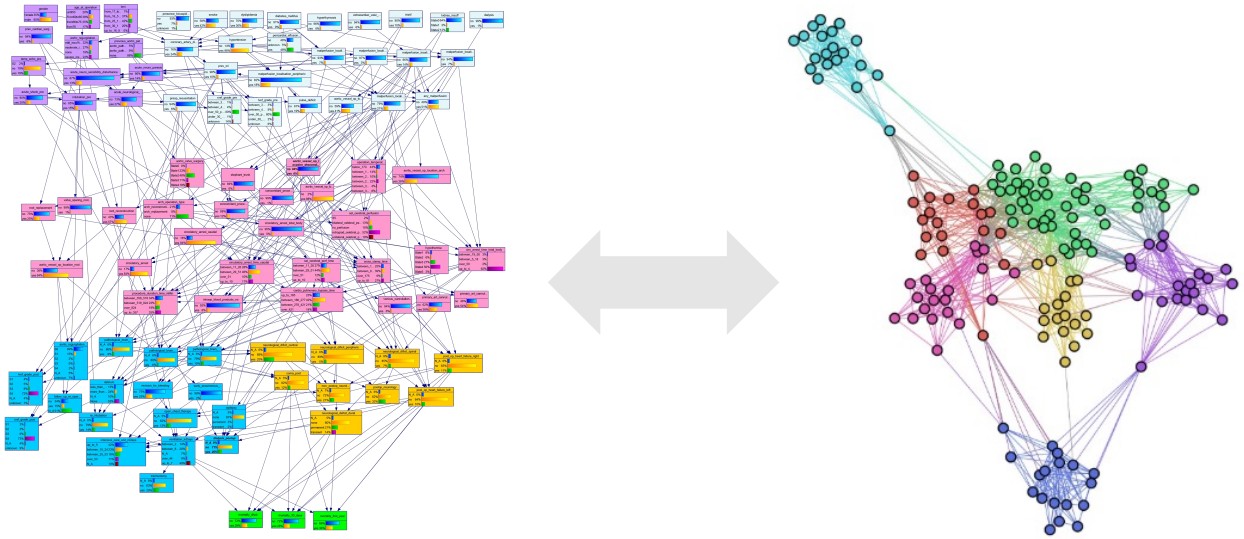


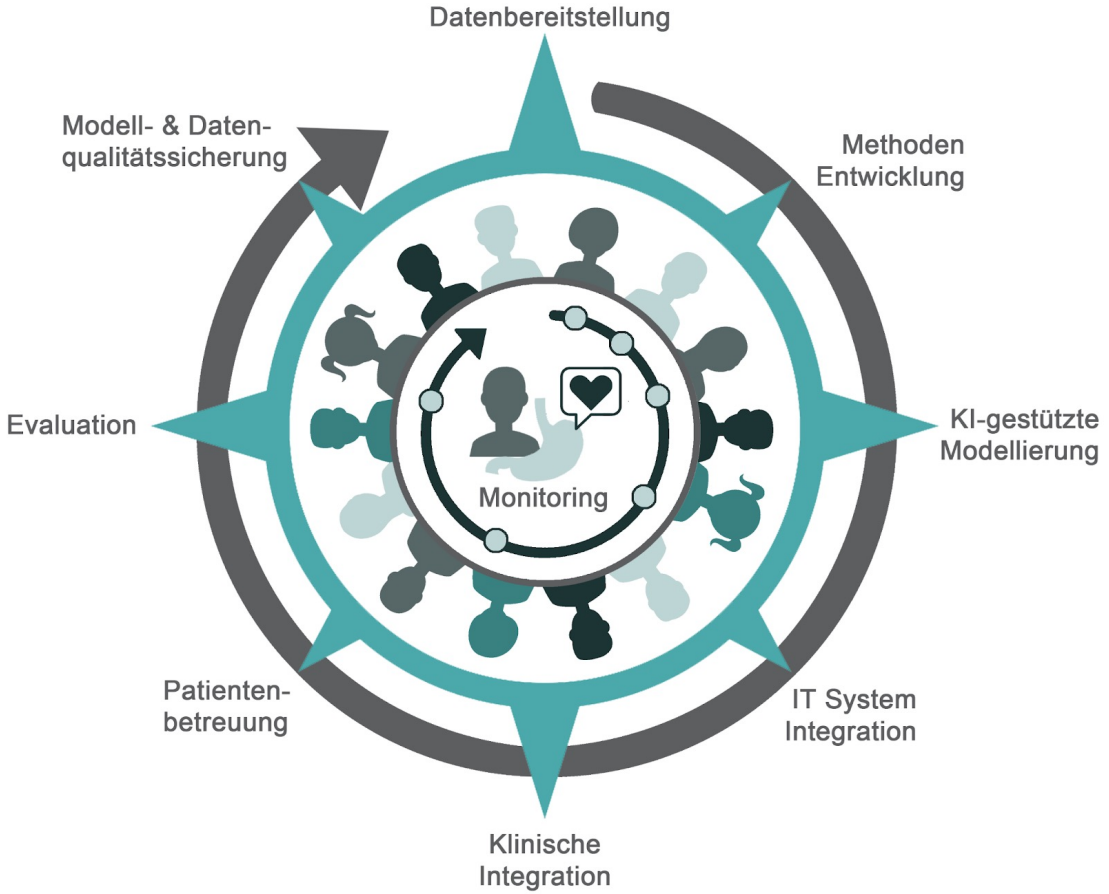
Muhammad-Agus Salim, Master thesis 2024

Bayesian network | Combing Expert Models with Embedding Models

KI COMPASS

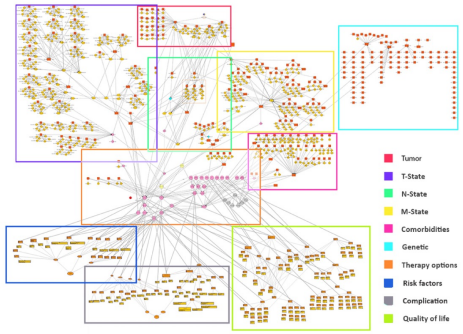
Knowledge-**I**ntegrated
Cardio-**O**ncological
Modelling and **P**rediction
for **A**dvanced **S**urvivorship **S**upport

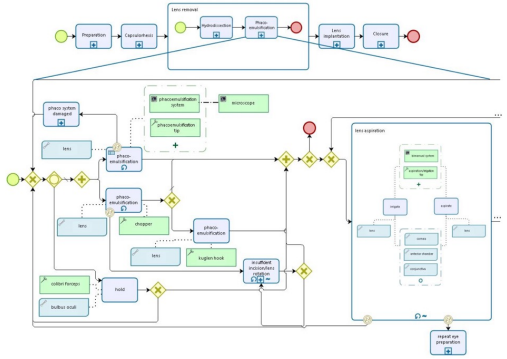


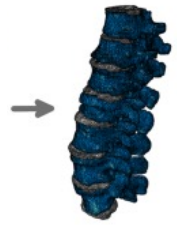


Data-driven vs. knowledge-based AI

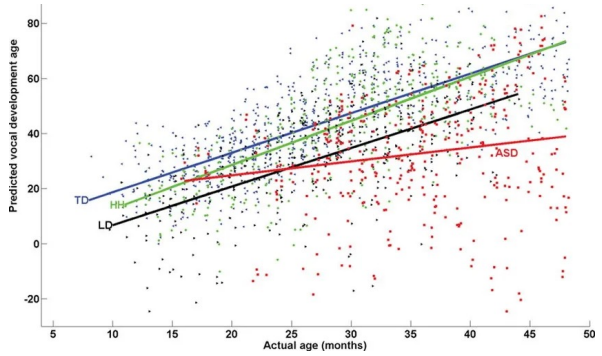
Understanding; Sharing; Improving

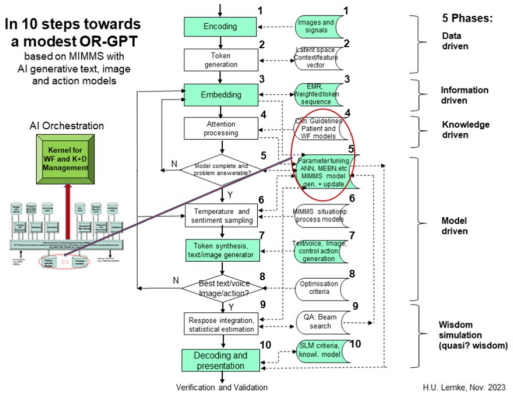


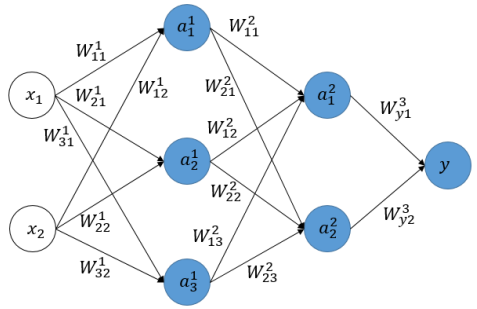




finite element mesh

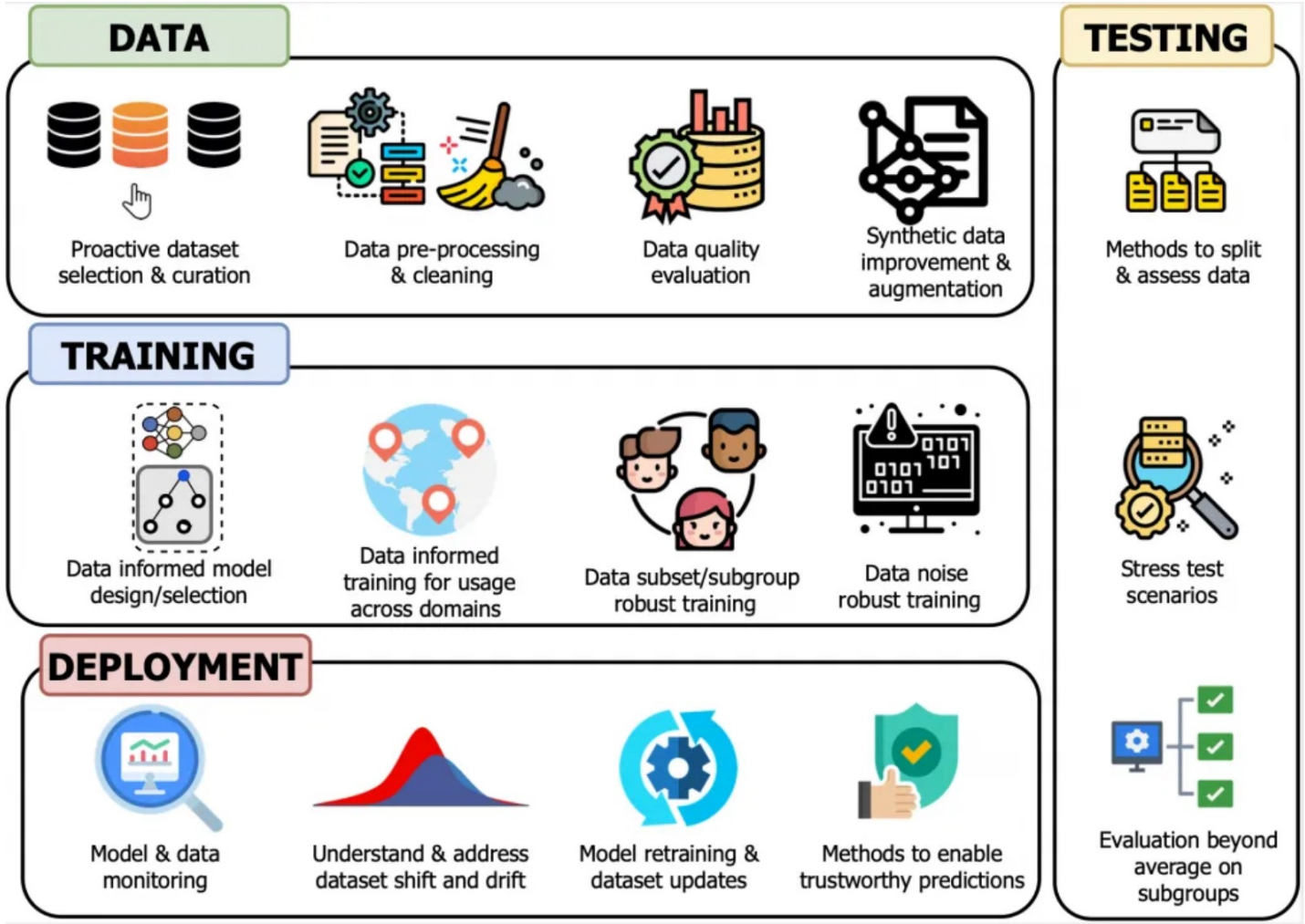






Data Centric AI

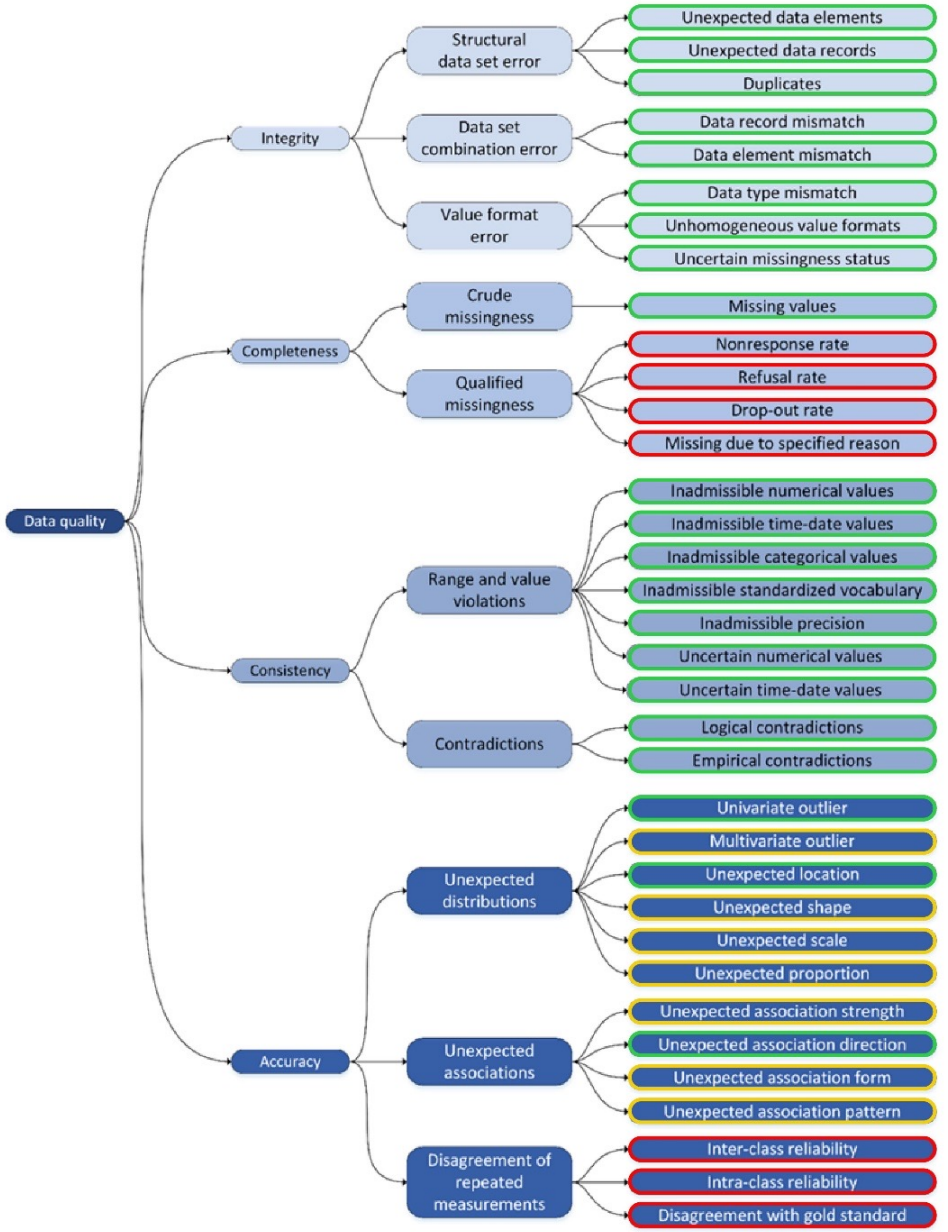
Data Centric AI



https://www.vanderschaar-lab.com/data-centric-ai/

Data Centric AI

Alban Bogdani; Master Project @IES Lab 2024



Source: https://dataquality.qihs.uni-greifswald.de/

End – Questions?

1. Explain the importance of personalization in AI-guided healthcare, especially when compared to population-based guidelines.
2. How can AI be used to support clinical decision-making in remote and extreme environments?
3. Describe the difference between CNNs and RNNs in the context of medical imaging and time-series data.
4. What are the key challenges in applying Bayesian Networks to clinical decision support systems?
5. Discuss the role of data-centric AI in improving healthcare applications. Provide examples of its impact.
6. How do LSTMs address the limitations of traditional RNNs when analyzing patient data?
7. Compare and contrast the use of GANs in data augmentation for medical imaging versus their use in molecular analysis for drug discovery.
8. What are the main limitations of LLMs when applied to medical data, and how can these be mitigated?
9. Describe a scenario where explainable AI is crucial for ensuring transparency in healthcare, and outline the methods used to achieve it.
10. How can time-series analysis with RNNs or LSTMs be applied to predict patient outcomes in an ICU setting?
11. Compare supervised learning with unsupervised learning in the context of healthcare applications. Provide examples where each approach would be most effective.
12. What are the ethical considerations when using AI for predictive modeling in clinical settings, particularly concerning patient privacy and bias?